Foundations of Sensitivity Analysis

Theory, Computation, and Applications

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Preface

This book develops sensitivity analysis (SA) as a mathematical discipline, not merely as a collection of computational recipes. SA asks the foundational question of quantitative modeling: which parameters most strongly determine the model's predictions?

Three ideas run through every chapter:

1. The sensitivity index is a normalized derivative.
2. The adjoint is the transpose, in whatever space you work.
3. Local SA is a linearization, with all the limitations that implies.

Students who understand these three ideas (not just their definitions but their implications and connections) will be equipped to design, implement, and critique SA studies in any application domain.

The SIR epidemic model serves as the running example throughout. Every method is demonstrated on this one model, so students can track increasing mathematical sophistication against a fixed biological context. All computational examples use MATLAB; each chapter includes a structured laboratory and a reusable code template. All code is freely available in the companion repository:

`https://github.com/mhyman/arriola2026sa`

The repository contains the complete `src/` library of production-quality MATLAB functions (one per major algorithm), three test suites with 48 automated checks, and this instructor's solutions manual.

The adjoint method (Chapters 5–7) is the conceptual centerpiece of the book. It is the same idea encountered three times in three different mathematical spaces: the matrix transpose in linear algebra, the reversed computation graph in algorithmic differentiation, and a backward ODE in dynamical systems. Chapter 8, *Caveat Emptor*, asks when sensitivity analysis can mislead, and is the chapter students most frequently cite as the one that changed how they think about mathematical models.

Relationship to existing literature. Cacuci (1981) established the adjoint formalism for nonlinear systems; Giles and Pierce (2000) developed it for aeronautical design; Cao et al. (2002) gave the computational treatment for ODE systems; and Griewank and Walther (2008) treat the computation-graph view comprehensively. The present book differs from each of these in its pedagogical aim: it is the only undergraduate text that introduces all three forms of the adjoint (matrix, graph, and ODE) as instances of a single unifying idea, and that teaches students when *not* to trust the results they compute.

For instructors. The book covers a one-semester undergraduate course. Students need calculus, linear algebra, and ODEs; no measure theory or functional analysis is assumed. A minimal one-semester course can omit the computational AD material in Chapter 7 and all appendices without loss of continuity. Chapters 1–4 and 8–9 form a self-contained global-SA course for students who will not use the adjoint in their research. Exercises tagged [Bloom L5] and [L6] are designed for semester projects and take-home examinations; those tagged [L2]–[L4] are suitable for weekly problem sets.

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The MATLAB code library in the companion GitHub repository (`https://github.com/mhyman/arriola2`
was developed with support from the Tulane University Department of Mathematics.

Dedication.

To our students, past and future:

May you always ask which parameters matter
before you run the model.

Part I

The Sensitivity Question

Chapter 1

Introduction and Overview

Sensitivity analysis begins when a model is forced to say what matters.

An infectious disease has just entered a new country. The public health response team has built a compartmental model with fourteen parameters: transmission rates, recovery rates, contact patterns, immunity duration, disease-induced mortality, and more. The team has funding to deploy exactly two interventions and time to measure exactly three parameters with high precision before a policy decision must be made.

The question due by tomorrow morning is this: *which three parameters are worth measuring, and which two interventions will most effectively reduce the spread?*

This is the problem that sensitivity analysis is designed to answer. Before the mathematics, we need a language for asking the question precisely.

1.1 The Central Question

Every mathematical model takes inputs and produces outputs. The inputs include parameters whose exact values may be uncertain or whose values might be changed through intervention. The outputs include whatever the model is designed to predict. Sensitivity analysis asks: *which inputs have the most effect on the outputs we care about?*

To answer this precisely, we adopt two terms used throughout the book.

A **Parameter of Interest** (POI) is any input to the model whose variation is worth studying. In practice, a parameter becomes a POI for one of three reasons, and the reason matters for how you interpret the sensitivity index.

1. **Epistemic uncertainty** (uncertainty from lack of knowledge). The parameter is fixed by nature but imperfectly known: different studies report different estimates, the data are sparse, or the measurement process is imprecise. Epistemic uncertainty is *reducible* — better data, improved experiments, or longer observation windows can sharpen the estimate. Sensitivity analysis identifies which epistemically uncertain parameters most affect the output, directing limited data-collection resources toward the measurements that would reduce output uncertainty the most. In the disease example, the transmission probability per contact is epistemically uncertain: studies estimate it from contact-tracing data, but estimates vary across populations and settings.

2. **Aleatory uncertainty** (inherent stochastic variability). The parameter genuinely varies from individual to individual, place to place, or time to time, and this variation is *irreducible*— it persists even with unlimited data. More observations characterize the distribution more precisely but do not eliminate the randomness. In the disease example, the contact rate varies stochastically across the population: individuals differ in how many contacts they make per day. Sensitivity analysis with aleatory uncertainty typically asks how much of the output variance is attributable to each stochastic source, motivating the global SA methods of Chapter 9.

3. **Control variable** (a lever the investigator can move). The parameter is known and could, in principle, be changed by a policy decision, intervention, or design choice. Sensitivity analysis in this context answers the question: which intervention has the most leverage over the outcome I care about? In the disease example, the contact rate can be reduced by social distancing policies, and the transmission probability can be reduced by mask requirements or vaccination. The sensitivity index tells the policy maker which reduction — contacts or transmission — produces the largest reduction in R_0.

Many parameters fall into more than one category simultaneously: the contact rate is both uncertain (its current value is imprecisely known) and controllable (it can be targeted by policy). In practice, the interpretation of a sensitivity index should always be read in light of which role the POI is playing. A large sensitivity index for an epistemically uncertain parameter is a call to collect better data; for an aleatory parameter, it is a call to characterize the distribution more carefully; for a control variable, it is a call to prioritize that intervention.

In the disease example above, the transmission probability per contact is primarily a POI because it is epistemically uncertain (different studies report different values) and because it can be targeted by an intervention (vaccines reduce it). The contact rate is a POI for both the same reasons: it varies across populations (epistemic and aleatory uncertainty) and can be reduced through social distancing policies (control variable).

A **Quantity of Interest** (QOI) is any model output that the investigator cares about predicting or controlling. The peak number of infectious individuals, the total number of deaths over a six-month period, and the basic reproduction number $\mathcal{R}_0$ are all plausible QOIs for an epidemic model. The choice of QOI is not neutral: different choices lead to different rankings of parameter importance.

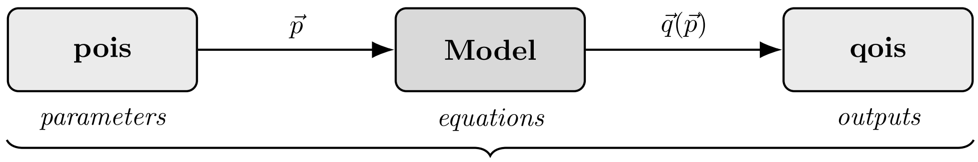
The relationship between POIs and QOIs is the subject of this book. Specifically, we want to know how much each POI affects each QOI, so that we can rank parameters by importance, design interventions efficiently, and allocate measurement resources wisely.

One of the most persistent misconceptions in mathematical modeling is the assumption that all parameters matter equally until proven otherwise. They do not. In virtually every model studied in this book, a small number of POIs account for most of the variation in the QOIs, while the remaining parameters have negligible effect. Identifying that small number is one of the primary goals of sensitivity analysis, and the answer is rarely obvious from the model equations alone.

Self-check. In the outbreak scenario above, classify each of the following as a POI, a QOI, or neither: the transmission probability per contact; the peak number of infectious individuals; the day the first case was reported; the mean number of days a patient is infectious.

> *Answer:* POI: transmission probability (uncertain; can be reduced by vaccines), mean infectious period (uncertain; can be reduced by treatment). QOI: peak number of infectious individuals (model output we care about). Neither: day of first case (a historical fact, not a model parameter or prediction). Note that the mean infectious period could also be a QOI in a different study; this classification depends on the question being asked.

Figure 1.1 illustrates the basic structure. The forward problem takes nominal POI values as input and produces the QOI as output. Sensitivity analysis adds one more step: it asks how the QOI changes as the POIs vary.



Sensitivity Analysis: $S_{p_i}^{q_j} = \dfrac{p_i}{q_j}\dfrac{\partial q_j}{\partial p_i}$

Figure 1.1: The forward problem takes nominal values for the input POIs and produces the associated output QOI. Sensitivity analysis quantifies how changes in the POIs propagate to the QOI by computing the normalized partial derivative $S_{p_i}^{q_j}$ for each POI–QOI pair.

1.2 Two Directions: Forward and Adjoint Sensitivity

There are two complementary ways to ask the sensitivity question, and they differ in a fundamental way.

Figure 1.2 captures the distinction geometrically. In panel (a), a perturbation originates at a specific input location and spreads forward through the model, influencing a broad region of the output. This is **forward sensitivity analysis**. Panel (b) reverses the direction: a specified variation in the output is traced backward to the input locations that could have caused it. This is **adjoint sensitivity analysis**.

The "What if?" and "How come?" labels are not merely rhetorical. They correspond to genuinely different computational strategies and genuinely different questions.

1.2.1 Forward Sensitivity Analysis: "What if?"

Consider a disease model for an emerging outbreak. The transmission probability β is estimated from early case data, but the estimate carries significant uncertainty. A public health officer asks: *if our estimate of β is wrong by 10%, how much does our predicted peak infection count change?*

This is a forward question. We are tracing the effect of a change in an input POI (the transmission probability) forward to an output QOI (the peak infection count). Forward sensitivity analysis answers this question by computing a sensitivity index: a number that tells us, for a small percentage change in a POI, what percentage change to expect in the QOI.

A sensitivity index of $+0.8$ would mean that a 10% increase in β produces approximately an 8% increase in the peak infection count. A sensitivity index of -1.0 for the recovery rate would mean that a 10% increase in the recovery rate produces a 10% decrease in the peak. The sign tells you the direction; the magnitude tells you the amplification. Chapter 2 develops this definition carefully and shows how to compute these indices for a wide range of models.

Self-check. A forward sensitivity analysis gives $S_{\tau}^{R_0} = +1.2$ (the notation is defined formally in Chapter 2), where τ is the mean infectious period (in days) and R_0 is the basic reproduction number.

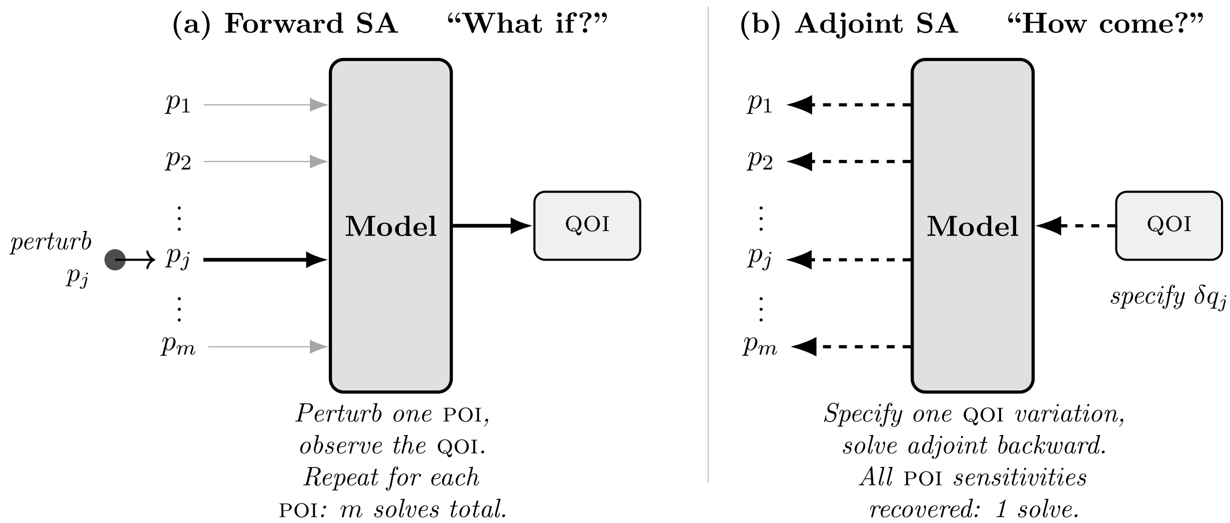


Figure 1.2: (a) Forward sensitivity analysis answers "What if?" questions. A perturbation to one input POI p_j propagates forward through the model and produces a change in the output QOI . Because the model must be solved once for each POI, forward sensitivity requires m solves for m POIs. (b) Adjoint sensitivity analysis answers "How come?" questions. A specified variation δq_j in the output QOI enters the model from the right and is propagated *backward* through it. A single adjoint solve simultaneously recovers the sensitivity of q_j with respect to every POI $p_1, \ldots, p_m$. The adjoint is developed in Chapters 5 and 6; for now, the key contrast is computational: forward SA costs m solves, adjoint SA costs 1.

If the mean infectious period increases by 5%, approximately what happens to R_0? Is this good news or bad news for disease control?

> *Answer:* R_0 increases by approximately $5\% \times 1.2 = 6\%$. This is bad news: a larger R_0 means more transmission. The positive sign tells us that POI and QOI move in the same direction, so longer infectious periods lead to more disease spread.

1.2.2 Adjoint Sensitivity Analysis: "How come?"

Now consider a different problem. City planners want to site a new industrial facility near a residential area. The facility emits pollutants, and regulators have set a standard: the average pollutant concentration at a specified monitoring station must not exceed a given threshold. A transport model predicts whether any candidate site meets this standard.

The planner's question is different in character from the previous one. Rather than asking "if this parameter changes, what happens to the output?", they want to know: *for a given allowed change in the output (the pollutant concentration), how much variation in the inputs (wind speed, emission rate, atmospheric stability) is permissible?*

This is an adjoint question. It runs the sensitivity analysis in reverse: from a specified variation in the QOI back to the POIs that could have caused it. For models with many POIs and only a few QOIs, adjoint sensitivity analysis is dramatically more efficient than computing forward sensitivities for every parameter individually. Chapters 5 and 6 explain why this is so and how to exploit it.

Self-check. An environmental regulator measures pollutant concentration at a monitoring station and finds it exceeds the legal threshold. She wants to determine which weather assumptions in the transport model are most likely responsible for the exceedance. Is this a forward or adjoint question? Explain in one sentence.

Answer: Adjoint. The regulator starts with a specified change in the QOI (exceedance of the threshold) and traces backward to identify which input POIs (weather assumptions) most plausibly caused it. A forward analysis would instead ask how changing a specific weather assumption would affect the predicted concentration, which runs in the opposite direction.

1.3 Global Sensitivity: When Local Is Not Enough

Both forward and adjoint sensitivity analysis are *local*: they measure the effect of small perturbations to the POIs near a fixed nominal value. This is powerful, but it assumes that the model behaves approximately linearly in the vicinity of that nominal value.

Many real models are not linear, and many POIs carry substantial uncertainty that cannot be called "small." In these cases, local sensitivity indices can be misleading. A POI with a small local sensitivity index may still have a large effect on the QOI when varied across its full range of plausible values, if the model is nonlinear in that parameter.

Example 1.1 (Local SI does not capture global behavior)**.** Consider $q(p) = p^2$ and suppose the nominal value is $p^* = 0.1$. The local sensitivity index is $S_p^q = (p/q)\,(dq/dp) = (p/p^2)(2p) = 2$, which looks moderate. But if p is uncertain and could range from 0.1 to 1.0, the QOI ranges from 0.01 to 1.0 — a hundredfold change. A local SI computed at $p^* = 0.1$ gives no information about this hundredfold range. Global SA, introduced in Chapter 9, is designed to capture exactly this kind of large-range nonlinear behavior.

Chapter 9 introduces **global sensitivity analysis**, which asks how the output varies when the POIs are varied across their entire plausible ranges simultaneously. The primary tool there, the Sobol index, measures how much of the total output variance is attributable to each POI. Global SA does not replace local SA; it complements it, answering questions that local SA cannot.

1.4 Sensitivity Analysis and Uncertainty Quantification

Sensitivity analysis and uncertainty quantification are related but distinct enterprises.

Sensitivity analysis, as developed in this book, is *deterministic*: we assign nominal values to the POIs and compute how the QOI changes when those values are perturbed. The POIs are treated as numbers with specific values, not as random variables. The sensitivity index is a derivative: a precise mathematical object that exists when the model is differentiable.

Uncertainty quantification treats the POIs as random variables with associated probability distributions. The goal is to determine the distribution of the QOI given the distributions of the POIs. This is a statistical enterprise, and its primary tools (Monte Carlo methods, Bayesian inference, polynomial chaos) are the subject of the companion volume by Smith [42]. Smith's Chapters 9 and 10 develop local and global sensitivity analysis at the graduate level, extending the material in this book with functional-analytic and measure-theoretic foundations. Smith's Chapter 6 treats parameter selection using the sensitivity Jacobian introduced in our Chapter 4, connecting it to the surrogate-model literature. The SA foundations developed here follow Arriola and Hyman [2, 3]. For the broader development of SA as a formal discipline, see Saltelli et al. [37], Razavi et al. [32], and the annotated historical timeline in Tarantola et al. [45].

The connection between the two fields is real and important. Global SA, covered in Chapter 9, bridges the gap: Sobol indices are defined in terms of variances, which requires probabilistic inputs, but they can be estimated by methods that build directly on the ideas in this book. Students who complete this course will find that the second semester's UQ material makes more sense because of the SA foundation established here.

1.5 Three Ideas That Run Through the Book

Every chapter of this book revisits one or more of three fundamental ideas. We name them here so that students can recognize them when they reappear, each time in a new setting.

Unifying Idea 1: The sensitivity index is a normalized derivative.

At its core, SA answers a single question: if this input changes by a small percentage, by what percentage does the output change? That ratio is the sensitivity index. It is not a mysterious new object: it is the derivative, normalized so that its magnitude is independent of units and scales.

This idea appears first in Chapter 2, where we define the sensitivity index formally. It reappears in Chapter 3 as a finite-difference approximation, in Chapters 4–6 as an analytic derivative, and in Chapter 9 as the foundation for Sobol indices. Whenever a sensitivity quantity appears in this book, it is an instance of this one idea. Smith [42] develops the same normalized-derivative framework in §9.1–9.2 and connects it to experimental design.

Unifying Idea 2: The adjoint is the transpose in whatever space you are working in.

The adjoint sensitivity method, which makes it possible to compute the sensitivities of one QOI with respect to hundreds of POIs at the cost of solving just two problems instead of hundreds, rests on a single mathematical operation: transposition. For a linear system $A\vec{u} = \vec{b}$, the adjoint involves the transposed matrix A^T. For a computation graph (the structure underlying every computer program), the adjoint is the chain rule applied in reverse order, which is the backpropagation operation familiar from machine learning. For an ordinary differential equation, the adjoint is a second ODE running backward in time.

These three forms look different on the surface. They are the same idea.

Chapter 5 introduces the adjoint through the matrix transpose, which is the simplest and most familiar form. Chapter 6 extends it to differential equations. At each transition, the book explicitly identifies the connection back to this unifying idea.

Unifying Idea 3: Local SA is a linearization.

Every sensitivity index in Chapters 2–8 is based on a derivative, and a derivative is a linear approximation to a generally nonlinear function. This approximation is valid when the perturbation is small and the model is smooth. It fails near bifurcation points, where the model changes qualitative behavior. It fails when the POI uncertainty is large. It fails when the model is strongly nonlinear in the POI of interest.

Chapter 8 (*Caveat Emptor*) documents the conditions under which local SA breaks down. Chapter 9 provides the global alternative. Both chapters are applications of this one principle: local SA is a linearization, and linearizations have limits.

1.6 A Note on Parameterization

One practical decision that profoundly affects the usefulness of SA results deserves mention before Chapter 2.

The sensitivity index depends on how a POI is defined. In a compartmental epidemic model, the recovery process can be parameterized either by the *recovery rate* γ (units: per day) or by the *mean infectious period* $\tau = 1/\gamma$ (units: days). These two parameterizations are mathematically equivalent, each determining the other. But their sensitivity indices are not equal: they have the same magnitude and opposite signs.

Which one should you use? The answer is: whichever one you can discuss in ordinary language without confusion. People who work with epidemic models (clinicians, public health officials, and policy makers) think naturally in terms of days. A clinician understands immediately what it means to say "the mean infectious period is 7 days." They do not think in recovery rates. Reporting $S_\gamma^{R_0} = -1.0$ requires the reader to invert γ mentally before the result makes intuitive sense. Reporting $S_\tau^{R_0} = +1.0$ is immediately clear: a longer infectious period increases transmission.

Throughout this book, we choose POIs that can be understood in everyday terms: durations rather than rates, probabilities rather than their complements, counts rather than densities when the distinction matters. This is not just a matter of aesthetics. Sensitivity indices that require mental inversion to interpret are sensitivity indices that will be misread.

Chapter 2 formalizes this principle and shows how to convert between parameterizations. Readers who find that their model's natural parameters are rates will learn how to restate their results in terms of the corresponding durations before reporting them.

1.7 Structure of the Book

The book is organized in five parts. Each part builds on the previous, and the SIR epidemic model [19, 47] serves as the recurring example throughout, so that students can track a single application as the mathematical tools become progressively more powerful.

Part I: The Sensitivity Question (Chapters 1–2) establishes the central problem and introduces the sensitivity index as the primary tool. Chapter 2 develops the formal definition, the sign interpretation, the tornado plot as a display format, and the principle of interpretable parameterization. By the end of Part I, students can compute and interpret sensitivity indices for any differentiable function of several variables.

Part II: Computing Sensitivity in Practice (Chapter 3) gives students an immediately usable computational method for any model, including models that cannot be differentiated analytically. Finite-difference approximations, step-size selection, the sensitivity Jacobian, and the treatment of correlated POIs are developed here. Students who work through Chapter 3 can perform SA on any model they can run, including inherited code and black-box simulation software.

Part III: Analytic Forward Sensitivity Analysis (Chapter 4) shows how to differentiate model equations directly to obtain exact sensitivity information at lower computational cost. The forward sensitivity equation is derived for algebraic systems, difference equations, and ordinary differential equations. The chapter ends by quantifying the computational bottleneck that motivates the adjoint approach.

Part IV: The Adjoint Approach (Chapters 5–7) introduces adjoint sensitivity analysis through three progressively more abstract settings: linear systems in Chapter 5, dynamical systems in Chapter 6, and computational tools including algorithmic differentiation in Chapter 7. The central theme is Unifying Idea 2: the adjoint is the matrix transpose in whatever mathematical space the problem inhabits.

Part V: Limitations and Extensions (Chapters 8–9) addresses the boundaries of local SA. Chapter 8 documents the conditions under which sensitivity indices give misleading results: bifurcation points, ill-conditioned systems, and method-question mismatch. Chapter 9 introduces global SA through variance decomposition, Sobol indices, and partial rank correlation coefficients. It closes with an explicit bridge to the second semester, where Smith's *Uncertainty Quantification* [42] takes over from the point where this book ends.

Computing environment. Every chapter includes a structured MATLAB laboratory that implements the chapter's core idea on the SIR epidemic model. Students who complete these laboratories will leave the course with a portable toolkit for performing SA on any model they encounter in research or practice. The complete MATLAB code library — production-quality functions, automated test suites, and usage examples — is available at:

`https://github.com/mhyman/arriola2026sa`

Each function in the repository corresponds directly to a concept or algorithm in the text. `sir_nominal.m` defines the SIR parameters used in every chapter; `sensitivity_jacobian.m` implements the centered-difference Jacobian from Chapter 3; `sobol_jansen.m` implements the Jansen estimators from Chapter 9. The `src/` directory is organized to mirror the book's structure, so students can find the code for any chapter without searching.

Notation. Throughout the book, the sensitivity index of a QOI q with respect to a POI p is written S_p^q. POIs are always chosen to be interpretable in everyday terms: durations, probabilities, counts. The relationship to the mathematical parameters in the model equations is made explicit wherever the two differ.

Scope note: scalar QOIs. Every sensitivity index in this book is computed for a *scalar* QOI — a single number such as the peak infection count, the total disease burden, or the value of R_0. Real applications sometimes require sensitivity of an entire output *trajectory*: how does the full time series $I(t)$ change when β changes? Extending SA to functional (trajectory-valued) QOIs requires either reducing the trajectory to a scalar (which loses information) or computing a time-dependent SI at each time point (as in Chapter 4's Figure 4.1). A rigorous treatment of functional sensitivity, using the adjoint ODE to compute sensitivity with respect to the entire output path, is developed in Smith [42] §9.4 and is a natural next step after mastering the scalar case.

1.8 Exercises

Each exercise carries a Bloom's Taxonomy tag indicating cognitive level: *L2* (understand), *L3* (apply), *L4* (analyze), *L5* (evaluate), *L6* (create and synthesize). Higher-level exercises require constructing an argument or designing a procedure, not just applying a formula.

Exercise 1.1 *([Bloom L2]).* Explain in your own words the difference between a POI and a QOI. For a model describing the spread of antibiotic-resistant bacteria in a hospital, give two examples of each. Then explain, in one sentence, what sensitivity analysis would tell you about this model that simply running the model would not.

Exercise 1.2 *([Bloom L2]).* What is the difference between forward and adjoint sensitivity analysis? Describe each in terms of the "What if?" and "How come?" framing. For each of the following questions, classify it as a *forward question* or an *adjoint question* based on the type of question being asked (not on which algorithm would be most efficient to use computationally — efficiency is the subject of Chapters 5 and 6).

(a) If the mean latent period increases by 10%, how does the peak infection count change?

(b) The peak infection count is higher than expected. Which parameter uncertainties are most likely to have caused this?

(c) Among all parameters in the model, which one has the largest effect on the basic reproduction number?

Exercise 1.3 *(*[Bloom L4]*)*. For each of the following modeling scenarios, state whether forward SA, adjoint SA, or global SA is most appropriate. Justify each choice in one or two sentences.

(a) A climate model has 500 tunable parameters. The investigator wants to know which 10 parameters most affect the mean surface temperature over a 100-year run. Each model run takes 8 hours.

(b) A pollution transport model has 6 input parameters. The regulator wants to know: if the pollution at a specific monitoring station exceeds the legal threshold, which input assumptions most likely caused this?

(c) A pharmacokinetic model has 4 parameters, each of which is known only to within a factor of 2. The investigator wants to know the range of predicted drug concentrations consistent with this uncertainty.

Exercise 1.4 *(*[Bloom L5]*)*. You are advising a team modeling the transmission of dengue fever. The model has 9 parameters. The team has funding to measure exactly 3 parameters precisely in the field and to implement exactly 2 vector-control interventions. Write a one-page plan describing how you would use SA to guide these decisions. Your plan should specify: what QOI you would focus on and why; how you would determine which 3 parameters to measure; what additional information (if any) you would need beyond the SA results to recommend interventions; and one important limitation of your approach that the team should be aware of.

Exercise 1.5 *(*[Bloom L6]*)*. A published report on a disease model states: "Sensitivity analysis was performed on all 22 model parameters simultaneously, demonstrating that the model is robust to parameter uncertainty." The analysis was conducted by computing local sensitivity indices at the best-fit parameter values.

Identify at least two substantive problems with this description or its conclusions. For each problem you identify, explain specifically what the authors should have done differently.

Chapter 2

Measuring Local Sensitivity

A sensitivity index is a normalized derivative with a practical job.

Calibration tells you which parameter values fit the data.
Sensitivity analysis tells you which parameters matter to the answer.

Two studies of the same epidemic model arrive at the following conclusions.

Study A reports: "Increasing the recovery rate by 1% decreases R_0 by 1%."

Study B reports: "Increasing the mean infectious period by 1% increases R_0 by 1%."

Both statements are mathematically correct. The recovery rate and the mean infectious period are reciprocals of each other, so the two studies analyzed the same model, used the same mathematics, and obtained the same numerical result. Yet they convey opposite signs and, more importantly, very different intuitions. Study A requires the reader to mentally invert a parameter before the public health implication becomes clear. Study B is immediately interpretable: patients who are infectious longer spread the disease more.

This chapter introduces the sensitivity index, the central quantitative tool of this book, and develops the conventions that make sensitivity analysis results readable rather than merely correct.

2.1 Ratios as Precursors to Derivatives

Comparison by ratio is one of the most fundamental quantitative habits in applied science. In finance, the price-to-earnings ratio tells an investor how many dollars they pay for one dollar of profit. In aeronautical engineering, the lift-to-drag ratio determines whether a wing design is viable. In experimental physics, the signal-to-noise ratio separates measurement from artifact. None of these ratios requires calculus. Each one gives a dimensionless number that can be compared across different systems and used to make decisions.

The sensitivity index is a ratio of this same type: it measures how much the output changes, per unit change in the input, normalized so that the result is dimensionless and comparable across parameters with different units and scales.

To see why normalization matters, consider the price-to-earnings ratio P/E. Suppose a stock trades at \$50 per share and has earned \$2.50 over the past year. The P/E ratio is $50/2.50 = 20$: you pay \$20 for each dollar of annual earnings. Now suppose earnings increase by \$0.50 to \$3.00. A raw comparison of the two earnings figures gives $\$3.00 - \$2.50 = \$0.50$, which tells you nothing useful without knowing the original value. The normalized comparison is $(3.00 - 2.50)/2.50 = 20\%$, a 20% increase in earnings, regardless of whether the numbers are in dollars, euros, or any other currency.

A derivative, in the same spirit, measures the rate of change of the output per unit change in the input. The sensitivity index normalizes this derivative so that it answers a specifically useful question: if the input changes by a small percentage, by what percentage does the output change?

Self-check. A second stock trades at \$45 per share and has earned \$3.00 per year. Calculate the P/E ratio for each stock. Which stock is a better value by this measure, and why?

Answer: Stock 1: $P/E = 50/2.50 = 20$. Stock 2: $P/E = 45/3.00 = 15$. Stock 2 has a lower P/E ratio, meaning you pay less per dollar of earnings, a better value by this measure. The ratio provides a dimensionless comparison that is fair across the two stocks despite their different prices and earnings.

2.2 Absolute, Relative, and Regularized Change

Before defining the sensitivity index, we need precise language for describing how quantities change. Three measures of change appear throughout the book, each appropriate in different circumstances.

2.2.1 Absolute Change

Let u denote the solution to a model, and suppose a perturbation δu is applied. The **absolute change** is simply the difference between the perturbed and original solutions:

$$\Delta u_{\text{Abs}} := \delta u = \hat{u} - u, \tag{2.1}$$

where $\hat{u} = u + \delta u$ is the perturbed solution. A 1-inch change in the length of a string has absolute change 1 inch, regardless of the original length.

2.2.2 Relative Change

The 1-inch change means something very different if the string is 6 inches long versus 1 mile long. The **relative change** accounts for this by normalizing against the original value:

$$\Delta u_{\text{Rel}} := \frac{\delta u}{u} = \frac{\hat{u} - u}{u}. \tag{2.2}$$

For the 6-inch string, the relative change is $1/6 \approx 16.7\%$. For the 1-mile string (63,360 inches), it is $1/63{,}360 \approx 0.0016\%$. The same absolute change has a very different significance depending on context, and relative change captures this.

2.2.3 Regularized Change

Relative change has one practical problem: if $u \approx 0$, the denominator in (2.2) is near zero and the ratio becomes unreliable. In dynamical models, the solution can pass through or near zero during its evolution: the infectious compartment $I(t)$ in an epidemic approaches zero as the outbreak resolves, for instance. The **regularized change** handles this case by adding a small correction to the denominator:

$$\Delta u_{\text{Reg}} := \frac{\delta u}{1 + u}. \tag{2.3}$$

When $u \approx 0$, this approximates absolute change: $\delta u/(1+0) = \delta u$. When $u \gg 1$, it approximates relative change: $\delta u/(1+u) \approx \delta u/u$. The regularized change interpolates smoothly between the two regimes.

Looking more carefully at regularized change reveals a pattern that recurs throughout applied mathematics: when a formula fails at a special value, adding a small stabilizing term to the denominator restores well-definedness without significantly changing the result away from the problematic point. The same idea appears in numerical linear algebra (Tikhonov regularization of ill-conditioned systems) and in Bayesian statistics (pseudocounts in categorical models). The regularized sensitivity index in Chapter 3 is one more instance.

Self-check. The infectious population in a disease model drops from $I = 0.04$ to $\hat{I} = 0.03$. Calculate the absolute, relative, and regularized change. Which measure gives the most informative description of this transition, and why?

> *Answer:* Absolute: $\delta I = 0.03 - 0.04 = -0.01$. Relative: $-0.01/0.04 = -0.25$, a 25% decrease. Regularized: $-0.01/(1+0.04) \approx -0.0096$, close to absolute.
> Since I is small relative to 1, relative change gives the most interpretable result: a 25% decrease in the infectious population is meaningful regardless of scale. For $I \ll 1$, absolute and regularized change are both small in magnitude and harder to interpret without the normalization that relative change provides.

2.3 The Sensitivity Index

We now define the central tool of local sensitivity analysis.

2.3.1 The Sign Rule

Before any formal definition, the most important thing to know about the sensitivity index is what its sign means.

The sign rule: A positive sensitivity index means that the QOI and POI move in the same direction: increase the POI and the QOI increases. A negative sensitivity index means they move in opposite directions: increase the POI and the QOI decreases. The magnitude tells you by how much: a sensitivity index of 2 means that a 1% change in the POI produces approximately a 2% change in the QOI.

Important: the sign belongs to the *parameterization*, not to the underlying biology. The recovery rate γ and the mean infectious period $\tau = 1/\gamma$ describe the same biological process, yet $S_{\gamma}^{R_0} < 0$ while $S_{\tau}^{R_0} > 0$. When two papers report opposite signs for what appears to be the same parameter, check whether they are using rate versus duration parameterizations before concluding there is a contradiction. Section 2.6 illustrates this with the SIR model.

Students sometimes interpret a negative sensitivity index as indicating an error in the calculation. It is not. A negative index is often the best possible news: if the sensitivity of R_0 with respect to the mean infectious period is positive, that means longer illness leads to more transmission, and the corresponding statement with the recovery rate would be negative: faster recovery reduces transmission. Both facts are correct; the sign simply tells you which direction the influence runs. With this intuition established, the formal definition follows naturally.

2.3.2 Formal Definition

Definition 2.1 (Sensitivity Index)**.** Let q denote a QOI that depends differentiably on a POI p. The **normalized sensitivity index** of q with respect to p is

$$S_p^q := \lim_{\delta p \to 0} \frac{\left(\dfrac{\delta q}{q}\right)}{\left(\dfrac{\delta p}{p}\right)} = \frac{p}{q}\frac{\partial q}{\partial p}, \qquad q, p \neq 0. \tag{2.4}$$

Three equivalent forms of this definition are in common use. All three give the same number; which one is most convenient depends on the problem at hand.

Ratio form (Definition 2.1): the ratio of relative changes in the QOI and the POI. This form is closest to the P/E ratio analogy from Section 2.1.

Derivative form: multiply the partial derivative by the ratio of nominal values:

$$S_p^q = \frac{p}{q}\frac{\partial q}{\partial p}. \tag{2.5}$$

This is the form used most often in computation. The factor p/q does the normalization work: it converts $\partial q/\partial p$, which has units of $[Q]/[P]$ and depends on the choice of units, into a dimensionless percent-for-percent ratio. Concretely, if $S_p^q = 2$, then a 1% increase in p produces approximately a 2% increase in q, regardless of whether p is measured in days, kilograms, or dollars.

Logarithmic form: for positive POIs and QOIs,

$$S_p^q = \frac{\partial \ln q}{\partial \ln p}. \tag{2.6}$$

This form reveals that the sensitivity index is the slope of the log-log plot of q versus p, a fact that becomes useful when comparing SI values across very different scales.

The three forms are equivalent whenever q and p are positive and q is differentiable with respect to p.

Unifying Idea 1. The sensitivity index is the central object of this book. It appears here as a ratio of relative changes. In Chapters 3 and 4 it reappears as a finite-difference approximation and as the solution of a sensitivity equation. In Chapters 5 and 6, it appears as the output of the adjoint calculation. In Chapter 9, it is the foundation for Sobol variance decomposition. Every one of these is the same normalized derivative, asked in the same way: *if the input changes by x%, by what percentage does the output change?*

Self-check. Use each of the three forms to compute the sensitivity index of the area $A = \pi r^2$ with respect to the radius r. Verify that all three give the same answer.

Answer: Ratio form: $\delta A/A = 2\pi r\, \delta r/(\pi r^2) = 2\delta r/r$, so $S_r^A = (\delta A/A)/(\delta r/r) = 2$.
Derivative form: $\partial A/\partial r = 2\pi r$, so $S_r^A = (r/\pi r^2)(2\pi r) = 2$.
Logarithmic form: $\ln A = \ln \pi + 2 \ln r$, so $\partial \ln A/\partial \ln r = 2$.
All three give $S_r^A = 2$: a 1% increase in radius produces a 2% increase in area. The circle's area is twice as sensitive to radius as to a parameter that appears linearly.

2.3.3 Interpreting the Magnitude

The sensitivity index answers a specific quantitative question: if the POI changes by x%, the QOI changes by approximately $x \cdot S_p^q$%. For small x, this estimate is accurate to first order. For larger

x, second-order corrections become relevant, and Chapter 3 discusses how to assess whether those corrections matter.

As a practical interpretive guide:

- $|S_p^q| > 1$: the QOI is more sensitive than the POI (the change in QOI exceeds the change in POI, percentage-for-percentage).
- $|S_p^q| = 1$: the QOI changes in exact proportion to the POI.
- $|S_p^q| < 1$: the QOI is less sensitive than the POI (the change in QOI is smaller than the change in POI).
- $|S_p^q| \approx 0$: the QOI is essentially insensitive to this POI near the nominal value.

2.4 Worked Examples

The following examples build in complexity, beginning with purely geometric functions and ending with the epidemic model that serves as the book's primary running example.

2.4.1 Volume of a Cylinder

The volume of a cylinder with radius r and height h is $V = \pi r^2 h$. We want to know which dimension (radius or height) has the greater effect on the volume.

Using the derivative form (2.5):

$$S_r^V = \frac{r}{V}\frac{\partial V}{\partial r} = \frac{r}{\pi r^2 h} \cdot 2\pi r h = 2. \tag{2.7}$$

$$S_h^V = \frac{h}{V}\frac{\partial V}{\partial h} = \frac{h}{\pi r^2 h} \cdot \pi r^2 = 1. \tag{2.8}$$

The volume is twice as sensitive to radius as to height. A 1% increase in radius produces a 2% increase in volume; the same percentage increase in height produces only a 1% increase. This follows from the fact that r appears squared in the volume formula while h appears linearly.

For a designer building a cylindrical tank with a fixed material budget, this result has a direct implication: small manufacturing errors in the radius matter twice as much as the same-sized errors in the height.

Self-check. For a cone with volume $V = \frac{1}{3}\pi r^2 h$, compute S_r^V and S_h^V. Compare with the cylinder results. Explain why the answer is the same despite the different formula.

Answer: $S_r^V = (r/V)(\partial V/\partial r) = (r/V)\cdot(2\pi r h/3) = 2$. $S_h^V = 1$. Identical to the cylinder. The reason: the sensitivity index depends on the *exponent* of each variable in the formula. Both $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ and $V_{\text{cylinder}} = \pi r^2 h$ have r to the power 2 and h to the power 1. The constant $\frac{1}{3}$ cancels in the logarithmic form $\partial \ln V/\partial \ln r = 2$. This is a general principle: for a monomial $q = c\,p_1^{a_1} p_2^{a_2}\cdots$, the sensitivity index with respect to p_i is simply a_i.

2.4.2 Arc Length of a Line

Let $u(t) = mt + b$ be a line with slope m and intercept b, defined on $[0, a]$. The arc length is

$$J(m, b) = a\sqrt{1 + m^2}.$$

Computing the sensitivity indices:

$$S_m^J = \frac{m}{J}\frac{\partial J}{\partial m} = \frac{m}{a\sqrt{1+m^2}} \cdot \frac{am}{\sqrt{1+m^2}} = \frac{m^2}{1+m^2}. \tag{2.9}$$

$$S_b^J = \frac{b}{J}\frac{\partial J}{\partial b} = 0, \tag{2.10}$$

since J does not depend on b.

The arc length is completely insensitive to the intercept. Geometrically, this is clear: shifting the line vertically does not change its length. Analytically, b does not appear in the formula for J, so its derivative is zero. A zero sensitivity index is not an error; it tells us that one POI has no local effect on the QOI, which is itself an informative finding.

Self-check. For the line $u(t) = 3t + 2$ on $[0, 4]$, compute S_m^J numerically. Verify: if m increases from 3 to 3.03 (a 1% increase), does J change by approximately S_m^J%?

Answer: $S_m^J = m^2/(1 + m^2) = 9/(1 + 9) = 0.9$.
At $m = 3$: $J = 4\sqrt{10} \approx 12.649$. At $m = 3.03$: $J = 4\sqrt{1 + 3.03^2} = 4\sqrt{10.1809} \approx 12.763$. Change: $(12.763 - 12.649)/12.649 \approx 0.90\% \approx 0.9 \times 1\%$. ✓

2.4.3 Area Under a Line

For the same line $u(t) = mt + b$ on $[0, a]$, the area is

$$J(m, b) = \int_0^a (mt + b)\,dt = \tfrac{1}{2}ma^2 + ba.$$

Both parameters now affect the area:

$$S_m^J = \frac{m}{\frac{1}{2}ma^2 + ba} \cdot \tfrac{1}{2}a^2 = \frac{am}{am + 2b}, \tag{2.11}$$

$$S_b^J = \frac{b}{\frac{1}{2}ma^2 + ba} \cdot a = \frac{2b}{am + 2b}. \tag{2.12}$$

Note that $S_m^J + S_b^J = 1$, so the two indices share a unit total for this particular linear functional. This is not a general property: sensitivity indices do not sum to any fixed value for arbitrary models or QOIs. Which parameter matters more depends on the values: $S_m^J > S_b^J$ when $am > 2b$, and the inequality reverses when $am < 2b$.

This example illustrates a general phenomenon: when two POIs both influence the same QOI, their relative importance depends on the nominal parameter values, not just on the model structure. SA must always be reported at specific nominal values.

2.4.4 Root of a Nonlinear Equation

Consider the nonlinear equation

$$f(u; \alpha) := \sin(\alpha u) - u = 0.$$

For $\alpha = 2$, the equation has three roots; for $\alpha > 1$, there is at least one nonzero root. We want to know how the positive root $u^*(\alpha)$ changes with α.

Differentiating $f(u; \alpha) = 0$ implicitly with respect to α:

$$u\cos(\alpha u) + \alpha\cos(\alpha u)\frac{\partial u}{\partial \alpha} - \frac{\partial u}{\partial \alpha} = 0.$$

Solving for $\partial u/\partial \alpha$:

$$\frac{\partial u}{\partial \alpha} = \frac{u\cos(\alpha u)}{1 - \alpha\cos(\alpha u)}.$$

At $\alpha = 2$, the positive root is $u^* \approx 0.9475$, giving

$$S_\alpha^{u^*} = \frac{\alpha}{u^*} \cdot \frac{u^* \cos(\alpha u^*)}{1 - \alpha \cos(\alpha u^*)} \approx -0.39.$$

A 1% increase in α decreases the positive root by approximately 0.39%. This result requires no numerical experiment once the derivative is known; the sensitivity index provides the estimate directly.

2.5 The Sensitivity Index for the SIR Model

The SIR epidemic model [19, 1] describes the spread of an infectious disease through a population divided into susceptible (S), infectious (I), and recovered (R) compartments. The basic reproduction number

$$R_0 = \frac{k\beta}{\gamma}$$

measures the average number of new infections produced by one infectious individual in an otherwise susceptible population, where k is the average number of contacts per day, β is the probability of transmission per contact, and γ is the recovery rate. The rigorous definition of R_0 via the next-generation operator is given in van den Driessche and Watmough [46] and Diekmann et al. [11].

This is the book's primary running example. In Chapter 4, we will compute time-dependent sensitivity indices for the full SIR model. Here we compute the static sensitivity indices for R_0: a closed-form calculation that introduces all the key ideas in a familiar setting.

2.5.1 Choosing Interpretable POIs

The parameters k, β, and γ appear directly in the model equations, which makes them a natural choice for POIs. However, we argued in Chapter 1 that POIs should be chosen to be interpretable in everyday terms.

The recovery rate γ (units: day^{-1}) is not directly interpretable. What a clinician understands and can measure is the *mean infectious period* $\tau = 1/\gamma$ (units: days): the average number of days a patient remains infectious. These two parameterizations are linked by:

$$S_\tau^q = -S_\gamma^q \qquad \text{for any QOI} q, \tag{2.13}$$

since $\partial q/\partial \tau = (\partial q/\partial \gamma)(\partial \gamma/\partial \tau) = -\gamma^2(\partial q/\partial \gamma)$, giving $S_\tau^q = (\tau/q)(\partial q/\partial \tau) = -(\gamma/q)\gamma(\partial q/\partial \gamma) = -S_\gamma^q$.

The magnitude is unchanged; the sign flips. Table 2.1 shows the effect of this reparameterization for R_0.

POI	Interpretation	Form	$S_p^{R_0}$
Recovery rate γ (day^{-1})	—	rate	-1
Mean infectious period $\tau = 1/\gamma$ (days)	"average days infectious"	duration	$+1$
Contact rate k (day^{-1})	"contacts per day"	rate	$+1$
Transmission probability β (dimensionless)	"chance of infection per contact"	probability	$+1$

Table 2.1: Sensitivity indices of R_0 with respect to rate and duration parameterizations of the SIR model. The mean infectious period $\tau = 1/\gamma$ has the same magnitude as γ but opposite sign. The duration form is easier to interpret: a longer infectious period increases transmission, and both the sign and the plain-language statement agree.

2.5.2 Analytic Computation

With the interpretable POIs $\{\tau = 1/\gamma,\ k,\ \beta\}$, the sensitivity indices are:

$$S_k^{R_0} = \frac{k}{R_0} \cdot \frac{\partial R_0}{\partial k} = \frac{k}{k\beta/\gamma} \cdot \frac{\beta}{\gamma} = 1, \tag{2.14}$$

$$S_\beta^{R_0} = \frac{\beta}{R_0} \cdot \frac{\partial R_0}{\partial \beta} = \frac{\beta}{k\beta/\gamma} \cdot \frac{k}{\gamma} = 1, \tag{2.15}$$

$$S_\tau^{R_0} = -S_\gamma^{R_0} = -\frac{\gamma}{R_0} \cdot \frac{\partial R_0}{\partial \gamma} = -\frac{\gamma}{k\beta/\gamma} \cdot \left(\frac{-k\beta}{\gamma^2}\right) = +1. \tag{2.16}$$

All three POIs have sensitivity index $+1$ with the interpretable parameterization. In plain language: a 1% increase in contact rate, transmission probability, *or* mean infectious period each increases R_0 by 1%. The three parameters are equally important for R_0, and all three promote disease spread when increased, as every positive sign and every positive biological interpretation confirms.

Self-check. An intervention reduces the contact rate k by 20%. Using the sensitivity indices above, estimate the change in R_0. If a second intervention reduces the mean infectious period τ by 15%, which intervention has the greater effect on R_0?

> *Answer:* First intervention: $\Delta R_0/R_0 \approx S_k^{R_0} \times (-20\%) = -20\%$. Second intervention: $\Delta R_0/R_0 \approx S_\tau^{R_0} \times (-15\%) = -15\%$. The first intervention has the greater effect. Both indices equal 1, so the percentage change in R_0 equals the percentage change in the POI. The intervention that changes its POI by the larger percentage has the larger effect.

2.6 Choosing Interpretable POIs

The SIR example illustrates a general principle that should guide every SA study before any calculation begins.

Choose pois that can be discussed in ordinary language, not merely because they appear in the model equations.

Model equations are written in the language that is mathematically convenient, which is not always the language that is scientifically or operationally natural. Rates (with units day^{-1}) are mathematically convenient in differential equations but not naturally interpretable; the corresponding mean times (with units days) are easier to discuss in a clinical or policy context.

The general rule for rate parameters in compartmental models: if a parameter p is a per-capita rate (units: time^{-1}), define the corresponding POI as $\tau = 1/p$ (the mean time spent in that state), compute $S^q_\tau = -S^q_p$, and report the result in terms of τ.

Two additional cases arise frequently:

Dimensionless probabilities. Transmission probabilities like β are already dimensionless and directly interpretable: "a 25% chance of transmission per contact" requires no conversion.

Composite parameters. In the SIR model, k and β always appear together as the product $k\beta$ in the expression for the force of infection $\lambda = k\beta I/N$. Neither can be identified separately from incidence data alone; only the product $k\beta$ is identifiable. In this case, $k\beta$ is a more natural single POI than either factor individually. A related issue arises whenever two POIs are not independently changeable, which leads to the topic of correlated POIs in Chapter 3.

A practical check: if you cannot explain what a 1% increase in a POI means in plain language to a colleague outside mathematics, reconsider the parameterization.

It is worth noting that even in highly cited published work, this distinction is sometimes overlooked. Section 4.1 of Chitnis, Hyman & Cushing [10], a paper that has become a standard reference for SA in mathematical epidemiology, reports sensitivity indices using rate parameters throughout. The SI values are mathematically correct, but several indices are negative for parameters that clearly promote disease spread, requiring a mental inversion before the public health interpretation is clear. The present book recommends always asking, before reporting any SI: would a policy maker understand this sign immediately, or does it require translation?

One further misconception deserves attention here. A sensitivity index near zero does *not* mean a parameter is globally unimportant. It means the model is approximately linear in that parameter near the nominal value, and the linear term happens to be small. If the model is nonlinear in that parameter, varying it over a wider range may reveal substantial effects that the SI misses entirely. Chapter 9 addresses this through global sensitivity analysis.

Self-check. A malaria model contains the parameter μ_v (mosquito mortality rate, units: day^{-1}). A colleague reports $S^{R_0}_{\mu_v} = -0.3$. Reparameterize in terms of mean mosquito lifespan $L = 1/\mu_v$ (units: days). What is $S^{R_0}_L$? Write a plain-language interpretation of each sensitivity index.

Answer: $S^{R_0}_L = -S^{R_0}_{\mu_v} = +0.3$.
Rate form: "A 10% increase in mosquito mortality rate decreases R_0 by 3%." This requires mental inversion: higher mortality → shorter lifespan → less transmission.
Duration form: "A 10% increase in mean mosquito lifespan increases R_0 by 3%." Direct: longer-lived mosquitoes transmit more malaria.

2.7 The Tornado Plot

When a model has several POIs, the collection of sensitivity indices becomes a list of numbers with signs and magnitudes. Reading such a list is manageable for three or four parameters; for ten or twenty, a visual display is essential.

The **tornado plot** is the standard display format for sensitivity indices. It is a horizontal bar chart with the following conventions:

- Bars extending to the *right* indicate positive sensitivity indices (POI and QOI move in the same direction).
- Bars extending to the *left* indicate negative sensitivity indices (opposite directions).
- Bars are sorted from largest to smallest by absolute magnitude, with the most influential parameter at the top.
- The length of each bar represents the magnitude of the index.

The sorted layout creates the "tornado" shape: wider at the top, narrower toward the bottom.

Figure 2.1 shows the tornado plot for R_0 in the SIR model with POIs $\{k, \beta, \tau\}$. All three bars extend to the right (all indices are +1), indicating that each parameter promotes disease spread when increased. All bars are the same length, confirming that the three parameters are equally important for R_0.

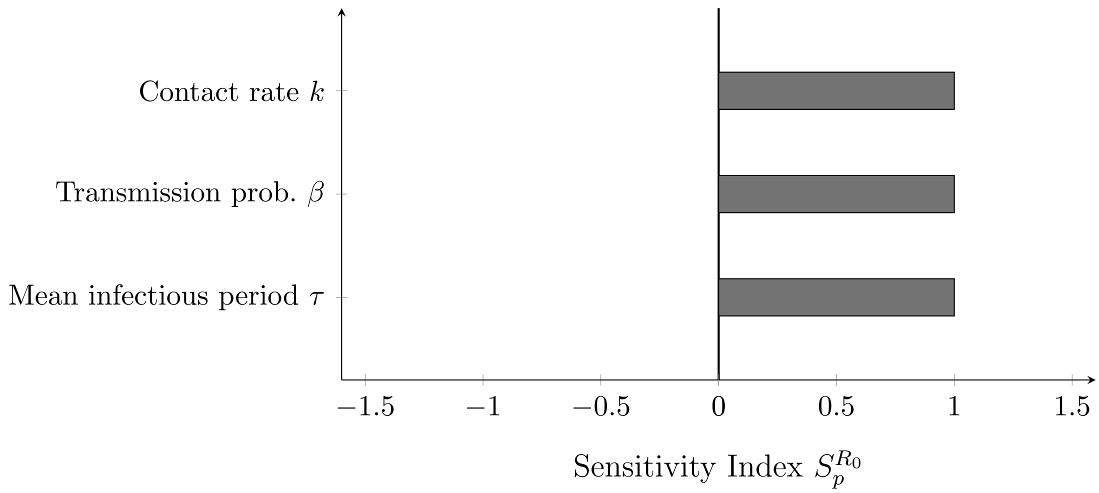


Figure 2.1: Tornado plot for the sensitivity of R_0 to the three POIs of the SIR model, parameterized using interpretable durations and probabilities. All three indices equal +1, so the bars are equal in length and all extend to the right. In models with more parameters and a wider range of index values, the sorted layout reveals immediately which parameters dominate.

MATLAB code to produce a tornado plot for any set of sensitivity indices:

```
function tornado_plot(SI_values, labels, QOI_name)
% SI_values : vector of sensitivity indices
% labels    : cell array of POI names (strings)
% QOI_name  : string label for the QOI

[~, idx] = sort(abs(SI_values), 'descend');
SI_sorted  = SI_values(idx);
lab_sorted = labels(idx);

barh(SI_sorted, 'FaceColor', [0.35 0.35 0.35]);
set(gca, 'YTick', 1:numel(SI_sorted), ...
         'YTickLabel', lab_sorted, ...
         'FontSize', 11);
xline(0, 'k', 'LineWidth', 1.2);
xlabel(['Sensitivity Index  S_p^{' QOI_name '}'], 'FontSize', 12);
title(['Tornado Plot: Sensitivity of ' QOI_name], 'FontSize', 13);
grid on; box off;
end
```

Self-check. A model for antibiotic-resistance spread has sensitivity indices $S_{\tau_{\text{inf}}}^{R_0} = +1.4$, $S_{\beta}^{R_0} = +0.9$, $S_{\tau_{\text{immune}}}^{R_0} = -0.6$, $S_k^{R_0} = +0.5$.

(a) Sketch the tornado plot by hand. (b) Which parameter is most important? Least important? (c) If τ_{immune} (mean time spent immune) increases by 5%, what happens to R_0?

Answer: (a) Sorted order by |SI|: τ_{inf} (1.4, right), β (0.9, right), τ_{immune} (0.6, *left*), k (0.5, right). (b) Most important: τ_{inf}. Least: k. (c) R_0 decreases by approximately $0.6 \times 5\% = 3\%$. The negative sign and the biological interpretation agree: a longer immune period keeps people protected longer, reducing transmission.

2.8 A Preview: Second-Order Terms

The sensitivity index is built on a first-order approximation. For a QOI q that depends on a single POI p, the Taylor expansion gives

$$q(p + \delta p) = q(p) + \underbrace{\frac{\partial q}{\partial p}\,\delta p}_{\text{first-order}} + \underbrace{\frac{1}{2}\frac{\partial^2 q}{\partial p^2}\,(\delta p)^2}_{\text{second-order}} + \cdots$$

The sensitivity index is derived from the first-order term alone. When the second-order term is small relative to the first-order term (that is, when

$$\left|\frac{1}{2}\frac{\partial^2 q}{\partial p^2}\,\delta p\right| \ll \left|\frac{\partial q}{\partial p}\right|,$$

the sensitivity index provides an accurate approximation. When this condition fails, the linear approximation is poor and the SI overstates or understates the true sensitivity.

Chapter 3 develops practical methods for estimating the second-order term numerically and for assessing when it is large enough to matter. This is the first instance of Unifying Idea 3 from Chapter 1: local SA is a linearization, and linearizations have limits. The second-order term is the first correction beyond those limits.

For two POIs p_i and p_j that are not varied independently (a situation addressed in Chapter 3), the cross term

$$\frac{\partial^2 q}{\partial p_i\,\partial p_j}\,\delta p_i\,\delta p_j$$

also enters the expansion. This cross derivative quantifies the interaction between the two POIs: it measures how the sensitivity to p_i changes as p_j varies. Chapter 9 shows how Sobol interaction indices generalize this idea to the full range of parameter values.

2.9 MATLAB Laboratory: Sensitivity Indices for the SIR Model

This laboratory implements and visualizes the sensitivity indices developed in the chapter, using the SIR model as the working example. Complete the five parts in order; each builds on the previous.

Part 1: Analytic SI via the Symbolic Toolbox. Define $R_0 = k\beta/\gamma$ symbolically in MATLAB. Compute $S_k^{R_0}$, $S_\beta^{R_0}$, $S_\gamma^{R_0}$ using the `diff` function and the derivative form of Definition 2.1. Simplify and confirm that all three equal ± 1 analytically.

Note: The code below requires the Symbolic Math Toolbox. If you do not have it, skip directly to Part 3 and verify the same results numerically. Parts 3–5 do not require the Symbolic Toolbox.

```
syms k beta gamma positive
R0   = k * beta / gamma;
SI_k = simplify( (k/R0) * diff(R0, k) )
SI_b = simplify( (beta/R0) * diff(R0, beta) )
SI_g = simplify( (gamma/R0) * diff(R0, gamma) )
```

Part 2: Reparameterize. Replace γ with the mean infectious period $\tau = 1/\gamma$. Rewrite $R_0 = k\beta\tau$ symbolically. Compute $S_\tau^{R_0}$ and verify that it equals $-S_\gamma^{R_0}$ as predicted by equation (2.13).

Part 3: Numerical verification. Use the nominal values $k = 5$, $\beta = 0.06$, $\tau = 7$ days (so $\gamma = 1/7$). Compute R_0 numerically. Estimate $S_\tau^{R_0}$ using a centered finite difference with $h = 10^{-5}\tau$. Compare with the analytic value of $+1$. How large is the relative error?

```
k = 5; beta = 0.06; tau = 7;
R0    = k * beta * tau;
h     = 1e-5 * tau;
R0_p  = k * beta * (tau + h);
R0_m  = k * beta * (tau - h);
SI_fd = (tau / R0) * (R0_p - R0_m) / (2*h);
fprintf('Analytic SI: 1.00000\n');
fprintf('FD estimate: %.5f\n', SI_fd);
```

Part 4: Tornado plots for both parameterizations. Using the `tornado_plot` function from Section 2.7, produce two tornado plots side by side: one with rate parameters $\{k, \beta, \gamma\}$ and one with interpretable parameters $\{k, \beta, \tau\}$. Observe how the sign of the γ/τ bar changes between the two plots and why.

MATLAB note: The `tornado_plot` function must be saved as a separate file named `tornado_plot.m` in your working directory (function files cannot be pasted into a script). Part 4 also uses `subplot` and cell array syntax for labels; the file `run_sir_example.m` in the companion repository demonstrates both if these are unfamiliar.

Part 5: Effect of nominal value. Compute $S_k^{R_0}$, $S_\beta^{R_0}$, $S_\tau^{R_0}$ at two different nominal transmission probabilities: $\beta_1 = 0.06$ and $\beta_2 = 0.6$. Do the SI values change? What does this tell you about the sensitivity of R_0 to each parameter? (Recall that $R_0 = k\beta\tau$ is a product of monomials; for such functions, the SI is constant regardless of the nominal value. This is a special property of multiplicative models that does not hold for more complex functions like those in Chapter 4.)

2.10 Exercises

Exercise 2.1 *(*[Bloom L2]*)*. Explain the sign rule in your own words. Without performing any calculation, state the sign of the sensitivity index for each of the following, and explain your reasoning: (a) the sensitivity of lung capacity to cigarette consumption, treating capacity as the QOI and cigarettes per day as the POI; (b) the sensitivity of average crop yield to annual rainfall, near a region of the parameter space where crops are mildly drought-stressed; (c) the sensitivity of R_0 to the mean infectious period τ.

Exercise 2.2 *(*[Bloom L2]*)*. For the function $q = \sqrt{p}$ (defined for $p > 0$), compute S_p^q using all three equivalent forms of Definition 2.1. Verify that all three give the same result. Interpret the answer: if p doubles, by approximately what percentage does q change?

Exercise 2.3 *(*[Bloom L3]*).* Let $u(t) = \frac{1}{2}\lambda t^2$ be a parabola on $[0, b]$. The arc-length functional is $J = \int_0^b \sqrt{1 + (\lambda t)^2}\, dt$. Show that the sensitivity indices are

$$S_b^J = \frac{2\mu}{\mu + (\operatorname{arcsinh}\mu)/\sqrt{1+\mu^2}}, \qquad S_\lambda^J = \frac{\mu(1+\mu^2) - \sqrt{1+\mu^2}\ \operatorname{arcsinh}\mu}{\mu(1+\mu^2) + \sqrt{1+\mu^2}\ \operatorname{arcsinh}\mu},$$

where $\mu = \lambda b$. For $\mu = 1$, compute both indices numerically and verify using the derivative form.

Exercise 2.4 *(*[Bloom L3]*).* Let $u(t) = at^2 + bt + c$ and let $J_\pm = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ denote the two real roots (assume $b^2 > 4ac$). Show that the sensitivity indices of the positive root J_+ with respect to a, b, c can be written

$$S_a^{J_+} = -\frac{aJ_+}{2aJ_+ + b}, \quad S_b^{J_+} = -\frac{b}{J_+(2aJ_+ + b)}, \quad S_c^{J_+} = -\frac{c}{J_+(2aJ_+ + b)}.$$

For $a = 1$, $b = -3$, $c = 2$, compute all three indices. Interpret each sign.

Exercise 2.5 *(*[Bloom L3]*).* A SEIR model (Susceptible-Exposed-Infectious-Recovered) has reproduction number

$$R_0 = \frac{k\beta\,\nu}{(\nu + \mu)(\gamma + \mu)},$$

where ν is the rate of progression from exposed to infectious, μ is the background mortality rate, and k, β, γ are as in the SIR model. Reparameterize using mean times: $\tau_E = 1/\nu$ (mean latent period), $\tau_I = 1/\gamma$ (mean infectious period), $L = 1/\mu$ (mean lifespan). Compute $S_{\tau_E}^{R_0}$, $S_{\tau_I}^{R_0}$, $S_L^{R_0}$, $S_k^{R_0}$, $S_\beta^{R_0}$ and produce a tornado plot. Which parameter is most important?

Exercise 2.6 *(*[Bloom L4]*).* A published paper reports $S_\gamma^{R_0} = -1.0$ for an SIR model, where γ is the recovery rate in units of day^{-1}. A second paper on the same model reports $S_\tau^{R_0} = +1.0$, where $\tau = 1/\gamma$ is the mean infectious period in days. A policy maker reading both papers is confused: one paper shows a negative index, the other shows a positive index for what appears to be the same parameter.

Write a one-paragraph explanation for the policy maker that (a) resolves the apparent contradiction, (b) explains which parameterization is more informative for policy, and (c) describes what each sensitivity index means in plain language.

Exercise 2.7 *(*[Bloom L4]*).* The following tornado plot data is given for a food-safety model with QOI J = mean number of illness cases per outbreak. The sensitivity indices are: $S_{\tau_c}^J = +2.1$ (mean contamination duration, days), $S_{\beta_f}^J = +1.8$ (food-to-person transmission probability), $S_{\tau_s}^J = -1.3$ (mean hand-washing interval, days), $S_{k_f}^J = +0.7$ (food contacts per day), $S_\mu^J = -0.2$ (background clearance rate).

(a) Produce the tornado plot in MATLAB using the code template from Section 2.7.

(b) Rank the five parameters by absolute importance.

(c) A public health program can achieve a 15% reduction in either τ_c or a 20% reduction in k_f. Which intervention reduces J more? By how much?

(d) The parameter μ has a small negative index. Write a one-sentence interpretation of what this means for disease control.

Exercise 2.8 *(*[Bloom L5]*)*. You are advising the World Health Organization on a vaccine allocation strategy for a new respiratory pathogen. A compartmental model has been fitted to early outbreak data. The model has six parameters; their sensitivity indices with respect to the QOI "total deaths at 6 months" are not yet known.

Design the SA study. Your design should specify:

(a) Which parameterization you will use (rates or durations) for each parameter, and why.

(b) What QOI or QOIs you would compute sensitivity indices for, and what additional QOI you might consider beyond the stated one.

(c) How you would present the results to a non-mathematical audience (what figure or table, and what the key message would be).

(d) One limitation of the local SA approach that the WHO advisors should be aware of in using these results.

Exercise 2.9 *(*[Bloom L6]*)*. A modeling study reports the following: "Sensitivity analysis of R_0 was performed with respect to all model parameters. The parameters are listed in Table 3 using the standard model notation. All sensitivity indices were positive, confirming that R_0 is an increasing function of all model parameters."

Table 3 lists recovery rate γ, mortality rate μ, and waning immunity rate ρ alongside transmission and contact parameters.

Identify the error in the study's statement and explain specifically what is wrong, referring to the mathematical relationship between rates and their reciprocals. Restate the conclusion correctly.

Part II

Computing Sensitivity in Practice

Chapter 3

Computing Sensitivity in Practice

A derivative is useful only if you can compute it honestly.

You have inherited a model from a collaborator. It is 500 lines of MATLAB code, produces reasonable epidemic projections, and has 11 parameters. Your collaborator has moved to another institution. The code runs correctly, but it uses conditional statements, lookup tables, and an internal adaptive ODE solver; none of which can be differentiated analytically. Your funding agency wants to know which parameters matter before committing to a three-year field measurement campaign.

How do you compute sensitivity indices for a model you cannot differentiate?

The answer is finite differences, and this chapter shows exactly how to do it, including how to avoid the mistakes that produce wrong results.

3.1 The Sensitivity Jacobian

Chapter 2 computed sensitivity indices one parameter at a time. For a model with n_p POIs and n_q QOIs, the complete local SA is captured by an $n_q \times n_p$ matrix; one entry per POI–QOI pair.

Definition 3.1 (Sensitivity Jacobian)**.** Let $\vec{q} = (q_1, \ldots, q_{n_q})$ denote the vector of QOIs and $\vec{p} = (p_1, \ldots, p_{n_p})$ denote the vector of POIs. The **normalized sensitivity Jacobian** is the $n_q \times n_p$ matrix $\mathbf{S}$ with entries

$$S_{ki} = S_{p_i}^{q_k} = \frac{p_i}{q_k} \frac{\partial q_k}{\partial p_i}. \tag{3.1}$$

For the SIR model with POIs $\{k, \beta, \tau, \mu\}$ (contact rate, transmission probability, mean infectious period, mortality rate) and QOIs $\{S(T), I(T), R(T)\}$ (the three compartment sizes at time T), the sensitivity Jacobian is a 3×4 matrix of twelve numbers. Each number answers one specific "What if?" question: how much does compartment i change if POI j changes by 1%?

The tornado plots in Chapter 2 displayed single rows or columns of this matrix. The full matrix captures all POI–QOI relationships simultaneously.

Computing this matrix efficiently, in both time and accuracy; is the central computational challenge of sensitivity analysis.

A common initial reaction is to treat the sensitivity Jacobian as nothing more than a collection of individual SI calculations; as if the matrix label were just a convenient way to organize numbers that are computed independently. It is more than that. The Jacobian has structure: its columns can be computed in parallel, many methods share a single baseline evaluation, and the adjoint method (Chapters 5 and 6) exploits the matrix structure to recover all columns at once from a single backward solve. Understanding the Jacobian as a matrix, not just a table, is what makes efficient SA possible.

Self-check. A climate model has $n_p = 30$ POIs and $n_q = 8$ QOIs. How many individual sensitivity indices does the full SA require? If you had to compute each one separately, how many model evaluations would a naive brute-force approach require?

> *Answer:* $30 \times 8 = 240$ sensitivity indices. A naive approach, perturb one POI at a time, re-run the model, compute the difference; requires $30 + 1 = 31$ model evaluations for all POIs simultaneously if only first-order (forward) differences are used, or $2 \times 30 + 1 = 61$ for centered differences. The key: you do not need one evaluation per SI entry. You need one evaluation per POI (plus one baseline), and then each evaluation contributes an entire column of the Jacobian.

3.2 Finite Difference Formulas

A finite difference approximates the partial derivative $\partial q/\partial p_i$ by computing how the QOI changes when p_i alone is shifted by a small but finite amount h. Three approximations are in common use, differing in accuracy and cost.

3.2.1 Forward Difference

The simplest approximation uses one additional model evaluation:

$$\frac{\partial q}{\partial p_i} \approx \frac{q(p^* + h\vec{e}_i) - q(p^*)}{h}, \tag{3.2}$$

where $\vec{e}_i$ is the unit vector in the p_i direction and p^* is the nominal parameter vector. This is a first-order approximation: the error is $O(h)$, meaning that halving h approximately halves the error.

3.2.2 Centered Difference

Using one evaluation on each side of the nominal value gives a second-order approximation:

$$\frac{\partial q}{\partial p_i} \approx \frac{q(p^* + h\vec{e}_i) - q(p^* - h\vec{e}_i)}{2h}. \tag{3.3}$$

The error is $O(h^2)$: halving h reduces the error by a factor of four. For the same step size, centered differences are substantially more accurate than forward differences.

3.2.3 Fourth-Order Formula (Rarely Used in Practice)

When exceptionally high accuracy is required and model evaluations are inexpensive, one can eliminate the leading-order error term by combining four evaluations at two step sizes:

$$\frac{\partial q}{\partial p_i} \approx \frac{-q(p^* + 2h\vec{e}_i) + 8q(p^* + h\vec{e}_i) - 8q(p^* - h\vec{e}_i) + q(p^* - 2h\vec{e}_i)}{12h}. \tag{3.4}$$

The error is $O(h^4)$: halving h reduces the error by a factor of sixteen. This requires four evaluations per POI rather than two. In the applied SA literature this formula is uncommon; the centered difference (3.3) is the standard workhorse. Knowing which formula to use and what step size to choose are the practical skills taken up next.

3.2.4 Practical Workflow: Step-Size Selection by Error Estimation

In practice, the centered difference (3.3) is the default choice. The forward difference (3.2) plays a secondary role not as a first approximation in its own right, but as a cheap error estimator: because it requires only the already-computed baseline $q(p^*)$ and the one-sided perturbed evaluation $q(p^*+h\vec{e}_i)$, no extra model runs are needed. The difference between the two approximations at the same h,

$$\text{estimated error} \approx \left| \frac{q(p^* + h\vec{e}_i) - q(p^*)}{h} - \frac{q(p^* + h\vec{e}_i) - q(p^* - h\vec{e}_i)}{2h} \right|, \tag{3.5}$$

provides a practical error indicator. If this indicator is large relative to the desired precision, the analyst reduces h (or increases ODE solver tolerances) and repeats until the error is acceptable. This adaptive strategy is more reliable in practice than deriving a theoretical optimal h from equations (3.8)–(3.9), because those formulas depend on unknown higher-order derivatives of the model output.

A complementary check uses the backward difference $[q(p^*)-q(p^*-h\vec{e}_i)]/h$ as a second one-sided estimate. Comparing both one-sided approximations with the centered result gives two independent error indicators, and agreement among all three signals that h lies in the reliable range.

3.2.5 Cost Comparison

Table 3.1 summarizes the cost of computing the complete sensitivity Jacobian by each method.

Method	Model evaluations	Error order	Best when
Forward difference	$n_p + 1$	$O(h)$	Error check only; not standalone
Centered difference	$2n_p + 1$	$O(h^2)$	Standard choice
Fourth-order FD	$4n_p + 1$	$O(h^4)$	Rare; cheap evaluations only
Analytic FSE (Ch. 4)	n_p lin. solves	Exact	Differentiable model
Adjoint (Ch. 5–6)	2 solves	Exact	Many POIs, few QOIs

Table 3.1: Cost of computing the complete sensitivity Jacobian by different methods, for a model with n_p POIs and n_q QOIs. The number of model evaluations does *not* depend on n_q: each evaluation contributes an entire column of the $n_q \times n_p$ Jacobian simultaneously. Adding more QOIs is free — the cost scales with the number of POIs n_p, not with the size of the output. The finite-difference counts assume a single baseline evaluation is reused across all POIs.

For the SIR model with $n_p = 4$ parameters and $n_q = 3$ state variables, centered differences require $2 \times 4 + 1 = 9$ model evaluations. If each takes 1 second, the full SA takes 9 seconds.

For the malaria model of Chitnis, Hyman & Cushing [10] with $n_p = 17$ parameters and $n_q = 7$ state variables, centered differences require $2 \times 17 + 1 = 35$ evaluations. If each takes 1 minute: approximately 35 minutes. If each takes 1 hour: approximately 35 hours.

For a climate model with $n_p = 500$ parameters and model runs taking 6 hours each, centered differences require $2 \times 500 + 1 = 1001$ evaluations, at 6 hours each, that is approximately 250 days of computation. This is why Chapter 6 introduces the adjoint method, which reduces the cost to 2 solves regardless of n_p.

Unifying Idea 1. Every entry of the sensitivity Jacobian is a normalized derivative: an instance of Unifying Idea 1 from Chapter 1. The finite-difference formulas in this chapter compute those derivatives numerically, for any model that can be evaluated, whether or not it can be differentiated analytically.

Self-check. For a model with $n_p = 20$ POIs and $n_q = 5$ QOIs, how many model evaluations does the centered-difference Jacobian require? (Remember: adding more QOIs is free — each evaluation contributes an entire column of the Jacobian.) If each evaluation takes 45 seconds, how long does the complete SA take? At what value of n_p would centered differences take more than 24 hours, assuming each evaluation takes 2 minutes?

Answer: $2 \times 20 + 1 = 41$ evaluations, regardless of $n_q = 5$; $41 \times 45 = 1{,}845$ seconds ≈ 31 minutes. For 24 hours = 1,440 minutes with 2-minute evaluations: $2n_p + 1 \leq 720$ gives $n_p \leq 359$, so for $n_p \geq 360$, centered differences take more than 24 hours.

3.3 Step Size Selection — The U-Curve

The step size h controls a fundamental accuracy tradeoff. There is an optimal value: too large and the approximation misses the local curvature; too small and the computer's limited numerical precision destroys the result.

3.3.1 Two Sources of Error

Truncation error arises because finite differences are Taylor-series approximations. For the centered difference:

$$\frac{q(p+h) - q(p-h)}{2h} = q'(p) + \underbrace{\frac{h^2}{6}\, q'''(p) + \cdots}_{\text{truncation error}}. \tag{3.6}$$

This error decreases as $h \to 0$, with slope -2 on a log-log plot of error vs. h.

Rounding error arises because the computed model output $\tilde{q}(p+h)$ differs from the true output $q(p+h)$ by a rounding error of order $\varepsilon_{\text{mach}}|q|$, where $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$ in IEEE 754 double precision. When h is small, subtracting $\tilde{q}(p-h)$ from $\tilde{q}(p+h)$ amplifies this rounding error:

$$\text{rounding error in FD} \approx \frac{2\varepsilon_{\text{mach}}|q|}{2h} = \frac{\varepsilon_{\text{mach}}|q|}{h}. \tag{3.7}$$

This error *increases* as $h \to 0$, with slope $+1$ on a log-log plot.

3.3.2 The U-Curve and the Optimal Step Size

Before improving the model, find out which knob actually moves the machine.

The total error in the centered-difference approximation is the sum of truncation and rounding errors. As h decreases from a large value, truncation error falls while rounding error rises. The minimum total error occurs at the crossover, as shown schematically in Figure 3.1.

Setting the two error contributions equal gives an approximate starting point for step-size selection:

$$h^*_{\text{forward}} \approx \sqrt{\varepsilon_{\text{mach}}} \cdot |p^*| \approx 1.5 \times 10^{-8} \cdot |p^*|, \tag{3.8}$$
$$h^*_{\text{centered}} \approx \varepsilon_{\text{mach}}^{1/3} \cdot |p^*| \approx 6 \times 10^{-6} \cdot |p^*|. \tag{3.9}$$

For a parameter with nominal value $p^* = 0.3$, this suggests a starting centered-difference step of approximately $h \approx 1.8 \times 10^{-6}$.

These formulas should be understood as rough guides, not precise prescriptions. The unknown third derivative $q'''(p)$ makes the exact truncation error inaccessible, so the true optimal h depends

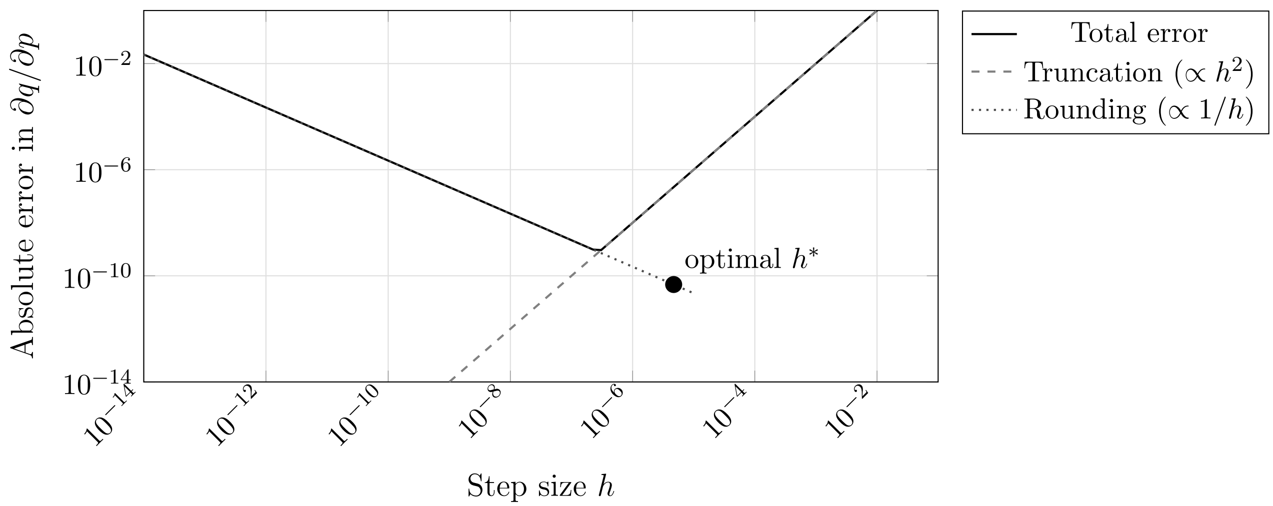


Figure 3.1: The U-curve for centered differences in double precision. Truncation error (dashed) decreases as h decreases; rounding error (dotted) increases. The total error (solid) is minimized at the optimal step size h^*. Using h much smaller than h^*: a common mistake; gives larger errors, not smaller ones.

on the specific model being analyzed. The practical workflow described in Section 3.2, computing the error indicator (3.5) and adjusting h until the one-sided and centered approximations agree to acceptable precision, is more reliable in practice than relying on these theoretical formulas alone.

The most common mistake in finite-difference SA is choosing h that is too small. Practitioners who set $h = 10^{-12}$ or $h = 10^{-15}$, reasoning that "smaller is more accurate," obtain results dominated by catastrophic cancellation (subtracting two nearly equal floating-point numbers, which destroys significant digits and amplifies rounding error). The U-curve explains why: below the optimal step size, rounding error grows as $1/h$ while truncation error continues to fall, so the total error increases rather than decreases. The error indicator (3.5) will detect this failure: when h is too small, the centered and forward approximations both carry large rounding errors and will disagree by a large relative amount, signaling that a larger step size is needed.

The U-curve also reveals a practical limitation for models implemented with adaptive ODE solvers. For a smooth analytic function the curve has a well-defined minimum, and the error indicator (3.5) decreases reliably as h is reduced toward the optimal step size. For a model that calls `ode45` internally, the output is not mathematically smooth: the solver makes discrete step-size decisions that can cause the model output to change non-smoothly as a parameter varies by a small amount. In these cases the error indicator may not decrease monotonically, and the one-sided and centered approximations may disagree by more than machine precision even at step sizes near $10^{-6}|p^*|$. The remedy is to tighten the ODE solver tolerances (`RelTol` and `AbsTol` in `ode45`) before reducing h further. After tightening tolerances, applying the adaptive workflow (compute the error indicator, check for agreement, reduce h if needed) usually locates a reliable step size.

Self-check. For the SIR model with transmission probability $\beta^* = 0.06$, the theoretical formula (3.9) suggests a starting centered-difference step of $h \approx 3.6 \times 10^{-7}$. You then compute the centered difference and a one-sided forward difference at this h and find they agree to four significant figures. (a) Is this step size acceptable? What does this agreement indicate? (b) Now suppose you repeat the check at $h = 10^{-12}$ and find the two approximations disagree by more than 50%. What does this signal, and what should you do?

Answer: (a) Agreement to four significant figures indicates that the rounding error is small relative to the truncation error at this h, and the centered approximation is reliable. The step size is acceptable for four-digit SI accuracy.

(b) Large disagreement at $h = 10^{-12}$ signals that rounding error dominates: both approximations suffer catastrophic cancellation. The remedy is to increase h toward the range suggested by the theoretical formula, then recheck the error indicator. Rounding error $\propto 1/h$: at $h = 10^{-12}$ the rounding error is $(3.6 \times 10^{-7}/10^{-12}) = 3.6 \times 10^{5}$ times larger than at $h^* \approx 3.6 \times 10^{-7}$. Moving to even larger h would re-introduce truncation error, so the reliable range lies in between.

3.4 Second-Order Terms

Chapter 2 previewed second-order terms as corrections to the first-order SI approximation. This section shows how to compute them.

3.4.1 Pure Second-Order Derivatives

The second derivative $\partial^2 q/\partial p_i^2$ measures the curvature of the QOI as a function of p_i. It can be estimated by reusing the three evaluations already computed for the centered first-order difference:

$$\frac{\partial^2 q}{\partial p_i^2} \approx \frac{q(p^* + h\vec{e}_i) - 2q(p^*) + q(p^* - h\vec{e}_i)}{h^2}. \tag{3.10}$$

No additional model evaluations are needed.

When the curvature is large, the first-order SI understates the true sensitivity for larger perturbations. Specifically, if

$$\left| \frac{h}{2} \frac{\partial^2 q/\partial p_i^2}{\partial q/\partial p_i} \right| \ll 1,$$

the linear approximation is valid for perturbations of size h. If this ratio is not small, the sensitivity index provides only a local estimate, and Chapter 9's global SA methods are needed for a complete picture.

Unifying Idea 3. The second-order terms computed in this section are the first quantitative correction beyond the linearization that defines every sensitivity index in this book. The first-order SI is valid when the model behaves approximately linearly near the nominal parameter values. The second-order term tells you how quickly that approximation degrades as perturbations grow. Chapter 9 provides the full global picture when the linearization is no longer adequate.

3.4.2 Mixed Second-Order Derivatives

The cross-derivative $\partial^2 q/\partial p_i\,\partial p_j$ measures how the sensitivity to p_i changes with p_j. It requires one additional model evaluation beyond those already computed:

$$\frac{\partial^2 q}{\partial p_i\,\partial p_j} \approx \frac{q(p^* + h\vec{e}_i + h\vec{e}_j) - q(p^* + h\vec{e}_i) - q(p^* + h\vec{e}_j) + q(p^*)}{h^2}. \tag{3.11}$$

For all $\binom{n_p}{2}$ pairs, this requires $\binom{n_p}{2}$ additional evaluations. For $n_p = 4$, that is 6 evaluations; for $n_p = 17$ (the malaria model), it is 136.

The cross-derivative $\partial^2 q/\partial p_i\,\partial p_j$ is large when the two POIs *interact*: varying them together is not the same as varying them separately. Chapter 9 shows that Sobol interaction indices S_{ij} are the global, variance-based analog of these local cross-derivatives.

Self-check. For the SIR model with $R_0 = k\beta\tau$, compute $\partial^2 R_0/\partial k\,\partial\beta$ analytically. Is the cross-derivative large or small relative to the individual first-order derivatives $\partial R_0/\partial k$ and $\partial R_0/\partial\beta$? What does this imply about the interaction between k and β for R_0?

> *Answer:* $\partial^2 R_0/\partial k\,\partial\beta = \tau$ (positive, since $R_0 = k\beta\tau$).
> The first-order derivatives are $\partial R_0/\partial k = \beta\tau$ and $\partial R_0/\partial\beta = k\tau$.
> The ratio $(\partial^2 R_0/\partial k\,\partial\beta)/(\partial^2 R_0/\partial k^2) = \tau/0$ is undefined (the pure second derivative is zero for a monomial). More informatively, the mixed partial is nonzero, meaning k and β interact in their effect on R_0: increasing both together increases R_0 more than the sum of the individual increases. This is expected because R_0 is multiplicative in k and β. For Sobol analysis, this interaction will produce a nonzero $S_{k\beta}$.

3.5 Correlated POIs — The Most Common Mistake

> *A parameter can be uncertain without being important, and important without being uncertain. Correlation between parameters adds a third possibility: a parameter can appear important in isolation while its true effect is inseparable from another.*

Before computing any sensitivity indices, there is a question that must be asked: can these parameters actually be varied independently? In many models, the answer is no.

Every finite-difference SA computation implicitly assumes that each POI can be perturbed while all others are held exactly fixed. If the POIs are correlated, meaning that in the real world, changing one tends to change another, then this assumption is violated and the resulting sensitivity indices can be misleading.

3.5.1 Diagnosing Correlation

Correlation among POIs arises from several sources.

Structural correlation occurs when parameters appear only as a product or ratio in the model equations. In the SIR model, the force of infection is $\lambda = k\beta I/N$. The parameters k and β appear only as the product $k\beta$. Observational data, incidence counts over time, can identify $k\beta$ as a unit but cannot separate the two factors. Computing $S_k^{R_0}$ and $S_\beta^{R_0}$ as independent indices is mathematically valid but can be operationally misleading, because any intervention must target the product $k\beta$ (by reducing contact rate, transmission probability, or both). Reparameterizing to use $k\beta$ as a single POI is more natural in this case.

This observation is the starting point for **structural identifiability** analysis: the question of whether the parameters of a model can, in principle, be recovered from the observable outputs, regardless of measurement noise or sample size. A model is *structurally identifiable* in a parameter p if distinct values of p produce distinct output trajectories. The SIR model is not structurally identifiable in (k, β) separately, but is identifiable in the product $k\beta$. Sensitivity analysis reveals this: when two parameters always have the same sensitivity index (here $S_k^q = S_\beta^q$ for any QOI q), they are structurally correlated and the model cannot distinguish them. A sensitivity index of zero for a parameter is a related signal: the model output does not respond to that parameter at all, which may indicate that the parameter cannot be estimated from data. Chapter 9 returns to identifiability in the context of global SA, where variance-based methods provide computational tools for detecting structural correlation across the full parameter range. A full treatment of structural and practical identifiability, including formal algebraic methods, is beyond the scope of this book; see Smith [42] §9.3 and the references therein.

Empirical correlation occurs when parameter values are estimated from data that naturally links them. In the malaria model studied by Chitnis, Hyman & Cushing [10], the mosquito biting rate σ_v and the human biting saturation σ_h both appear in the force of infection and are often positively correlated across field studies. The paper introduces the composite parameter $\zeta = \sigma_v\sigma_h/(\sigma_v N_v^* + \sigma_h N_h^*)$ precisely to address this correlation.

Constraint correlation occurs when a physical or biological requirement links parameters. In models where stage durations must sum to a total lifecycle, or where transition probabilities must sum to one, no parameter can change without forcing compensating changes in others.

3.5.2 The Impact of Ignoring Correlation

To see concretely what goes wrong, consider a model whose QOI is $q = p_1 + p_2$ where the two POIs satisfy $p_2 = 2p_1$ (a linear constraint). The standard sensitivity indices are:

$$S^q_{p_1} = \frac{p_1}{q}\cdot 1 = \frac{p_1}{3p_1} = \frac{1}{3}, \qquad S^q_{p_2} = \frac{p_2}{q}\cdot 1 = \frac{2p_1}{3p_1} = \frac{2}{3}.$$

These look reasonable: the second parameter is twice as important as the first. But in reality, p_1 and p_2 cannot be varied independently. Any change to p_1 forces a change to $p_2 = 2p_1$. The total effect of a 1% increase in p_1 is:

$$\Delta q = \frac{\partial q}{\partial p_1}\,\delta p_1 + \frac{\partial q}{\partial p_2}\,\delta p_2 = 1\cdot(0.01p_1) + 1\cdot(0.02p_1) = 0.03p_1.$$

The normalized total-derivative SI for p_1 is then $(p_1/q)\cdot(0.03p_1/0.01p_1) = (1/3)\cdot 3 = 1$, not $1/3$. The standard index understates the true sensitivity by a factor of three.

3.5.3 Four Practical Responses

Response 1: Diagnose first. Before any SA computation, examine the correlation structure of the POIs. Compute the Pearson correlation matrix from any available data on parameter values across studies or populations. Plot scatter matrices (`plotmatrix` in MATLAB). A correlation $|\rho_{ij}| > 0.5$ warrants attention.

Response 2: Reparameterize structurally correlated pairs. When two parameters appear only as a product or ratio, define that product or ratio as the natural POI. For the SIR model: use $k\beta$ as a single POI, with the composite sensitivity index $S^{R_0}_{k\beta} = S^{R_0}_k + S^{R_0}_\beta = 1 + 1 = 2$. (The additive rule for sensitivity of composite parameters is developed in the exercises.)

Response 3: Total derivative for empirically correlated pairs. When empirical data establishes that a proportional change in p_i induces a proportional change in p_j with coefficient c_{ij} (i.e., $\Delta p_j/p_j \approx c_{ij}\,\Delta p_i/p_i$), the total derivative SI is:

$$S^q_{p_i}\Big|_{\text{tot}} = S^q_{p_i}\Big|_{\text{std}} + \sum_{j\neq i} S^q_{p_j}\,c_{ij}. \tag{3.12}$$

In plain language: c_{ij} is the induced proportional change in p_j per unit proportional change in p_i. If a 1% increase in k tends to be accompanied by a 0.6% increase in β (from field data), then $c_{k\beta} = 0.6$. Estimate c_{ij} from regression of observed parameter pairs, or from domain knowledge about shared drivers. Report both the standard and total-derivative SIs together with the correlation coefficients so readers can assess the impact.

Response 4 (Report transparently). When correlations exist but cannot be fully quantified, state explicitly in the SA report that the independence assumption was made, identify the correlated pairs, and flag which SI values are most likely affected.

Self-check. In the SIR model, the product $k\beta$ represents the effective transmission rate. If a field study establishes that k and β are positively correlated with coefficient $c_{k\beta} = 0.4$ (i.e., $\Delta\beta \approx 0.4\,\Delta k$ for proportional changes), use equation (3.12) to compute the total-derivative SI of R_0 with respect to k at the nominal values. Compare with the standard SI $S_k^{R_0} = 1$.

Answer: $S_k^{R_0} = 1$, $S_\beta^{R_0} = 1$. Here $c_{k\beta} = 0.4$ means $(\Delta\beta/\beta)/(\Delta k/k) = 0.4$, so $p_\beta c_{k\beta}/p_k = \beta \cdot 0.4/k$. But in proportional terms: total SI $\approx 1 + 1 \times 0.4 = 1.4$. A 1% increase in k (with the correlated 0.4% increase in β it entails) increases R_0 by approximately 1.4%, not 1%. The standard SI underestimates the true effect by 40%.

3.6 Black-Box Models and the MATLAB Template

When the model is inherited code, a compiled simulation, or a package you did not write, analytical differentiation is not possible. Finite differences are then the only option for local SA, and they work for *any* model that accepts a parameter vector and returns a QOI vector.

Students sometimes treat finite differences as an inferior approximation to be used only when analytic derivatives are unavailable. For black-box models, this framing is backwards: finite differences are not a fallback; they are the correct and complete tool. The accuracy they provide at the optimal step size is typically more than sufficient for SA decisions, and their applicability to models of arbitrary complexity makes them indispensable in practice.

The following MATLAB function computes the full normalized sensitivity Jacobian by centered differences, using the optimal step size from (3.9). Students should copy this template for every black-box SA problem.

```
%% From: src/core/sensitivity_jacobian.m (see GitHub repository)
p     = sir_nominal();
R0_fn = @(pv) pv(1)*pv(2)*pv(3);   %% R0 = k*beta*tau
p_vec = [p.k; p.beta; p.tau; p.L];
[S, info] = sensitivity_jacobian(R0_fn, p_vec, 1);
%% S(j) = SI_{p_j}^{R0};  info.n_evals = 2*n_p+1 = 9

%% If n_q is wrong, the function catches it immediately:
%% >> sensitivity_jacobian(R0_fn, p_vec, 3)
%% Error: MODEL returned 1 value but N_Q = 3.
```

The function handles the near-zero QOI case using the regularized change from Chapter 2. The `max` in the step-size formula prevents h from being identically zero when $p_{\text{nom},i} = 0$.

To validate this template on any model, call it with the analytic sensitivity indices known from Chapter 2 and verify agreement. For the SIR model:

```
%% SIR R0 as function of p = [k, beta, tau]
R0_model = @(p) p(1)*p(2)*p(3);    %% R0 = k * beta * tau

p_nom = [5, 0.06, 7];     %% nominal: k, beta, tau
S = sensitivity_jacobian(R0_model, p_nom, 1);
```

```
fprintf('Sensitivity indices for R0:\n')
fprintf('  SI_k    = %6.4f  (exact: 1.0000)\n', S(1))
fprintf('  SI_beta = %6.4f  (exact: 1.0000)\n', S(2))
fprintf('  SI_tau  = %6.4f  (exact: 1.0000)\n', S(3))
```

Self-check. The template above uses $h_{\min} = 10^{-10}$ as a floor for the step size. Why is this floor necessary, and what happens if a POI has nominal value exactly zero?

Answer: If $p_{\text{nom},i} = 0$, then $6 \times 10^{-6}|p_{\text{nom},i}| = 0$, giving $h = 0$ and a division-by-zero error. The floor $h_{\min} = 10^{-10}$ prevents this. For a POI with nominal value zero, the step size is not scaled by the parameter magnitude, it is just the fixed small value 10^{-10}. In this case, the SI is also not well-defined in the ratio form (it would require dividing by $p_i = 0$), so the code computes the unnormalized derivative instead. If a POI is exactly zero at its nominal value, the investigator should either reparameterize or report the absolute (unnormalized) derivative rather than the normalized SI.

3.7 MATLAB Laboratory: SA of the SIR Model by Finite Differences

This laboratory puts the chapter's tools together and provides a complete reference implementation for black-box SA. Use the SIR ODE model with the four POIs $\{k, \beta, \tau, \mu\}$ and the three QOIs $\{S(T), I(T), R(T)\}$ at time $T = 60$ days.

Setup. Implement `SIR_model(p)` as a MATLAB function that takes the parameter vector $p = [k, \beta, \tau, \mu]$ and returns $[S(60), I(60), R(60)]$ by calling `ode45`. Use nominal values $k = 5$, $\beta = 0.06$, $\tau = 7$ days, $\mu = 0.0001$ day^{-1}, and initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$.

Part 1: Compute the full sensitivity Jacobian. Call `sensitivity_jacobian(SIR_model, p_nom, 3)` to produce the 3×4 sensitivity matrix. Display it as a labeled table with QOI names as row labels and POI names as column labels.

Part 2: Plot the U-curve. Fix all parameters at nominal values except β. Estimate $\partial I(60)/\partial\beta$ using centered differences for $h = 10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}, 10^{-11}, 10^{-13}$. Compute the absolute error using the fourth-order formula (3.4) as a reference value (computed at $h = 10^{-4}$ where it is accurate). Also plot the one-sided forward difference error at each h. Plot centered-difference error, forward-difference error, and the discrepancy between the two (the practical error indicator (3.5)) vs. h on log-log axes. Identify the range of h where the error indicator is small and the centered approximation is reliable.

Part 3: Tornado plots. Using the Jacobian from Part 1, produce separate tornado plots for each QOI: the sensitivity of $S(60)$, $I(60)$, and $R(60)$ with respect to all four POIs. Interpret: which POI matters most for each QOI? Do the three plots give the same ranking?

Part 4: Correlation check. The contact rate k and transmission probability β appear only as the product $k\beta$ in the SIR force of infection. Confirm numerically that $S_k^{I(60)} \approx S_\beta^{I(60)}$ at the nominal values. Then reparameterize: define a new POI $\lambda_0 = k\beta$ and compute $S_{\lambda_0}^{I(60)}$. Verify that $S_{\lambda_0}^{I(60)} = S_k^{I(60)} + S_\beta^{I(60)}$.

Part 5: Correlation check from literature. The SIR force of infection is $\lambda = k\beta I/N$, so k and β appear only as the product $k\beta$ and are structurally correlated. Use the four-parameter SIR model. Confirm numerically that $S_k^{I(60)} \approx S_\beta^{I(60)}$ at nominal values. Then search the epidemiological literature for any reported correlations between contact rates k and transmission probabilities β across outbreak settings. Construct a 4×4 correlation matrix using whatever estimates you find, or assume a moderate positive correlation $\rho_{k\beta} = 0.6$ and zero for all other pairs. Use equation (3.6) to compute the total-derivative SI of $I(60)$ with respect to k accounting for this correlation. Compare with the standard SI and discuss the practical difference.

Extension. Using equation (3.10), compute the pure second-order sensitivities $\partial^2 I(60)/\partial p_i^2$ for each of the four POIs. For each parameter, assess whether the second-order term is large enough to matter for 10% perturbations. (Reuse evaluations from Part 1 to keep the additional cost minimal.)

3.8 Exercises

Exercise 3.1 *([Bloom L2])*. Explain in your own words why making the step size h smaller does not always improve the accuracy of a finite-difference sensitivity estimate. Describe the two sources of error and explain why they act in opposite directions as h changes.

Exercise 3.2 *([Bloom L3])*. A model has $n_p = 15$ POIs and $n_q = 4$ QOIs.

(a) Fill in Table 3.1 for this model: how many model evaluations does each method require?

(b) If each evaluation takes 30 seconds, how long does the centered-difference Jacobian take?

(c) At what point (in terms of n_p and evaluation time) would the adjoint method from Chapter 6 be strictly necessary?

Exercise 3.3 *([Bloom L3])*. For the function $q(p) = \sin(p^2)$ at $p^* = 1.3$, compute $\partial q/\partial p$ using centered differences for $h \in \{10^{-1}, 10^{-2}, \ldots, 10^{-14}\}$. Compare each estimate to the exact value $q'(p) = 2p\cos(p^2)$. Plot error vs. h on log-log axes and identify the optimal h. Does it agree with the formula in equation (3.9)?

Exercise 3.4 *([Bloom L3])*. For the SIR model $R_0 = k\beta\tau$:

(a) Compute the pure second derivatives $\partial^2 R_0/\partial k^2$, $\partial^2 R_0/\partial \beta^2$, $\partial^2 R_0/\partial \tau^2$ analytically. What do you notice?

(b) Compute the cross derivatives $\partial^2 R_0/\partial k\,\partial \beta$, $\partial^2 R_0/\partial k\,\partial \tau$, $\partial^2 R_0/\partial \beta\,\partial \tau$ analytically.

(c) For a 10% perturbation to k and a simultaneous 10% perturbation to β, estimate the true change in R_0 using both the first-order and the full second-order Taylor approximations. How large is the correction?

Exercise 3.5 *([Bloom L4])*. In the SIR model, suppose field data suggests that the contact rate k and transmission probability β are positively correlated: across outbreak settings, when k increases by 10%, β tends to increase by 6%.

(a) Compute the standard sensitivity indices $S_k^{R_0}$ and $S_\beta^{R_0}$.

(b) Using equation (3.12), compute the total-derivative SI of R_0 with respect to k, accounting for the correlation.

(c) Which is larger: the standard SI or the total-derivative SI? What does this mean for a policy maker who can reduce contact rates (through social distancing) but not transmission probability directly?

Exercise 3.6 *(*[Bloom L4]*).* A colleague provides you with the function `seir_model(p)` that implements a SEIR epidemic model with $n_p = 6$ parameters and returns the peak infection count as a single QOI. Each model evaluation takes approximately 0.2 seconds.

(a) How long will it take to compute the complete sensitivity Jacobian using centered differences?

(b) Apply the `sensitivity_jacobian` template to this model. What additional information do you need before interpreting the results? (Think about parameterization.)

(c) You find that two of the six parameters have $|\rho_{ij}| = 0.78$. What should you do before reporting the sensitivity indices?

Exercise 3.7 *(*[Bloom L5]*).* A government agency funds you to perform SA on a dengue fever transmission model with 9 parameters and 3 QOIs (peak infections, total deaths, days until outbreak peak). Each model evaluation takes 4 minutes.

Design the complete computational SA study. Your design should specify:

(a) Which finite-difference method you will use and why.

(b) The total computation time.

(c) How you will choose the step size for each parameter.

(d) What checks you will run before reporting any results.

(e) How you will present the results to a non-technical audience.

(f) One scenario in which the standard SA would give misleading results, and what you would do instead.

Exercise 3.8 *(*[Bloom L6]*).* A published sensitivity analysis reports: "All finite-difference computations used $h = 10^{-12}$ to ensure maximum precision. The code was run in MATLAB using double-precision arithmetic."

The authors report that for several parameters, the sensitivity indices are exactly zero to all reported decimal places.

Identify what is most likely wrong with these results and explain your reasoning quantitatively, using the U-curve analysis from Section 3.3. What should the authors have done, and how would you determine whether the zero SI values are genuine or numerical artifacts?

Part III

Analytic Forward Sensitivity Analysis

Chapter 4

Analytic Forward Sensitivity Analysis

Differentiate the model, and the important parameters identify themselves.

Chapter 3 showed how to compute sensitivity indices for any model by finite differences. For the malaria model with 17 POIs and 7 state variables, the centered-difference Jacobian requires 239 model evaluations, about 4 hours if each run takes a minute, or 10 days if each takes an hour.

There is a faster and more accurate alternative, provided the model can be differentiated analytically: differentiate the governing equations directly. The result is a new system of equations the *forward sensitivity equations*, whose solution is the exact derivative at the cost of one linear solve per POI. For the malaria model: 17 linear solves of a 7×7 system, each taking milliseconds.

The catch is that the model must be differentiable and the analyst must be patient enough to derive the sensitivity equations. This chapter develops that derivation systematically, shows that it always produces a *linear* system no matter how nonlinear the original model is, and ends by quantifying the remaining computational bottleneck that motivates Chapters 5 and 6.

4.1 The FSE Derivation Pattern

The forward sensitivity equation (FSE) for any model is obtained by a single operation: differentiate the governing equation with respect to a POI and apply the chain rule. The result is always a linear equation in the sensitivity variables, regardless of whether the original model is linear or nonlinear.

4.1.1 The Pattern in Three Steps

Step 1. Write the governing equation in the form $\mathcal{F}(u;p) = 0$ or $\mathcal{F}(u;p) = f$, where u is the solution and p is the POI of interest.

Step 2. Differentiate both sides with respect to p. Apply the chain rule to every term that depends on u. Collect terms. The result is an equation for $\partial u/\partial p$.

Step 3. Recognize that the equation from Step 2 is *linear in $\partial u/\partial p$*, even if $\mathcal{F}$ is nonlinear in u.

The linearity in Step 3 is not a coincidence. Differentiation is a linear operation: applying it to a nonlinear equation in u produces a linear equation in $\partial u/\partial p$. This structural fact is what makes analytic SA computationally tractable.

Misconception to confront. A common reaction to the FSE is: "Deriving and solving these equations must be at least as hard as solving the original model." This is wrong in two ways. First, the FSE is always *linear* in the sensitivity variable, even when the original model is highly nonlinear; a linear system is categorically easier to solve than a nonlinear one. Second, the FSE for an ODE uses the *same* coefficient matrix F_u that the original solver already computes, so the additional cost per parameter is one linear solve, not one full nonlinear solve. The FSE is not a second model; it is a linear companion to the first.

4.1.2 Application to Three Model Types

The same three-step pattern applies to algebraic systems, difference equations, and ordinary differential equations. Sections 4.2–4.3 work through each type in detail. Here we display the pattern side by side.

Model type	Governing equation	Forward sensitivity equation
Algebraic	$f(u;p) = 0$	$f_u \frac{\partial u}{\partial p} = -f_p$
Linear system	$A\vec{u} = \vec{b}$	$A \frac{\partial \vec{u}}{\partial p} = \frac{\partial \vec{b}}{\partial p} - \frac{\partial A}{\partial p}\vec{u}$
ODE IVP	$\dot{u} = F(u;p)$	$\frac{d}{dt}\left[\frac{\partial u}{\partial p}\right] = F_u \frac{\partial u}{\partial p} + F_p$
O∆E	$u_{n+1} = G(u_n;p)$	$\frac{\partial u_{n+1}}{\partial p} = G_u \frac{\partial u_n}{\partial p} + G_p$

In each case, the coefficient multiplying the unknown sensitivity (f_u, A, F_u, G_u) is the Jacobian of the model with respect to the solution u, evaluated at the nominal solution. The right-hand side contains only known quantities. The sensitivity variable ($\partial u/\partial p$) always appears linearly on the left.

Unifying Idea 1. Every entry of the sensitivity Jacobian that this chapter computes is a normalized derivative: an instance of Unifying Idea 1 from Chapter 1. The FSE computes those derivatives exactly, without the $O(h^2)$ truncation error of the finite-difference approach. The FSE is not a different kind of SA result; it is the same sensitivity index, obtained by a more accurate and often faster computational route.

Self-check. For the logistic ODE $\dot{u} = ru(1 - u/K)$, write down the FSE for $\partial u/\partial r$ and for $\partial u/\partial K$. Identify the coefficient F_u in each case.

Answer: $F(u; r, K) = ru(1-u/K)$. Computing partial derivatives: $F_u = r(1-2u/K)$, $F_r = u(1-u/K)$, $F_K = ru^2/K^2$.
FSE for $w_r = \partial u/\partial r$: $\dot{w}_r = r(1 - 2u/K)\, w_r + u(1 - u/K)$.
FSE for $w_K = \partial u/\partial K$: $\dot{w}_K = r(1 - 2u/K)\, w_K + ru^2/K^2$.
Both FSEs have the same coefficient $F_u = r(1 - 2u/K)$ on the left; only the right-hand-side forcing term differs. The coefficient depends on the forward solution $u(t)$, so the FSE must be solved simultaneously with the original ODE.

4.2 Algebraic Systems and Equilibria

For static problems, equations whose solutions do not change in time, the FSE is a linear system that can be solved by standard linear algebra.

4.2.1 A Single Nonlinear Equation

For the scalar equation $f(u;p) = 0$, differentiating with respect to p gives

$$f_u \frac{\partial u}{\partial p} + f_p = 0,$$

so $\partial u/\partial p = -f_p/f_u$ provided $f_u \neq 0$. This is exactly the implicit function theorem. When $f_u = 0$, the derivative is undefined, which corresponds to a bifurcation point, a situation studied in Chapter 8.

4.2.2 Linear Systems: The FSE for $A\vec{u} = \vec{b}$

For the system $A\vec{u} = \vec{b}$, differentiating with respect to any scalar parameter $q \in \{a_{ij}, b_i\}$ gives:

$$A \frac{\partial \vec{u}}{\partial q} = \frac{\partial \vec{b}}{\partial q} - \frac{\partial A}{\partial q}\vec{u}. \tag{4.1}$$

The right-hand side is a known vector once the forward solution $\vec{u}$ is known. Solving equation (4.1) requires one additional linear solve with the same coefficient matrix A for each parameter q. Since the LU factorization of A is computed only once, the cost for m parameters is one LU factorization plus m back-substitutions.

Important: Do not solve (4.1) by computing A^{-1}. Explicit matrix inversion is numerically unstable and computationally wasteful. In MATLAB, use the backslash operator (`A\b`), which solves the system by LU factorization without forming the inverse.

4.2.3 The Symbiotic Population Model

Consider the two-species symbiotic population model [28]:

$$\frac{du_1}{dt} = u_1(1 - u_1 - au_2), \tag{4.2}$$

$$\frac{du_2}{dt} = bu_2(1 - u_2 - cu_1), \tag{4.3}$$

with parameters a, b, $c > 0$ satisfying $0 < a, c < 1$. The coexistence equilibrium $(\bar{u}_1, \bar{u}_2)$ satisfies the nonlinear system

$$\bar{u}_1 = \frac{1-a}{1-ac}, \qquad \bar{u}_2 = \frac{1-c}{1-ac}. \tag{4.4}$$

To find how this equilibrium depends on a and c, we differentiate the equilibrium equations

$$f_1(\bar{u}_1, \bar{u}_2; a, c) := \bar{u}_1(1 - \bar{u}_1 - a\bar{u}_2) = 0,$$
$$f_2(\bar{u}_1, \bar{u}_2; a, c) := b\bar{u}_2(1 - \bar{u}_2 - c\bar{u}_1) = 0,$$

with respect to a to obtain the FSE

$$D_{\vec{u}}[\vec{f}]\, \frac{\partial \vec{u}}{\partial a} = -\frac{\partial \vec{f}}{\partial a}, \tag{4.5}$$

where $D_{\vec{u}}[\vec{f}]$ is the Jacobian matrix of $\vec{f}$ with respect to $\vec{u}$, evaluated at the equilibrium:

$$D_{\vec{u}}[\vec{f}]\big|_{\text{eq}} = \begin{pmatrix} 1 - 2\bar{u}_1 - a\bar{u}_2 & -a\bar{u}_1 \\ -bc\bar{u}_2 & b(1 - 2\bar{u}_2 - c\bar{u}_1) \end{pmatrix}, \tag{4.6}$$

and

$$\frac{\partial \vec{f}}{\partial a} = \begin{pmatrix} -\bar{u}_1\bar{u}_2 \\ 0 \end{pmatrix}.$$

Solving the 2×2 linear system (4.5) gives $\partial\bar{u}_1/\partial a$ and $\partial\bar{u}_2/\partial a$ exactly. The sensitivity indices follow from Definition 2.1.

A key observation: although the equilibrium equations (4.2)–(4.3) are nonlinear, the FSE (4.5) is a *linear system* in the sensitivity variables. This is always true for equilibrium problems, as the following result shows.

Theorem 4.1 (Linearity of the FSE)**.** *Let $\vec{f}(\vec{u};\vec{p}) = \vec{0}$ be a system of nonlinear equations with $\vec{f}$ differentiable in $\vec{u}$ and $\vec{p}$. The forward sensitivity equation for $\partial\vec{u}/\partial p_j$ is the linear system*

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} = -\frac{\partial \vec{f}}{\partial p_j},$$

provided the Jacobian $D_{\vec{u}}[\vec{f}]$ is nonsingular.

Proof. Differentiating $\vec{f}(\vec{u}(\vec{p});\vec{p}) = \vec{0}$ with respect to p_j and applying the chain rule gives

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} + \frac{\partial \vec{f}}{\partial p_j} = \vec{0},$$

where $D_{\vec{u}}[\vec{f}]$ denotes the Jacobian matrix of $\vec{f}$ with respect to $\vec{u}$, evaluated at the nominal solution $(\vec{u}(\vec{p});\vec{p})$. Rearranging yields

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} = -\frac{\partial \vec{f}}{\partial p_j}.$$

The right-hand side $-\partial\vec{f}/\partial p_j$ is a known vector once $\vec{u}(\vec{p})$ is available. Linearity in $\partial\vec{u}/\partial p_j$ follows from the fact that differentiation is itself a linear operation: the quantity $\partial\vec{u}/\partial p_j$ appears in the left-hand side only through the matrix-vector product $D_{\vec{u}}[\vec{f}]\,(\partial\vec{u}/\partial p_j)$. No powers, products, or compositions of $\partial\vec{u}/\partial p_j$ appear, regardless of the nonlinearity of $\vec{f}$ in $\vec{u}$. Nonsingularity of $D_{\vec{u}}[\vec{f}]$ guarantees a unique solution. □

This theorem is one of the most practically useful facts in SA. Regardless of how complex or nonlinear the model is, its FSE at an equilibrium is always a linear system, and linear systems can be solved efficiently.

Self-check. For the symbiotic population model with $a = 0.4$, $b = 1$, $c = 0.3$: (a) Compute the coexistence equilibrium using equation (4.4). (b) Assemble the Jacobian (4.6) at this equilibrium. (c) Solve the FSE for $\partial\bar{u}_1/\partial a$ and $\partial\bar{u}_2/\partial a$. (d) Compare with the analytic derivatives of (4.4).

Answer: (a) $\bar{u}_1 = (1-0.4)/(1-0.12) = 0.6/0.88 \approx 0.6818$; $\bar{u}_2 = (1-0.3)/0.88 = 0.7/0.88 \approx 0.7955$.

(b) $D_u f = \begin{pmatrix} 1-2(0.6818)-0.4(0.7955) & -0.4(0.6818) \\ -1(0.3)(0.7955) & 1-2(0.7955)-0.3(0.6818) \end{pmatrix} \approx \begin{pmatrix} -0.6818 & -0.2727 \\ -0.2386 & -0.7955 \end{pmatrix}$.

(c) RHS: $-\partial\vec{f}/\partial a = (0.6818\times0.7955, 0)^T \approx (0.5424, 0)^T$. Solve 2×2 system.

(d) Analytic: $\partial\bar{u}_1/\partial a = -(1-c)/((1-ac)^2) = -(0.7)/(0.88)^2 \approx -0.904$. Compare with FSE solution.

4.2.4 The Car Rental Problem: OΔE Sensitivity

Consider the car rental Markov chain: two airports A and B with transition probabilities $\alpha = \Pr(A \to A)$ and $\beta = \Pr(B \to B)$. The discrete-time model is

$$A_{n+1} = \alpha A_n + (1-\beta)B_n, \tag{4.7}$$
$$B_{n+1} = (1-\alpha)A_n + \beta B_n. \tag{4.8}$$

The fixed point satisfies (A^*, B^*):

$$A^* = \frac{1-\beta}{2-\alpha-\beta}, \qquad B^* = \frac{1-\alpha}{2-\alpha-\beta}.$$

The FSE for the sensitivity of the fixed point to α is exactly equation (4.1) with the transition matrix playing the role of A. The result is:

$$S_\alpha^{A^*} = \frac{\alpha}{2-\alpha-\beta}, \qquad S_\alpha^{B^*} = -\frac{\alpha(1-\beta)}{(1-\alpha)(2-\alpha-\beta)}. \tag{4.9}$$

These indices tell management which transition probability has the greater effect on the long-run car distribution. At nominal values $\alpha = 0.7$ and $\beta = 0.3$, the formula gives $S_\alpha^{A^*} = 0.7/(2-0.7-0.3) = 0.70$: a 1% increase in the probability of keeping a car at airport A produces a 0.70% increase in the long-run fraction of cars at A.

4.3 Forward Sensitivity for ODE Systems

For a dynamical model governed by an ODE, the sensitivity $\partial u/\partial p$ changes over time as the solution evolves. The FSE is itself an ODE that must be solved alongside the original model.

4.3.1 Derivation of the ODE FSE

Consider the scalar, autonomous ODE with parameter p:

$$\frac{du}{dt} = F(u;p), \qquad u(0) = u_0(p). \tag{4.10}$$

Define the sensitivity variable $w(t) = \partial u(t)/\partial p$. Differentiating (4.10) with respect to p and exchanging the order of $\partial/\partial t$ and $\partial/\partial p$:

$$\frac{d}{dt}\left[\frac{\partial u}{\partial p}\right] = \frac{\partial F}{\partial u}\frac{\partial u}{\partial p} + \frac{\partial F}{\partial p}. \tag{4.11}$$

In terms of w:

$$\dot{w} = F_u(u;p)\, w + F_p(u;p), \qquad w(0) = \frac{du_0}{dp}. \tag{4.12}$$

The coefficient $F_u(u(t);p)$ depends on the forward solution $u(t)$, so the FSE (4.12) must be solved simultaneously with the original ODE (4.10). Together they form an *augmented system* of two ODEs.

For the vector ODE $\dot{\vec{u}} = \vec{F}(\vec{u};\vec{p})$ with n state variables and m parameters:

$$\frac{d}{dt}\left[\frac{\partial \vec{u}}{\partial p_j}\right] = J(\vec{u};\vec{p})\,\frac{\partial \vec{u}}{\partial p_j} + \frac{\partial \vec{F}}{\partial p_j}, \tag{4.13}$$

where $J = \partial\vec{F}/\partial\vec{u}$ is the $n \times n$ Jacobian of $\vec{F}$ with respect to $\vec{u}$. One equation (4.13) is needed per parameter p_j, each an n-dimensional ODE system. For m parameters: the augmented system has $n + mn = n(1 + m)$ equations.

Theorem 4.1 extends to the dynamical case: each FSE is *linear in the sensitivity variable* $\partial\vec{u}/\partial p_j$, even though $\vec{F}$ is nonlinear in $\vec{u}$. The coefficient $J(\vec{u}(t); \vec{p})$ is time-varying, but the structure is linear.

Self-check. Write down the augmented ODE system (original ODE + FSE) for the logistic equation $\dot{u} = ru(1 - u/K)$ with respect to both parameters r and K simultaneously. How many total ODEs does the augmented system have?

Answer: The augmented system consists of: Original: $\dot{u} = ru(1 - u/K)$. FSE for $w_r = \partial u/\partial r$: $\dot{w}_r = r(1 - 2u/K)w_r + u(1 - u/K)$. FSE for $w_K = \partial u/\partial K$: $\dot{w}_K = r(1 - 2u/K)w_K + ru^2/K^2$. Total: 3 ODEs. In general, for n state variables and m parameters, the augmented system has $n + mn = n(1 + m)$ equations.

4.3.2 Time-Dependent Sensitivity Indices

For dynamical models, the sensitivity index is time-dependent:

$$S_p^u(t) = \frac{p}{u(t)}\, w(t) = \frac{p}{u(t)} \frac{\partial u(t)}{\partial p}. \tag{4.14}$$

The relative importance of POIs can change over time. A parameter that strongly influences the early phase of an epidemic may become less important as the system approaches equilibrium.

4.3.3 The SIR Model: Full FSE Treatment

The SIR model with demography is:

$$\frac{dS}{dt} = \mu N - \frac{k\beta SI}{N} - \mu S, \tag{4.15}$$
$$\frac{dI}{dt} = \frac{k\beta SI}{N} - (\gamma + \mu)I, \tag{4.16}$$
$$\frac{dR}{dt} = \gamma I - \mu R. \tag{4.17}$$

With interpretable POIs $\vec{p} = (k, \beta, \tau, L)$ where $\tau = 1/\gamma$ (mean infectious period, days) and $L = 1/\mu$ (mean lifespan, days), the augmented system for a single POI p_j adds 3 sensitivity ODEs:

$$\frac{dw_{Sj}}{dt} = J_{11}w_{Sj} + J_{12}w_{Ij} + J_{13}w_{Rj} + \frac{\partial F_S}{\partial p_j}, \tag{4.18}$$
$$\frac{dw_{Ij}}{dt} = J_{21}w_{Sj} + J_{22}w_{Ij} + J_{23}w_{Rj} + \frac{\partial F_I}{\partial p_j}, \tag{4.19}$$
$$\frac{dw_{Rj}}{dt} = J_{31}w_{Sj} + J_{32}w_{Ij} + J_{33}w_{Rj} + \frac{\partial F_R}{\partial p_j}, \tag{4.20}$$

where J_{ij} are entries of the Jacobian of (F_S, F_I, F_R) with respect to (S, I, R). For $m = 4$ POIs, the full augmented system has $3 + 4 \times 3 = 15$ ODEs.

The initial conditions for the FSE are zero: $w_{Sj}(0) = w_{Ij}(0) = w_{Rj}(0) = 0$ (assuming the initial conditions do not depend on the parameters).

Figure 4.1 shows the time-dependent SI for $I(t)$ with respect to k, β, τ, and L. In the early phase of the outbreak (first 30 days), the contact rate k and transmission probability β dominate.

As the outbreak progresses and the susceptible pool depletes, the mean infectious period τ becomes increasingly important. These insights are only available through time-dependent SA; a static SI computed at a single time point would miss the qualitative change in parameter importance.

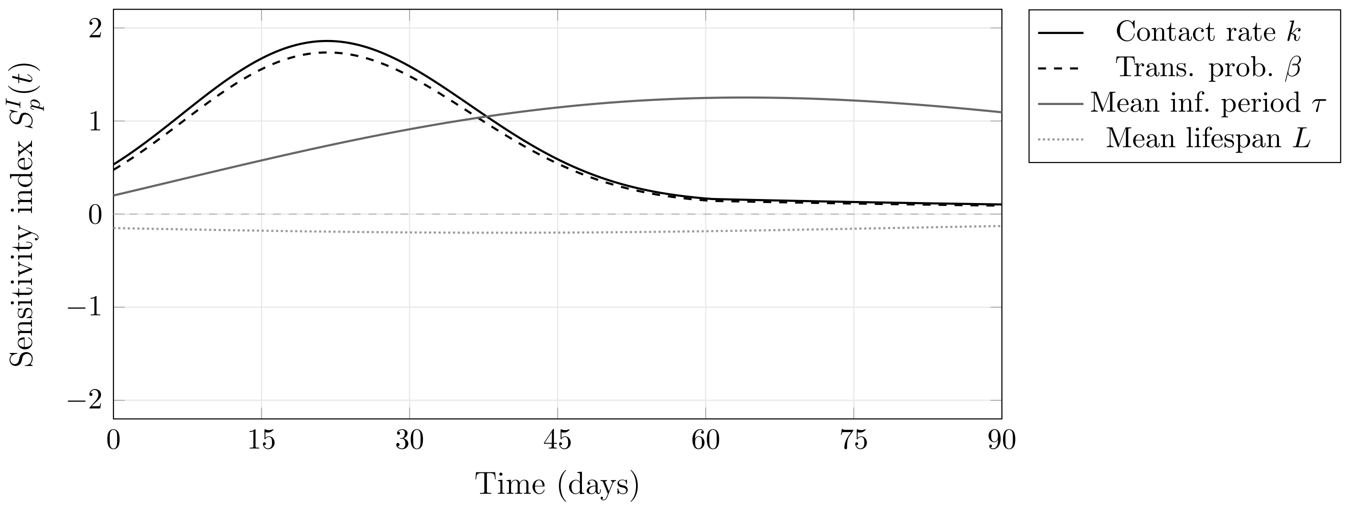


Figure 4.1: Schematic of time-dependent sensitivity indices $S_p^I(t)$ for the SIR model. In the early outbreak phase, the contact rate k and transmission probability β (nearly equal because $S_k^{R_0} = S_\beta^{R_0}$) dominate. The mean infectious period τ grows in importance as the outbreak progresses. The mean lifespan L has a small negative influence throughout. Curve shapes are representative; the MATLAB laboratory (Section 4.6) and the function `plot_time_si.m` in the companion repository produce exact computed curves at the nominal parameters.

4.4 The Proliferation Problem and Its Consequences

Before improving the model, find out which knob actually moves the machine.

The FSE is exact and efficient for a single parameter. For m parameters, the cost scales linearly: m augmented ODE solves, each requiring n additional state variables. Table 4.1 quantifies this for representative models.

Counting convention. Throughout this section, "solve counts" refer to the total number of ODE solves, including the one baseline forward solve (of the original n-equation system) required before any sensitivity computation begins. The FSE adds m augmented solves on top of this baseline, for a total of $m + 1$ solves.

For the SIR model, 4 solves is extremely fast in practice. For the climate model, 200 solves at 6 hours each is 50 days. This is the computational wall that motivates the adjoint approach.

Unifying Idea 2. Table 4.1 reveals the bottleneck that Chapters 5 and 6 resolve. No matter how the sensitivity Jacobian is computed, by finite differences, by analytic FSE, or by the adjoint; the answer is the same matrix of normalized derivatives. The adjoint (Unifying Idea 2, introduced in Chapter 5) reduces the cost from m solves to 2 solves by exploiting the transposed structure of the problem. The cost reduction is proportional to m; for $m = 200$, the adjoint is 100 times faster than the FSE.

Unifying Idea 3. Every row of Table 4.1 is a linearization. The FSE computes the exact derivative of a nonlinear model, which is a linear approximation to the model's response. That approximation

Model	n states	m POIs	Augmented size	FSE cost
SIR	3	4	$3 + 12 = 15$	4 ODE solves
SEIR	4	6	$4 + 24 = 28$	6 ODE solves
Malaria model	7	17	$7 + 119 = 126$	17 ODE solves
Climate model	500	200	$500 + 100{,}000$	200 ODE solves

Table 4.1: Augmented system size and FSE cost for representative models. For the climate model, 200 ODE solves each taking 6 hours equals 50 days of computation. This is the cost that motivates the adjoint method in Chapter 6.

is valid when parameters are near their nominal values. The proliferation problem scales with m; the validity limitation is independent of m. Chapter 8 addresses the second problem; Chapters 5 and 6 address the first.

Self-check. For the malaria model with $n = 7$ state variables and $m = 17$ POIs, and one response functional (say, the endemic equilibrium fraction i_h): (a) How many ODE solves does the FSE approach require? (b) If each ODE solve takes 2 minutes, how long does the complete FSA take? (c) How many ODE solves will the adjoint method in Chapter 6 require?

> *Answer:* (a) 17 FSE solves (one per parameter), plus 1 forward solve = 18 total. (b) $18 \times 2 = 36$ minutes. (c) The adjoint always requires exactly 2 solves (1 forward + 1 adjoint), regardless of m. For 17 parameters, the adjoint is 9 times faster than the FSE.

4.5 The Decision Table: When to Use Which Method

With three computational methods now available (finite differences, analytic FSE, and the adjoint previewed above), the right choice depends on the model and the problem.

For this course, the default progression is:

1. Start with finite differences (Chapter 3) to get SA results quickly.
2. Switch to analytic FSE (this chapter) when the model is differentiable and accuracy matters.
3. Use the adjoint (Chapters 5–6) when m is large.
4. Check for correlated POIs before reporting any results (Chapter 3).
5. Consider global SA (Chapter 9) when parameter uncertainty is large.

A common assumption is that finite differences are adequate for any SA study once the step size is chosen correctly. For black-box models, this is true: FD is the only option and produces reliable results at the optimal step size. For differentiable models, the analytic FSE is simultaneously more accurate (exact rather than $O(h^2)$) and often faster (no repeated model evaluations at perturbed parameter values). The two methods give the same numbers when FD is implemented correctly; the FSE gives those same numbers without approximation error and without the step-size selection problem.

4.6 MATLAB Laboratory: Analytic FSA of the SIR Model

This laboratory implements the augmented ODE system for the SIR model and compares the analytic FSE approach with the finite-difference method from Chapter 3.

Setup. Use the SIR model (4.15)–(4.17) with POIs $\vec{p} = (k, \beta, \tau, L)$ (interpretable parameterization), nominal values $k = 5$, $\beta = 0.06$, $\tau = 7$ days, $L = 10{,}000$ days, and initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$, $N = 1000$.

Situation	Method	Accuracy	Cost	When to use
Black-box model	Finite differences	$O(h^2)$	$2n_p + 1$ evals	Ch 3
Analytic, small n_p	FSE	Exact	n_p solves	Ch 4
Analytic, large n_p	Adjoint	Exact	2 solves	Ch 5–6
Correlated POIs	Reparameterize first	—	—	Ch 3
Model near bifurcation	FSE + caution	Exact*	n_p solves	Ch 8
Large parameter uncertainty	Global SA	Variance-based	$N(m + 2)$ evals	Ch 9

Table 4.2: Decision table for selecting the SA computational method. For a systematic benchmarking of these methods on biological ODE models, see Mester et al. [26]. (*Near a bifurcation, the FSE solution is exact but the sensitivity index diverges, limiting its interpretability.)

Part 1: Implement the augmented ODE system. Write a MATLAB function `sir_augmented(t, y, k, beta, tau, L)` that returns the derivatives for the full 15-equation augmented system: 3 original state equations plus $4 \times 3 = 12$ FSE equations. The FSE equations require the Jacobian of (F_S, F_I, F_R) with respect to (S, I, R).

State vector layout. The 15-component state vector `y` is organized as follows:

Index	Meaning
1	$S(t)$
2	$I(t)$
3	$R(t)$
4–6	$(\partial S/\partial k,\ \partial I/\partial k,\ \partial R/\partial k)$
7–9	$(\partial S/\partial \beta,\ \partial I/\partial \beta,\ \partial R/\partial \beta)$
10–12	$(\partial S/\partial \tau,\ \partial I/\partial \tau,\ \partial R/\partial \tau)$
13–15	$(\partial S/\partial L,\ \partial I/\partial L,\ \partial R/\partial L)$

The initial condition is `y0 = [S0; I0; 0; zeros(12,1)]`. The sensitivity variables all start at zero because the initial conditions do not depend on the parameters. The file `sir_augmented.m` in the companion repository implements this system and may be used as a reference after completing your own implementation.

Part 2: Solve and extract sensitivities. Solve the augmented system from $t = 0$ to $t = 90$ days using `ode45`. Extract the time-dependent sensitivity indices $S^I_k(t)$, $S^I_\beta(t)$, $S^I_\tau(t)$, $S^I_L(t)$.

Part 3: Plot time-dependent SIs. Plot all four $S^I_{p_j}(t)$ curves on a single axis, alongside the epidemic curve $I(t)$. Identify: (a) at $t = 10$ days, which parameter matters most? (b) at $t = 50$ days, has the ranking changed? (c) which parameter has the largest absolute SI at the epidemic peak?

Part 4: Compare with finite differences. At $t = 30$ days and $t = 70$ days, compare the FSE-based SI values against the centered-difference estimates from Chapter 3. Are they equal to 4 significant figures? Report any discrepancies and explain their likely source.

Part 5: Cost comparison. Time the two approaches: the augmented ODE (all 4 sensitivities at once) versus 4 separate centered-difference calls to `SIR_model`. Which is faster? Which is more accurate? At what value of m (number of POIs) would you expect the two approaches to take approximately equal time?

4.7 Exercises

Exercise 4.1 *(*[Bloom L2]*)*. State the three-step FSE derivation pattern in your own words, without using any symbols. Then explain, in one sentence, why the FSE is always linear in the sensitivity variable even when the original equation is nonlinear.

Exercise 4.2 *(*[Bloom L3]*)*. For the nonlinear equation $g(u; p) := u^3 - 3pu + 2 = 0$:

(a) Write the FSE for $\partial u/\partial p$.

(b) At $p = 1$, find all real roots u numerically.

(c) Compute S_p^u at each root.

(d) At $p = 1$, for which root is the sensitivity largest in magnitude? Verify your answer using a centered finite difference with optimal h.

Exercise 4.3 *(*[Bloom L3]*)*. For the logistic ODE $\dot{u} = ru(1 - u/K)$, $u(0) = u_0$:

(a) Write the FSE for $w_r = \partial u/\partial r$.

(b) Verify that the FSE is linear in w_r by identifying the coefficient (which depends on $u(t)$) and the forcing term.

(c) Using MATLAB, implement the augmented system (u, w_r, w_K) and solve from $t = 0$ to $t = 20$.

(d) Plot $S_r^u(t)$ and $S_K^u(t)$. Interpret: near $t = 0$ vs. near the carrying capacity.

Exercise 4.4 *(*[Bloom L3]*)*. For the symbiotic population model (4.2)–(4.3) with $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute the coexistence equilibrium.

(b) Assemble and solve the FSE for $\partial \vec{u}/\partial c$.

(c) Compute the normalized sensitivity indices $S_c^{\bar{u}_1}$ and $S_c^{\bar{u}_2}$.

(d) Using equation (4.4), compute the same indices analytically. Verify agreement.

Exercise 4.5 *(*[Bloom L4]*)*. The SEIRS model (Susceptible-Exposed-Infectious-Recovered-Susceptible) has 4 state variables and 7 parameters.

(a) How many ODEs are in the augmented FSE system for all 7 parameters?

(b) If each ODE solve takes 3 minutes, how long does the full FSA take using the FSE approach? The adjoint approach (Chapter 6)?

(c) Suppose a modeler decides to reduce computation time by computing sensitivities only for the 3 parameters they expect to be most important, based on biological intuition. Identify one scenario where this strategy would lead to a misleading conclusion.

Exercise 4.6 *(*[Bloom L4]*)*. Using your MATLAB implementation from the laboratory:

(a) At three time points, $t = 5$ (early epidemic), $t = t_{\text{peak}}$ (peak infections), and $t = 80$ (late); report the tornado plot of SIs for $I(t)$.

(b) Does the parameter ranking change across the three time points?

(c) A public health officer says: "We should target the parameter with the largest SI at the epidemic peak, since that is when the most people are affected." Critique this statement using your results.

Exercise 4.7 *(*[Bloom L5]*)*. A colleague is modeling cholera transmission with a SEIR-type model having 8 parameters and 4 state variables. She asks for advice on the SA methodology. Design a complete FSA study, specifying:

(a) Whether to use finite differences or analytic FSE, and why.

(b) How you would validate the implementation.

(c) What QOIs you would compute sensitivity indices for, and at what time points.

(d) How you would present the results to identify the two most critical intervention targets.

(e) One situation in which the FSE results might be misleading, and what additional analysis you would recommend.

Exercise 4.8 *(*[Bloom L6]*)*. A published paper reports: "We performed sensitivity analysis of the SEIR model using the forward sensitivity equations. All sensitivity indices were computed at $t = 365$ days, giving a single set of values for the entire epidemic."

The model simulates a disease whose outbreak dynamics play out over approximately 60 days, but the model runs for a year.

Identify at least two methodological issues with this approach. For each issue, state what the authors should have done instead and explain what information they likely missed.

Part IV

The Adjoint Approach

Chapter 5

The Adjoint in Linear Algebra

The adjoint is the same sensitivity question, asked from the other end.

Chapter 4 showed that analytic forward sensitivity analysis for a model with $m = 50$ parameters requires 50 linear solves, one per parameter. Each solve uses the same coefficient matrix A, so we pay once for the LU factorization and 50 times for back-substitution. That is still 50 operations.

Now suppose we care about a single response functional, not the entire solution $\vec{u}$. The quantity of interest is $J = \vec{c}^T\vec{u}$: the pollution level at one monitoring station, the total disease burden, the profit at equilibrium. We want $\partial J/\partial q$ for all 50 parameters q.

Is there a way to compute all 50 derivatives simultaneously, in the cost of a *single* linear solve?

Yes. The answer is the adjoint. The adjoint approach to sensitivity analysis was developed for nonlinear systems by Cacuci [6] and has since become a standard tool in optimal design and computational science [12].

5.1 A Preview: The Same Answer, One Solve Instead of Fifty

Before the general theory, work through a small example by hand to see exactly what the adjoint does and why it is efficient.

The Setup

Consider the 2×2 linear system

$$A\vec{u} = \vec{b}(q_1, q_2), \qquad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}, \tag{5.1}$$

where q_1 and q_2 are two parameters of interest. The scalar response functional is $J = u_1 + u_2 = \vec{c}^T\vec{u}$ with $\vec{c} = (1,1)^T$. We want $\partial J/\partial q_1$ and $\partial J/\partial q_2$.

The Direct (Forward) Approach: Two Solves

Differentiating $A\vec{u} = \vec{b}$ with respect to q_1: $A\,(\partial\vec{u}/\partial q_1) = \partial\vec{b}/\partial q_1 = (1,0)^T$. Differentiating with respect to q_2: $A\,(\partial\vec{u}/\partial q_2) = \partial\vec{b}/\partial q_2 = (0,1)^T$. Two linear solves (one per parameter) with the same matrix A:

$$A^{-1} = \tfrac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad \text{so} \quad \frac{\partial \vec{u}}{\partial q_1} = \tfrac{1}{5}\begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad \frac{\partial \vec{u}}{\partial q_2} = \tfrac{1}{5}\begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

The derivatives of $J = \vec{c}^T\vec{u}$ are:

$$\frac{\partial J}{\partial q_1} = \vec{c}^T\frac{\partial \vec{u}}{\partial q_1} = (1,1)\cdot\tfrac{1}{5}(2,-1)^T = \tfrac{1}{5}, \qquad \frac{\partial J}{\partial q_2} = \vec{c}^T\frac{\partial \vec{u}}{\partial q_2} = (1,1)\cdot\tfrac{1}{5}(-1,3)^T = \tfrac{2}{5}.$$

Two solves, two derivatives. For 50 parameters: 50 solves.

The Adjoint Approach: One Solve

Instead of solving for $\partial\vec{u}/\partial q_j$ separately for each j, introduce the **adjoint vector** $\vec{v}$ satisfying

$$A^T\vec{v} = \vec{c}. \tag{5.2}$$

This is one linear solve with the transposed matrix. Since A is symmetric here, $A^T = A$, so $\vec{v} = A^{-1}\vec{c}$:

$$\vec{v} = A^{-1}\vec{c} = \tfrac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \tfrac{1}{5}\begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

What does $\vec{v}$ represent? The adjoint vector answers the question: *if the state $\vec{u}$ were perturbed by a small amount $\delta\vec{u}$, how much would J change?* A direct perturbation gives $\delta J \approx \vec{c}^T\delta\vec{u}$, but not all $\delta\vec{u}$ are achievable — the system constraint $A\vec{u} = \vec{b}$ means only perturbations satisfying $A\,\delta\vec{u} = \delta\vec{b}$ are reachable. The adjoint vector $\vec{v}$ encodes how much each component of the state contributes to J, after accounting for the coupling imposed by the matrix A.

In the 2×2 example, $\vec{v} = \frac{1}{5}(1,2)^T$ means that a perturbation to u_2 matters *twice as much* to $J = u_1 + u_2$ as a perturbation to u_1 — not because $\vec{c} = (1,1)^T$ treats them equally, but because the system A couples u_1 and u_2 differently. This coupling is what makes the adjoint more than a convenient shortcut: $\vec{v}$ is the correct sensitivity weighting for the constrained system, not the naive weighting $\vec{c}$ for the unconstrained response.

Now observe: for any parameter q_j,

$$\frac{\partial J}{\partial q_j} = \vec{c}^T\frac{\partial \vec{u}}{\partial q_j} = (A^T\vec{v})^T\frac{\partial \vec{u}}{\partial q_j} = \vec{v}^T\underbrace{A\frac{\partial \vec{u}}{\partial q_j}}_{=\partial\vec{b}/\partial q_j} = \vec{v}^T\frac{\partial \vec{b}}{\partial q_j}.$$

This is the key step: $A\,(\partial\vec{u}/\partial q_j) = \partial\vec{b}/\partial q_j$ is the FSE right-hand side, a known vector. No additional solve is needed — just a dot product.

Checking against our direct answers:

$$\frac{\partial J}{\partial q_1} = \vec{v}^T\frac{\partial \vec{b}}{\partial q_1} = \tfrac{1}{5}(1,2)\cdot(1,0)^T = \tfrac{1}{5}\ \checkmark \qquad \frac{\partial J}{\partial q_2} = \vec{v}^T\frac{\partial \vec{b}}{\partial q_2} = \tfrac{1}{5}(1,2)\cdot(0,1)^T = \tfrac{2}{5}\ \checkmark$$

One solve for $\vec{v}$, then one dot product per parameter. For 50 parameters: still one solve, then 50 dot products. The cost advantage is a factor of 50.

Self-check. Add a third parameter q_3 to the system above by setting $\vec{b} = (q_1 + q_3,\ q_2)^T$. Without solving any new linear systems, use the adjoint vector $\vec{v}$ already computed to find $\partial J/\partial q_3$.

> *Answer:* $\partial\vec{b}/\partial q_3 = (1,0)^T$. $\partial J/\partial q_3 = \vec{v}^T(1,0)^T = \frac{1}{5}(1,2)\cdot(1,0)^T = \frac{1}{5}$. No new solve needed. Adding parameters costs only one dot product each once $\vec{v}$ is known. This is the computational power of the adjoint.

The pattern above is the entire idea of the adjoint, stripped to its essentials. Section 5.2 states it precisely.

5.2 The Strategy, Stated Explicitly

The cost analysis above raises the key question: can we recover all m sensitivities without solving m separate systems? It turns out we can, through a trick that is worth understanding before seeing the symbols.

The adjoint is not a new kind of problem. It is a deliberate trick: introduce an auxiliary vector, choose it to cancel the unknown sensitivity variable, and read off the result. Here is the trick in plain language before any symbols.

Step 1. We want $\partial J/\partial q_j$ for all parameters q_j. Since $J = \vec{c}^T\vec{u}$, this is $\partial J/\partial q_j = \vec{c}^T(\partial\vec{u}/\partial q_j)$. But $\partial\vec{u}/\partial q_j$ requires a separate FSE solve for every j (50 solves for 50 parameters).

Step 2. Introduce an auxiliary vector $\vec{v}$ and observe that for any vector $\vec{v}$:

$$\vec{c}^T\frac{\partial\vec{u}}{\partial q_j} = \vec{v}^T\left[A\frac{\partial\vec{u}}{\partial q_j}\right] + \left(\vec{c}^T - \vec{v}^TA\right)\frac{\partial\vec{u}}{\partial q_j}.$$

The first term on the right is $\vec{v}^T\vec{r}_j$, where $\vec{r}_j = A(\partial\vec{u}/\partial q_j)$ is the FSE right-hand side (a known vector once $\vec{u}$ is known). The second term vanishes if we choose $\vec{v}$ so that $\vec{v}^TA = \vec{c}^T$.

Step 3. Choose $\vec{v}$ to satisfy the **adjoint equation**:

$$A^T\vec{v} = \vec{c}. \tag{5.3}$$

Then $\vec{c}^T - \vec{v}^TA = \vec{0}$ and the sensitivity formula reduces to:

$$\frac{\partial J}{\partial q_j} = \vec{v}^T\vec{r}_j \qquad \text{for all } j = 1,\ldots,m. \tag{5.4}$$

One solve for $\vec{v}$ (equation (5.3)) recovers all m sensitivities through formula (5.4). The trick works because the adjoint equation does not depend on j: $\vec{v}$ is determined by $\vec{c}$ and A alone.

The adjoint is not a separate, unrelated problem. It is constructed entirely from the coefficient matrix A (via transposition) and the response vector $\vec{c}$ of the forward problem. Everything about the adjoint is determined by the forward problem.

Self-check. For the system $A\vec{u} = \vec{b}$ with $A = \begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}$ and response functional $J = u_1$ (so $\vec{c} = (1,0)^T$): write down the adjoint equation (5.3) and solve for $\vec{v}$.

Answer: $A^T\vec{v} = \vec{c}$ becomes $\begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}\vec{v} = \begin{pmatrix}1\\ 0\end{pmatrix}$. Solving: $v_1 = 1/2$, $v_2 = -1/6$. This single solve (one back-substitution with A^T) gives us the sensitivity of $J = u_1$ to any number of parameters, via formula (5.4).

5.3 The Commutative Diagram

The three-step strategy and the 2×2 preview both express the same mathematical fact: the FSE route and the adjoint route reach the same sensitivity, at different cost. Figure 5.1 shows this as a diagram — read it as a summary of the algebra already developed, not as an introduction to the idea.

The diagram commutes: both paths from the forward problem to the sensitivity give identical results. This can be verified directly from the algebra in Section 5.2.

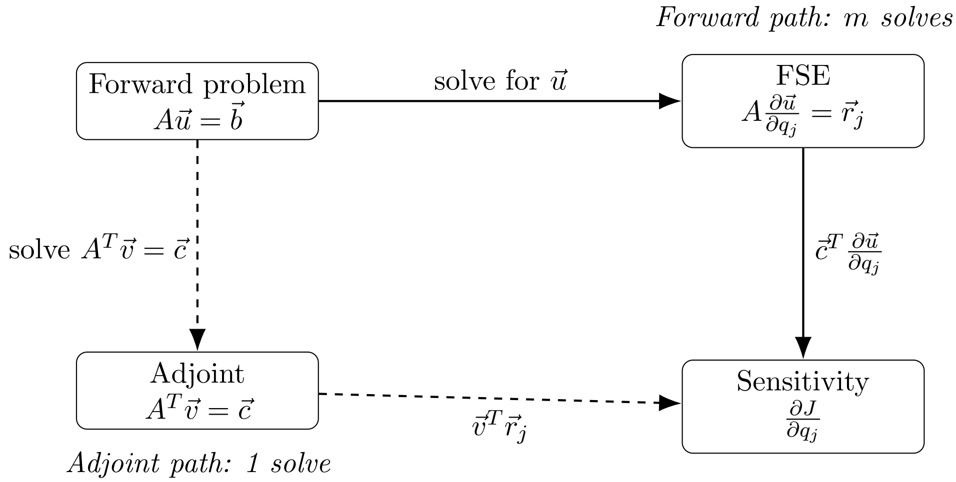


Figure 5.1: The commutative diagram for sensitivity analysis. The forward path (solid arrows) computes $\partial J/\partial q_j$ by solving the FSE for each parameter, m solves total. The adjoint path (dashed arrows) solves the adjoint equation once, then evaluates the sensitivity formula for each parameter; 1 solve total. Both paths deliver the same result.

Theorem 5.1 (Forward–Adjoint Equivalence). *Let $A\vec{u} = \vec{b}$ with A nonsingular, $J = \vec{c}^T\vec{u}$, and FSE $A\,\partial\vec{u}/\partial q_j = \vec{r}_j$. Let $\vec{v}$ solve the adjoint equation $A^T\vec{v} = \vec{c}$. Then*

$$\frac{\partial J}{\partial q_j} = \vec{c}^T\frac{\partial \vec{u}}{\partial q_j} = \vec{v}^T\vec{r}_j$$

for every parameter q_j.

Proof. $\vec{c}^T(\partial\vec{u}/\partial q_j) = (A^T\vec{v})^T(\partial\vec{u}/\partial q_j) = \vec{v}^T A(\partial\vec{u}/\partial q_j) = \vec{v}^T\vec{r}_j$. □

The adjoint gives *exactly* the same sensitivity values as the FSE. It is not an approximation. Students sometimes expect that a method requiring fewer solves must sacrifice accuracy; here it does not.

Self-check. Using Theorem 5.1, state in words why the adjoint equation does not need to be re-solved when a new parameter q_k is added to the sensitivity study.

Answer: The adjoint equation $A^T\vec{v} = \vec{c}$ depends only on A (the system coefficient matrix) and $\vec{c}$ (the response vector), neither of which changes when we add a new parameter. Adding parameter q_k changes only the FSE right-hand side $\vec{r}_k = \partial\vec{b}/\partial q_k - (\partial A/\partial q_k)\vec{u}$, which is a known vector once the forward solution $\vec{u}$ is computed. The sensitivity $\partial J/\partial q_k = \vec{v}^T\vec{r}_k$ is then obtained by a single dot product, no additional linear solve.

5.4 Cost Comparison and the Decision Rule

5.4.1 When to Use the Adjoint

Both the FSE and the adjoint produce the same sensitivity Jacobian. Their costs differ dramatically when m and r are unequal, where m is the number of POIs and r is the number of response functionals.

For the FSE: one solve per POI → m solves total.

For the adjoint: one solve per response functional → r solves total.

$$\boxed{m > r \implies \text{use adjoint}; \qquad r > m \implies \text{use FSE.}} \tag{5.5}$$

In most SA applications, there are many POIs (m large) and a small number of QOIs of primary interest (r small), so the adjoint is preferred. The canonical situation is $r = 1$: one scalar response functional (peak infection count, total burden, profit at equilibrium) and many parameters.

5.4.2 Cost Comparison Plot

Figure 5.2 shows the number of linear solves required by each method as a function of the number of parameters.

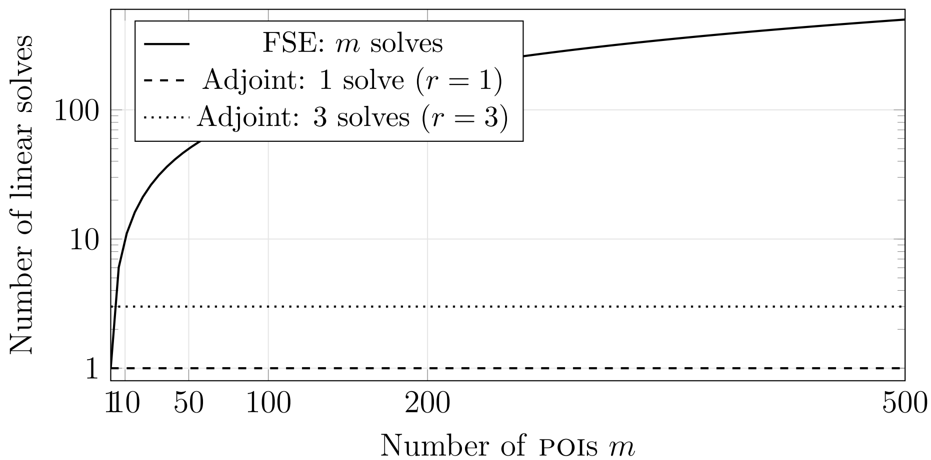


Figure 5.2: Cost comparison for FSE vs. adjoint sensitivity analysis, measured in number of linear solves. For $r = 1$ response functional (dashed), the adjoint always requires exactly 1 solve regardless of m. For $r = 3$ (dotted), the adjoint requires 3 solves. The FSE requires m solves. The crossover occurs at $m = r$; for $m > r$, the adjoint wins.

For the symbiotic population model ($m = 3$ parameters, $r = 2$ response functionals $\{\bar{u}_1, \bar{u}_2\}$): FSE requires 3 solves; adjoint requires 2. The advantage is modest here. For the malaria model ($m = 17$, $r = 1$): FSE requires 17 solves; adjoint requires 1. For a climate model ($m = 200$, $r = 2$): FSE requires 200 solves; adjoint requires 2: a factor of 100 improvement.

Self-check. A model has $m = 30$ POIs and $r = 5$ response functionals. Each linear solve takes 2 minutes. How long does the FSE approach take? How long does the adjoint approach take? At what value of r would the two approaches have equal cost?

Answer: FSE: $30 \times 2 = 60$ minutes. Adjoint: $5 \times 2 = 10$ minutes. Savings: 50 minutes. Equal cost when $r = m = 30$.

5.5 The Symbiotic Population: Full Adjoint Treatment

We now apply the adjoint to the symbiotic population equilibrium from Chapter 4, comparing the forward and adjoint approaches side by side. The forward problem is $A\vec{u} = \vec{b}$ at equilibrium, with $A = D_{\vec{u}}[\vec{f}]$ as in equation (4.7). The comparison makes the cost advantage concrete.

5.5.1 FSE Approach: Three Parameters, Three Solves

For parameters a, b, c, the FSE requires solving $A(\partial\vec{\bar{u}}/\partial q_j) = -\partial\vec{f}/\partial q_j$ for $j = a, b, c$, three linear solves with the same matrix A, one LU factorization and three back-substitutions.

5.5.2 Adjoint Approach: Two Solves for Two Response Functionals

Now suppose we want $\partial J_1/\partial q_j$ and $\partial J_2/\partial q_j$ for all three parameters, where $J_1 = \bar{u}_1$ and $J_2 = \bar{u}_2$. The response vectors are $\vec{c}_1 = (1, 0)^T$ and $\vec{c}_2 = (0, 1)^T$.

The two adjoint equations are:

$$A^T\vec{v}_1 = \vec{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad A^T\vec{v}_2 = \vec{c}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{5.6}$$

Two solves (one per response functional) give $\vec{v}_1$ and $\vec{v}_2$.

A note on correlated QOIs. Even when the POIs are independent (uncorrelated inputs), the QOIs are almost always correlated outputs: $J_1 = \bar{u}_1$ and $J_2 = \bar{u}_2$ both depend on the same equilibrium $\vec{\bar{u}}$, so perturbations to any parameter affect both simultaneously. This is not a problem for the adjoint method; it is simply a property of the model. The adjoint computes each $\partial J_k/\partial q_j$ independently by solving $A^T\vec{v}_k = \vec{c}_k$ for each response functional. Any statistical correlation among the J_k's is a property of the model, not an issue for the adjoint method — no special handling is required. The adjoint naturally produces correct, consistent sensitivities for all r response functionals by virtue of operating through the same forward solution $\vec{\bar{u}}$.

The sensitivities follow from formula (5.4):

$$\frac{\partial J_k}{\partial q_j} = \vec{v}_k^T\vec{r}_j, \qquad \vec{r}_j = -\frac{\partial\vec{f}}{\partial q_j}, \tag{5.7}$$

giving all $2 \times 3 = 6$ sensitivities from two adjoint solves rather than three FSE solves.

5.5.3 Verification

Theorem 5.1 guarantees the two approaches give identical results. At nominal values $a = 0.4$, $b = 1$, $c = 0.3$:

	$S_a^{\bar{u}_1}$	$S_b^{\bar{u}_1}$	$S_c^{\bar{u}_1}$	$S_a^{\bar{u}_2}$	$S_b^{\bar{u}_2}$	$S_c^{\bar{u}_2}$
FSE (3 solves)	-0.530	0	0.136	$+0.136$	0	-0.292
Adjoint (2 solves)	-0.530	0	0.136	$+0.136$	0	-0.292

The rows are identical. The adjoint is not an approximation.

A structural observation: the two adjoint solutions $\vec{v}_1, \vec{v}_2$ are the columns of $(A^T)^{-1}$, which equals $(A^{-1})^T$. Collecting all adjoint solutions into a matrix therefore gives the transpose of the inverse. This connection is intellectually interesting but not computationally useful: we never form A^{-1} explicitly, we extract the product $\vec{v}_k^T\vec{r}_j$ for each required sensitivity, which is far cheaper than computing the full inverse matrix.

5.6 Reverse Mode and Backpropagation

The adjoint strategy of the preceding sections applies to linear systems. This section extends it to arbitrary differentiable computations via the computation graph, revealing the connection to algorithmic differentiation and, through it, to the training of neural networks.

5.6.1 The Computation Graph

Every differentiable computation can be represented as a directed acyclic graph in which each node holds an intermediate value and each edge represents an elementary operation (addition, multiplication, division, elementary function).

For $R_0 = k\beta/\gamma$, the computation graph is:

$$k, \beta \xrightarrow{\times} v_1 = k\beta \xrightarrow{\div\gamma} v_2 = k\beta/\gamma = R_0. \tag{5.8}$$

Reading left to right is the forward evaluation: given k, β, γ, compute R_0.

5.6.2 Forward Mode: One Pass Per Input

In **forward mode** differentiation, we propagate derivatives $\dot{v}_i = \partial v_i/\partial q$ through the graph for one input q at a time.

For $q = k$: set $\dot{k} = 1$, $\dot{\beta} = 0$, $\dot{\gamma} = 0$. Propagate forward:

$$\dot{v}_1 = \dot{k}\beta + k\dot{\beta} = \beta, \qquad \dot{R}_0 = \dot{v}_1/\gamma - v_1\dot{\gamma}/\gamma^2 = \beta/\gamma.$$

So $\partial R_0/\partial k = \beta/\gamma$. Repeat with $\dot{\beta} = 1$ to get $\partial R_0/\partial\beta$. Repeat with $\dot{\gamma} = 1$ to get $\partial R_0/\partial\gamma$. Total: 3 forward passes for 3 parameters.

5.6.3 Reverse Mode: One Pass for All Inputs

In **reverse mode** differentiation, we propagate the adjoint variable $\bar{v}_i = \partial J/\partial v_i$ backward through the graph.

Set $\bar{R}_0 = \partial J/\partial R_0 = 1$ (we are computing $\partial R_0/\partial(\cdot)$, so $J = R_0$). Propagate backward:

$$\bar{v}_1 = \bar{R}_0 \cdot \frac{\partial R_0}{\partial v_1} = 1 \cdot (1/\gamma) = 1/\gamma, \tag{5.9}$$

$$\bar{k} = \bar{v}_1 \cdot \frac{\partial v_1}{\partial k} = (1/\gamma) \cdot \beta = \beta/\gamma, \tag{5.10}$$

$$\bar{\beta} = \bar{v}_1 \cdot \frac{\partial v_1}{\partial \beta} = (1/\gamma) \cdot k = k/\gamma, \tag{5.11}$$

$$\bar{\gamma} = \bar{R}_0 \cdot \frac{\partial R_0}{\partial \gamma} = 1 \cdot (-v_1/\gamma^2) = -k\beta/\gamma^2. \tag{5.12}$$

One backward pass gives all three derivatives simultaneously:

$$\frac{\partial R_0}{\partial k} = \bar{k} = \frac{\beta}{\gamma}, \quad \frac{\partial R_0}{\partial \beta} = \bar{\beta} = \frac{k}{\gamma}, \quad \frac{\partial R_0}{\partial \gamma} = \bar{\gamma} = -\frac{k\beta}{\gamma^2}.$$

These match the analytic results. The reverse pass cost is proportional to the graph size, one pass regardless of the number of inputs.

Table 5.1 summarizes the comparison.

	Forward mode	Reverse mode
Propagates	$\dot{v}_i = \partial v_i/\partial q$ (one q)	$\bar{v}_i = \partial J/\partial v_i$ (one J)
Direction	Input $\to$ output	Output $\to$ input
Cost	1 pass per input	1 pass per output
Preferred when	$m \leq r$ (few inputs)	$m \geq r$ (few outputs)
Matrix analog	FSE: $A\,\partial\vec{u}/\partial q_j = \vec{r}_j$	Adjoint: $A^T\vec{v} = \vec{c}$

Table 5.1: Forward mode and reverse mode differentiation: a side-by-side comparison. The matrix analogs confirm that the two modes are the computation-graph generalizations of the FSE and adjoint approaches from Sections 5.2–5.4.

5.6.4 The Backpropagation Connection

Note for readers without machine learning background: this subsection connects the adjoint to neural network training. If backpropagation is unfamiliar, skip this subsection and proceed to Chapter 6 — the adjoint ODE does not depend on it. Return here after completing Chapter 6 if you want the connection.

If you have encountered training a neural network, you have already used the adjoint under a different name.

A neural network is a composition of parameterized functions. Training minimizes a loss function $J(\theta)$ over parameters θ using gradient descent, which requires $\partial J/\partial\theta_i$ for every parameter θ_i. The loss function is a scalar ($r = 1$) and modern networks have millions of parameters ($m \gg 1$), so reverse mode is overwhelmingly preferred. Backpropagation is exactly the reverse-mode computation described above, applied to the computation graph of the network.

To map notation explicitly: each computation-graph node v_i corresponds to the i-th standard basis vector $\vec{e}_i$, and the adjoint variable $\lambda_i = \partial J/\partial v_i$ records how much J changes per unit change at node i — exactly the role played by $\bar{v}_i$ in reverse mode.

The adjoint equation for the ODE case, derived in Chapter 6, is the continuous limit of this discrete backward accumulation. The backward ODE for $\lambda(t)$ is the functional-analytic version of the backward pass through the computation graph.

Unifying Idea 2. The adjoint is the transpose in whatever space you are working in.

In this chapter: the adjoint equation is $A^T\vec{v} = \vec{c}$, the transposed coefficient matrix applied to the adjoint vector.

In the computation graph: reverse mode is the chain rule applied in reverse order, accumulating $\bar{v}_i = \partial J/\partial v_i$ backward through the graph.

In Chapter 6: the adjoint will be an ODE running backward in time, with a terminal condition at $t = T$. The integration-by-parts identity that produces this ODE is the functional-analytic analog of the matrix transposition that produced A^T here.

One idea. Three settings. One cost advantage: when $m \gg r$, the adjoint/reverse-mode is the preferred computational strategy.

Self-check. Use the reverse-mode computation of equations (5.9)–(5.12) to compute $S_k^{R_0}$, $S_\beta^{R_0}$, and

$S_\gamma^{R_0}$. Express each as a pure number and verify using the Chapter 2 results.

Answer: $\partial R_0/\partial k = \beta/\gamma$. $S_k^{R_0} = (k/R_0)(\partial R_0/\partial k) = (k/(k\beta/\gamma)) \cdot (\beta/\gamma) = 1$.
$\partial R_0/\partial \beta = k/\gamma$. $S_\beta^{R_0} = (\beta/(k\beta/\gamma))(k/\gamma) = 1$.
$\partial R_0/\partial \gamma = -k\beta/\gamma^2$. $S_\gamma^{R_0} = (\gamma/(k\beta/\gamma))(-k\beta/\gamma^2) = -1$.
All three match the Chapter 2 results. The reverse-mode computation recovers the full sensitivity Jacobian in one pass.

5.7 Bridge to Chapter 6

Students sometimes expect the adjoint for differential equations to require advanced functional analysis (Hilbert spaces, adjoint operators, and formal operator theory. Appendix A provides the L^2 inner product and orthogonality needed for that formalism; Appendix B develops the full abstract operator framework that makes the connection rigorous. The preceding sections show why this expectation is misplaced: the core idea is the matrix transpose. Every formal construct in Chapter 6 is the natural extension of the matrix algebra developed here. Reading Chapter 6 with this chapter in mind, the abstract notation should feel like familiar territory.

Unifying Idea 1. Every sensitivity value computed in Section 5.5, via the adjoint, is a normalized derivative, an instance of Unifying Idea 1 from Chapter 1. The adjoint is a computational strategy; the quantity it produces is still the SI defined in Chapter 2. The two commutative paths in Figure 5.1 produce the same sensitivity index by different routes.

Chapter 5 has established the adjoint for linear systems. Chapter 6 extends every result to ordinary differential equations. The sensitivity Jacobian computed in this chapter is the same object that Smith [42] uses in §6.1 to drive parameter selection and optimal experimental design; students who complete Chapter 6 will find §6.1 of Smith immediately accessible.

The translation is direct:

Chapter 5 (linear algebra)	Chapter 6 (dynamical systems)
Coefficient matrix A	Differential operator $d/dt - J(t)$
Transpose A^T	Adjoint operator $-(d/dt + J(t)^T)$
Adjoint equation $A^T\vec{v} = \vec{c}$	Adjoint ODE $-\dot{\lambda} = J^T\lambda + g_u$
Terminal condition: none	Terminal condition: $\lambda(T) = h_u^T$
Sensitivity formula $\vec{v}^T\vec{r}_j$	Sensitivity formula $\int_0^T \lambda^T F_{p_j}\, dt$
Cost: 1 solve for any m	Cost: 2 ODE solves for any m

The integration-by-parts identity in Chapter 6 plays the role of the algebraic transposition here. The terminal condition on $\lambda(T)$ is the time-dependent analog of the right-hand side $\vec{c}$ in $A^T\vec{v} = \vec{c}$. The cost advantage is identical: one solve (or two, for the ODE case where a forward solve is also needed) recovers all m sensitivities regardless of m.

In Chapter 6 we extend every result of this chapter to ordinary differential equations. The matrix A becomes a differential operator. The transpose A^T becomes the adjoint differential operator. The linear solve $A^T\vec{v} = \vec{c}$ becomes an ODE for $\lambda(t)$ with a terminal condition at $t = T$. The sensitivity formula $\vec{v}^T\vec{r}_j$ becomes an integral over $[0, T]$. The cost advantage is the same: 2 ODE solves instead of m.

5.8 MATLAB Laboratory: Forward vs. Adjoint

This laboratory compares the FSE and adjoint computations on the symbiotic population model, then implements reverse-mode differentiation for R_0.

Part 1: Symbiotic population via FSE. At nominal values $a = 0.4$, $b = 1$, $c = 0.3$, compute the equilibrium using equation (4.6). Assemble the Jacobian A and compute the LU factorization using MATLAB's `lu`. Solve the three FSEs ($\partial\vec{\bar{u}}/\partial a$, $\partial\vec{\bar{u}}/\partial b$, $\partial\vec{\bar{u}}/\partial c$) using `U\(L\r)` for each. Record the time.

Part 2: Symbiotic population via adjoint. Solve the two adjoint equations $A^T\vec{v}_1 = \vec{c}_1$ and $A^T\vec{v}_2 = \vec{c}_2$ using the same LU factorization (with transposed triangular solves). Compute all 6 sensitivities via $\vec{v}_k^T\vec{r}_j$. Record the time and verify that all 6 values agree with Part 1 to machine precision.

MATLAB note on the transposed solve. MATLAB's `lu` gives $A = P^T LU$, so $A^T = U^T L^T P$. Solving $A^T\vec{v} = \vec{c}$ therefore proceeds in three steps: $U^T\vec{w}_1 = P\vec{c}$, then $L^T\vec{w}_2 = \vec{w}_1$, then $\vec{v} = \vec{w}_2$. In MATLAB this is written compactly as `L' \ (U' \ (P'*c))`, reading right to left: first multiply by `P'`, then solve the two transposed triangular systems.

```
%% Assemble Jacobian at equilibrium (see Chapter 4)
A = jac_symp(ubar, a, b, c);
[L, U, P] = lu(A);

%% FSE: 3 solves
for j = 1:3
    rj = rhs_symp(ubar, j);          %% FSE right-hand side
    dwdp(:,j) = U \ (L \ (P * rj));
end

%% Adjoint: 2 solves
c1 = [1;0]; c2 = [0;1];
v1 = L' \ (U' \ (P' * c1));   %% solve A^T v1 = c1
v2 = L' \ (U' \ (P' * c2));   %% solve A^T v2 = c2
for j = 1:3
    rj = rhs_symp(ubar, j);
    dJdp(1,j) = v1' * rj;     %% dJ1/dp_j
    dJdp(2,j) = v2' * rj;     %% dJ2/dp_j
end
```

Part 3: Timing as a function of system size. For random systems of size $n = 5, 10, 50, 100$ with $m = 20$ parameters and $r = 1$ response functional, time both approaches. Plot computation time vs. n on log-log axes. Does the ratio of times approach $m/r = 20$ as expected?

Part 4: Reverse-mode R_0 by hand. Implement the reverse-mode computation (5.9)–(5.12) in MATLAB for $R_0 = k\beta/\gamma$. Verify all three sensitivities against the MATLAB Symbolic Toolbox. Count the number of arithmetic operations in forward mode vs. reverse mode for this graph.

Part 5: Verify that adjoint = 50 FSE solves. Set up a random 10×10 system with $m = 50$ parameters and $r = 1$. Compute $\partial J/\partial q_j$ for all j by: (a) 50 FSE solves (forward path); (b) 1 adjoint solve (adjoint path). Verify that the maximum absolute difference is less than 10^{-12}.

5.9 Exercises

Exercise 5.1 *([Bloom L2])*. State the three-step adjoint strategy from Section 5.2 in your own words, using no symbols. Then explain: why does the adjoint equation $A^T\vec{v} = \vec{c}$ not depend on which parameter q_j we are differentiating with respect to? Why does this independence make the adjoint efficient?

Exercise 5.2 *([Bloom L3])*. For the linear system

$$A\vec{u} = \vec{b}, \qquad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} b_1(q) \\ b_2(q) \end{pmatrix} = \begin{pmatrix} q \\ 2q \end{pmatrix},$$

with response functional $J = u_1 + u_2$ (so $\vec{c} = (1, 1)^T$):

(a) Solve the forward problem to find $\vec{u}(q)$ at $q = 1$.

(b) Write and solve the FSE for $\partial\vec{u}/\partial q$.

(c) Write and solve the adjoint equation.

(d) Use formula (5.4) to compute $\partial J/\partial q$.

(e) Verify by differentiating $J(q) = u_1(q) + u_2(q)$ directly.

Exercise 5.3 *([Bloom L3])*. For each scenario below, state whether you would use the FSE or the adjoint, and calculate the number of linear solves each approach requires.

(a) $m = 5$ parameters, $r = 20$ response functionals.

(b) $m = 100$ parameters, $r = 3$ response functionals.

(c) $m = 8$ parameters, $r = 8$ response functionals.

(d) $m = 1{,}000$ parameters, $r = 1$ response functional. Each solve takes 0.1 seconds. How much faster is the preferred method?

Exercise 5.4 *([Bloom L4])*. For the symbiotic population equilibrium with parameters $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute all 6 sensitivities ($S_a^{\bar{u}_1}$, $S_b^{\bar{u}_1}$, $S_c^{\bar{u}_1}$, $S_a^{\bar{u}_2}$, $S_b^{\bar{u}_2}$, $S_c^{\bar{u}_2}$) using the FSE approach (3 solves).

(b) Compute the same 6 sensitivities using the adjoint approach (2 solves).

(c) Verify that the two methods agree.

(d) Which approach requires fewer solves? By how much? At what values of m and r would the FSE be preferred?

Exercise 5.5 *([Bloom L4])*. The SEIR basic reproduction number is $R_0 = k\beta\nu/[(\nu + \mu)(\gamma + \mu)]$ with parameters k, β, ν (exposure rate), γ (recovery rate), μ (mortality rate).

(a) Draw the computation graph for R_0 (intermediate nodes and edges).

(b) Implement the forward-mode computation to find $\partial R_0/\partial\nu$. Count the number of arithmetic operations.

(c) Implement the reverse-mode computation to find all five derivatives $\partial R_0/\partial k$, $\partial R_0/\partial\beta$, $\partial R_0/\partial\nu$, $\partial R_0/\partial\gamma$, $\partial R_0/\partial\mu$ in one backward pass.

(d) Compute the five normalized sensitivity indices and produce a tornado plot using interpretable POIs (mean times, not rates).

Exercise 5.6 *(* [Bloom L5] *)*. A water-quality model of a river system has $m = 40$ parameters (pollution source rates, degradation rates, flow speeds, mixing coefficients) and $r = 2$ response functionals: (1) pollutant concentration at a downstream monitoring station, and (2) total pollutant load entering a protected estuary.

Design a complete SA study using the adjoint method. Specify:

(a) The form of the adjoint equation for this model.

(b) How many adjoint solves are required and why.

(c) How you would validate the implementation.

(d) How you would use the sensitivity results to advise regulators on which pollution sources to prioritize.

(e) One situation where the adjoint results might be misleading, and what you would do to detect and correct it.

Exercise 5.7 *(* [Bloom L6] *)*. A colleague states: "The adjoint always gives the same answer as forward SA, just faster. So if we already have the FSE code working, there is no reason to implement the adjoint."

Evaluate this claim carefully.

(a) Is the first sentence (same answer, just faster) correct? Give a precise mathematical justification.

(b) Is the second sentence (no reason to implement the adjoint) correct? Describe at least two situations where switching from FSE to adjoint would be strongly advisable.

(c) For the FSE to be preferred over the adjoint, what relationship between m and r must hold? Give a specific numerical example where FSE is strictly better.

Chapter 6

The Adjoint for Dynamical Systems

To understand the effect of an outcome, trace its influence backward through time.

The forward model tells you where you end up.
The adjoint tells you how you got there.

Climate models routinely have 500 or more tunable parameters. Forward sensitivity analysis using the FSE approach requires one ODE solve per parameter: 500 solves on a modern workstation, each taking 6 hours. Total: 3,000 hours, or 125 days.

The adjoint approach requires exactly two ODE solves: one forward, one backward. Total: 12 hours.

The mathematics behind this 125-to-1 speedup is a direct extension of the matrix adjoint from Chapter 5. In that chapter, the key operation was transposing the coefficient matrix: $A \to A^T$. Here, the key operation is transposing the differential operator, which turns a forward ODE into a backward ODE. This chapter derives that operation step by step, explains in plain language why the backward direction arises, and implements the full calculation for the SIR epidemic model. The notation is one step heavier than Chapter 5 — integration by parts and continuous-time functionals replace matrix transposes — but the underlying idea is the same transposition argument in a new space. The adjoint sensitivity equations for ODE systems, including their software implementation, are treated in depth by Cao et al. [8]; a recent comparison of forward and adjoint methods on biological models is given by Mester et al. [26].

6.1 The Timeline Figure

The most important idea in this chapter: the reason the adjoint ODE runs backward, is captured in Figure 6.1 before any equations appear.

The forward ODE begins at $t = 0$ and moves right, building up the state $\vec{u}(t)$ until it reaches the response J at time T. The adjoint ODE begins at $t = T$ and moves left, distributing the response backward through time.

This opposite direction is not a numerical trick. It is a mathematical necessity, explained in Section 6.2.

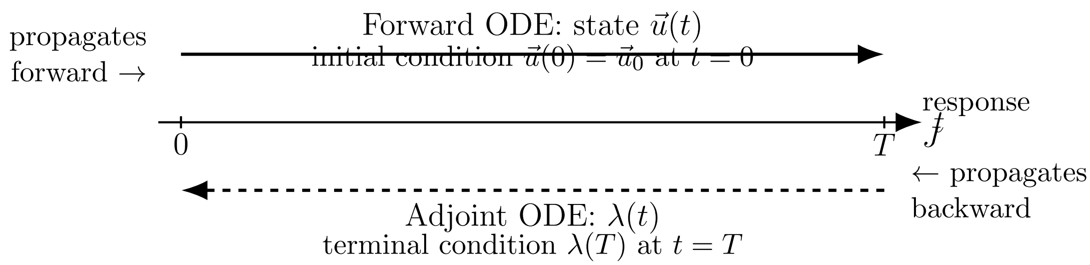


Figure 6.1: The timeline for forward and adjoint sensitivity analysis. The forward ODE propagates the model state $\vec{u}(t)$ from the initial condition at $t = 0$ toward the response J at $t = T$. The adjoint ODE propagates the sensitivity weight $\lambda(t)$ backward from a terminal condition at $t = T$, in the direction opposite to the forward problem. The two ODEs are not independent: the adjoint coefficient depends on the forward solution $\vec{u}(t)$ at every time step.

6.2 Why Backward? Intuition Before Derivation

The adjoint ODE runs backward because the question it answers is naturally backward: how much did each moment in time contribute to the final value of J?

Think of the epidemic. A perturbation applied to the infectious compartment at $t = 0$ has the entire simulation period to propagate: it changes the dynamics at $t = 1$, $t = 2$, and all the way to T. Its influence on J is large. A perturbation applied at $t = T - \epsilon$ has almost no time to propagate before the simulation ends. Its influence on J is negligible.

So the "weight" we should assign to perturbations decreases as we move from left to right along the time axis, and increases as we move from right to left. The adjoint variable $\lambda(t)$ is precisely this weight: $\lambda(t)$ measures how sensitive the response J is to a perturbation of the state at time t.

At $t = T$, the weight is determined by the response functional (how does J change if the final state changes?). At earlier times, $\lambda(t)$ is computed by propagating this weight backward, accounting for how perturbations at time t have time to amplify before reaching T.

The adjoint variable is large near $t = 0$ and small near $t = T$. The adjoint ODE starts at the only time where we know λ directly at $t = T$, and integrates backward to find $\lambda(t)$ at all earlier times.

This must be understood before the derivation. The backward direction of the adjoint is not a consequence of the algebra; the algebra is a consequence of this backward structure.

The adjoint and forward problems are not independent: the coefficient in the adjoint ODE at every time t involves $\vec{u}(t)$, the forward solution. The adjoint can only be solved after the forward problem is solved and $\vec{u}(t)$ is stored at every time point needed.

Self-check. Suppose the response functional is $J = S(T)$ (the susceptible population at the final time) and the epidemic has a peak at $t = 30$ in a 90-day simulation. At which time ($t = 5$, $t = 30$, or $t = 80$ would you expect $\lambda_I(t)$ to be largest in magnitude? Explain without any calculation.

> *Answer:* $\lambda_I(t)$ at time t measures how sensitive $J = S(T)$ is to a perturbation of I at time t. A perturbation to I at $t = 5$ has 85 days to propagate and affect $S(90)$: a large window of time for the perturbation to alter the epidemic trajectory. A perturbation at $t = 80$ has only 10 days. So $\lambda_I(t)$ should be largest at $t = 5$.

6.3 The Lagrange Multiplier Derivation

6.3.1 A Warm-Up: Exponential Decay

Before the five-step derivation, work through the simplest possible case to see exactly what the adjoint ODE looks like and how it gives the answer.

Let $\dot{u} = -pu$ with $u(0) = 1$ and $J = u(T)$. The forward solution is $u(t) = e^{-pt}$, so $J = e^{-pT}$ and $dJ/dp = -Te^{-pT}$ (by direct differentiation — the "correct" answer).

The adjoint approach finds the same result differently. For $F(u;p) = -pu$ and $g = 0$, $h(u) = u$:

- Adjoint ODE: $-\dot{\lambda} = F_u\,\lambda = -p\lambda$, so $\dot{\lambda} = p\lambda$.
- Terminal condition: $\lambda(T) = -h'(u(T)) = -1$.
- Solution (running backward from T): $\lambda(t) = -e^{p(t-T)}$.
- Sensitivity formula: $dJ/dp = -\int_0^T \lambda(t)\,F_p\,dt = -\int_0^T (-e^{p(t-T)})(-u(t))\,dt = -\int_0^T e^{p(t-T)}e^{-pt}\,dt = -\int_0^T e^{-pT}\,dt = -Te^{-pT}$. ✓

The adjoint gave the same derivative using only two ODE solves (forward, then backward), with no dependence on m (how many parameters there are). If p were joined by 49 other parameters, the cost would still be two ODE solves plus 49 integrals.

Now we derive this procedure in full generality.

6.3.2 The Five-Step Derivation

The derivation proceeds through five numbered steps, each announced before it is executed. The scalar ODE case is treated fully; the vector case follows by replacing scalars with vectors and matrices.

6.3.3 Setup

Consider the initial value problem

$$\dot{u} = F(u;p), \qquad u(0) = u_0, \tag{6.1}$$

with a response functional

$$J = \int_0^T g(u;p)\,dt + h(u(T)). \tag{6.2}$$

Here g is the running cost (an integrand depending on the solution and parameters) and h is a terminal cost (depending only on the final state). We want dJ/dp without solving the FSE.

The strategy is identical to Chapter 5: introduce an auxiliary variable, choose it to cancel the unknown sensitivity $\partial u/\partial p$, and read off the result from what remains. In Chapter 5 the auxiliary variable was a vector $\vec{v}$ satisfying $A^T\vec{v} = \vec{c}$; here it is a function $\lambda(t)$ satisfying a backward ODE.

Step 1: Introduce the Lagrange multiplier.

We introduce an auxiliary function $\lambda(t)$, called the adjoint variable or Lagrange multiplier, and observe that, because $u(t)$ satisfies the ODE (6.1):

$$\int_0^T \lambda(t)\big[\dot{u} - F(u;p)\big]\,dt = 0.$$

This is the ODE constraint written as a vanishing integral. Define the augmented functional

$$\tilde{J} = J + \int_0^T \lambda(t) \big[\dot{u} - F(u; p)\big] \, dt. \tag{6.3}$$

Since the bracketed term is zero, $\tilde{J} = J$. The strategy, carried over from Chapter 5, is to choose $\lambda(t)$ so that $\partial u/\partial p$ disappears when we differentiate $\tilde{J}$ with respect to p.

Physical analogy. Think of a building inspector who records the strain in each beam throughout the day. The inspector's record is $\lambda(t)$: it does not change the building, but it carries a running account of how much each moment in time contributes to the structure's overall stress J. Adding the constraint integral is like attaching a running tally that is identically zero as long as the beams obey the equilibrium equations — it costs nothing, but it opens the door to measuring sensitivity.

Step 2: Differentiate with respect to p.
Denote $w = \partial u/\partial p$. Differentiating (6.3):

$$\frac{d\tilde{J}}{dp} = \int_0^T g_u \, w \, dt + h'(u(T)) \, w(T) + \int_0^T \lambda \big[\dot{w} - F_u \, w - F_p\big] \, dt. \tag{6.4}$$

The unknown $w = \partial u/\partial p$ appears in three places. The next step eliminates it by choosing λ appropriately.

Physical analogy. Differentiating with respect to p is like asking: if we nudge a design parameter — say, the stiffness of one beam — how does the overall stress J respond? The sensitivity $w = \partial u/\partial p$ represents the ripple that nudge sends through the entire structure over time. It appears everywhere because the system is coupled; $\lambda(t)$ will be chosen precisely to absorb those ripples.

Step 3: Integrate by parts.
The term $\int_0^T \lambda \dot{w} \, dt$ contains $\dot{w}$. We remove the time derivative from w by integrating by parts: this is the key step, and it is the ODE analog of the matrix transposition $v^T A = (A^T v)^T$ in Chapter 5.

$$\int_0^T \lambda \, \dot{w} \, dt = \big[\lambda \, w\big]_0^T - \int_0^T \dot{\lambda} \, w \, dt = \lambda(T) \, w(T) - \lambda(0) \, w(0) - \int_0^T \dot{\lambda} \, w \, dt. \tag{6.5}$$

Substituting into (6.4):

$$\begin{aligned} \frac{d\tilde{J}}{dp} = &\int_0^T g_u \, w \, dt + h'(u(T)) \, w(T) \\ &+ \lambda(T) \, w(T) - \lambda(0) \, w(0) - \int_0^T \dot{\lambda} \, w \, dt - \int_0^T \lambda \, F_u \, w \, dt - \int_0^T \lambda \, F_p \, dt. \end{aligned} \tag{6.6}$$

Collecting the w-terms (the terms involving the unknown sensitivity):

$$\frac{d\tilde{J}}{dp} = \underbrace{\big[h'(u(T)) + \lambda(T)\big] w(T) - \lambda(0) \, w(0) + \int_0^T \big[g_u - \dot{\lambda} - \lambda F_u\big] w \, dt}_{\text{terms involving } w = \partial u/\partial p} - \int_0^T \lambda \, F_p \, dt. \tag{6.7}$$

Integration by parts is not merely an algebraic trick. It is exactly the operation that moves the time derivative from w to λ, transforming the FSE into the adjoint ODE, just as matrix transposition moved the linear operator from u to v in Chapter 5.

Physical analogy. Think of two runners passing a baton around a track. Forward mode carries $\dot{w}$ — the baton is held by the sensitivity variable. Integration by parts hands the baton to λ: now $\dot{\lambda}$ carries the time derivative instead, and w appears without any derivative. This exchange is what makes it possible to choose λ to cancel w in the next step; without it, we could not get rid of $\dot{w}$ by specifying an ODE for λ alone.

Step 4: Choose λ to eliminate w.

For the sensitivity formula (6.7) to be free of w, the coefficient of w must vanish everywhere. This requires three conditions to hold simultaneously.

First, the integrand coefficient of $w(t)$ must vanish for all $t \in (0, T)$:

$$-\dot{\lambda} - \lambda F_u + g_u = 0, \quad \text{i.e.,} \quad \dot{\lambda} = -F_u\,\lambda + g_u. \tag{6.8}$$

Written in standard form (with the sign convention consistent with backward integration):

$$\boxed{-\dot{\lambda} = F_u\,\lambda - g_u.} \tag{6.9}$$

Second, the coefficient of $w(T)$ must vanish:

$$h'(u(T)) + \lambda(T) = 0, \quad \text{i.e.,} \quad \lambda(T) = -h'(u(T)). \tag{6.10}$$

Third, the term $\lambda(0)w(0)$ must vanish. Since $u(0) = u_0$ does not depend on p, we have $w(0) = \partial u_0/\partial p = 0$, so this term vanishes automatically.

The terminal condition (6.10) does not arise because we are running the ODE backward. It arises because the response functional involves the final state $h(u(T))$, and differentiating this with respect to p produces a boundary term at $t = T$. This is a fundamental distinction: the terminal condition is imposed by the response functional, not by the direction of integration.

Physical analogy. Choosing λ to kill every w-term is like tuning the damping in a bridge so that any vibration induced by a parameter perturbation is immediately cancelled. The terminal condition $\lambda(T) = -h'(u(T))$ is the inspector's final reading: at the end of the day the ledger must close, recording exactly how sensitive the last measured outcome is to the final state. Once $\lambda(t)$ is tuned, w has nowhere to appear in the sensitivity formula — it has been designed out.

Step 5: Read off the sensitivity formula.

With $\lambda(t)$ satisfying (6.9)–(6.10), all terms involving w in (6.7) vanish and:

$$\boxed{\frac{dJ}{dp} = -\int_0^T \lambda(t)\,F_p(u(t);p)\,dt - \lambda(0)\,\frac{du_0}{dp}.} \tag{6.11}$$

When the initial condition does not depend on p, the second term vanishes. The sensitivity with respect to any parameter p_j is then:

$$\frac{dJ}{dp_j} = -\int_0^T \lambda(t)\,\frac{\partial F}{\partial p_j}\,dt. \tag{6.12}$$

One backward ODE solve for $\lambda(t)$ gives all m sensitivities via m scalar integrals, exactly the cost advantage we sought.

Physical analogy. The inspector's record $\lambda(t)$ is now complete: one backward sweep through time, consulting the forward solution $u(t)$ at each moment, accumulates everything needed. Reading off dJ/dp_j is then as simple as computing a running total — one scalar integral per parameter, with no additional solves. The backward sweep is the price paid once; the m integrals are the dividend collected once for every parameter in the model.

6.3.4 The Vector Case

For the vector system $\dot{\vec{u}} = \vec{F}(\vec{u};\vec{p})$ with $\vec{u} \in \mathbb{R}^n$ and functional $J = \int_0^T g(\vec{u};\vec{p})\,dt + h(\vec{u}(T))$, the derivation is identical with scalars replaced by vectors and F_u replaced by the Jacobian $J_F = \partial\vec{F}/\partial\vec{u}$. The adjoint ODE and terminal condition become:

$$-\dot{\vec{\lambda}} = J_F^T\,\vec{\lambda} - \nabla_u g^T, \tag{6.13}$$

$$\vec{\lambda}(T) = -\nabla_u h^T\big|_{t=T}. \tag{6.14}$$

Linearity of the adjoint ODE. Just as Theorem 4.1 showed that the FSE is always linear in the forward sensitivity variable $\partial\vec{u}/\partial p_j$, the adjoint ODE is always linear in $\vec{\lambda}(t)$ — even when the forward ODE is nonlinear in $\vec{u}(t)$. The coefficient $J_F^T(\vec{u}(t))$ is a matrix that depends on the forward solution, but not on $\vec{\lambda}$ itself. This linearity is the ODE counterpart of the algebraic linearity from Chapter 5, and it is why the adjoint is tractable: regardless of how complex the forward dynamics are, the adjoint ODE is a linear ODE that can be solved efficiently.

Sign convention. We write the adjoint in backward-time form as $-\dot{\vec{\lambda}} = \ldots$ to match the MATLAB implementation: `ode45` with `tspan = [T, 0]` integrates backward automatically. In the sensitivity formula (6.15), the minus sign is already absorbed: no additional sign change is needed when reading off dJ/dp_j from the integral.

The sensitivity formula becomes:

$$\frac{dJ}{dp_j} = -\int_0^T \vec{\lambda}^T\,\frac{\partial\vec{F}}{\partial p_j}\,dt - \vec{\lambda}(0)^T\,\frac{\partial\vec{u}_0}{\partial p_j}. \tag{6.15}$$

Unifying Idea 2. The derivation is now complete, and this is Unifying Idea 2 in its full generality. The adjoint in Chapter 5 was $A^T\vec{v} = \vec{c}$: the transposed coefficient matrix A^T applied to $\vec{v}$. The adjoint here is $-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} - \nabla_u g^T$: the transposed Jacobian J_F^T applied to $\vec{\lambda}$, with the sign reversal arising from the integration by parts (6.5). The integration by parts is the exact functional-analytic analog of the matrix transposition $A \to A^T$ from Chapter 5. Appendix A.3 makes this analogy precise through the Lagrange identity, and Appendix B.2 derives the sensitivity formula (6.15) rigorously from it. The terminal condition (6.14) is the time-dependent analog of the right-hand side $\vec{c}$ in $A^T\vec{v} = \vec{c}$. This is one idea.

Self-check. For the scalar ODE $\dot{u} = -pu$ with $J = u(T)$ (no running cost, $g = 0$, $h(u) = u$): (a) Write the adjoint ODE and terminal condition using equations (6.9) and (6.10). (b) Solve the adjoint ODE analytically. (c) Use formula (6.12) to compute dJ/dp. (d) Verify by solving the forward ODE analytically and differentiating.

Answer: (a) $F = -pu$, so $F_u = -p$. Adjoint ODE: $-\dot{\lambda} = -p\lambda$, i.e., $\dot{\lambda} = p\lambda$. Terminal condition: $\lambda(T) = -h'(u(T)) = -1$.
(b) $\dot{\lambda} = p\lambda$ with $\lambda(T) = -1$. Integrating forward from T (backward in time): $\lambda(t) = -e^{p(t-T)}$.
(c) $F_p = -u$. Sensitivity formula: $dJ/dp = -\int_0^T \lambda(t)(-u(t))\,dt = \int_0^T \lambda(t)u(t)\,dt$. The forward solution is $u(t) = u_0e^{-pt}$. $dJ/dp = \int_0^T(-e^{p(t-T)})(u_0e^{-pt})\,dt = -u_0e^{-pT}\int_0^T dt = -u_0Te^{-pT}$.
(d) $J = u(T) = u_0e^{-pT}$, so $dJ/dp = -u_0Te^{-pT}$. ✓

6.4 The SIR Model — Full Adjoint Treatment

We apply the derivation to the SIR model, writing all adjoint equations explicitly and implementing the full calculation in MATLAB.

6.4.1 Setup and Response Functional

The SIR model (4.4)–(4.6) with interpretable POIs $\vec{p} = (k, \beta, \tau, L)$:

$$\begin{aligned} F_S &= \mu N - \frac{k\beta SI}{N} - \mu S, \\ F_I &= \frac{k\beta SI}{N} - (\gamma + \mu) I, \\ F_R &= \gamma I - \mu R, \end{aligned}$$

where $\gamma = 1/\tau$ and $\mu = 1/L$. The response functional is

$$J = \int_0^T I(t)\, dt, \tag{6.16}$$

the total number of infectious person-days: a measure of the total disease burden. Here $g(\vec{u}) = I$, $h = 0$.

6.4.2 The Jacobian

The Jacobian $J_F = \partial(F_S, F_I, F_R)/\partial(S, I, R)$ evaluated at the forward solution $(S(t), I(t), R(t))$:

$$J_F = \begin{pmatrix} -\frac{k\beta I}{N} - \mu & -\frac{k\beta S}{N} & 0 \\ \frac{k\beta I}{N} & \frac{k\beta S}{N} - \gamma - \mu & 0 \\ 0 & \gamma & -\mu \end{pmatrix}. \tag{6.17}$$

6.4.3 Adjoint ODEs

With $g_S = 0$, $g_I = 1$, $g_R = 0$ (since $g = I$), the adjoint system $-\dot{\vec{\lambda}} = J_F^T \vec{\lambda} - \nabla g^T$ written out fully is:

$$-\dot{\lambda}_S = \left(-\frac{k\beta I}{N} - \mu\right) \lambda_S + \frac{k\beta I}{N} \lambda_I, \tag{6.18}$$

$$-\dot{\lambda}_I = -\frac{k\beta S}{N} \lambda_S + \left(\frac{k\beta S}{N} - \gamma - \mu\right) \lambda_I + \gamma \lambda_R - 1, \tag{6.19}$$

$$-\dot{\lambda}_R = -\mu \lambda_R. \tag{6.20}$$

Terminal conditions (since $h = 0$):

$$\lambda_S(T) = \lambda_I(T) = \lambda_R(T) = 0. \tag{6.21}$$

These three ODEs are integrated backward from $t = T$ to $t = 0$, using the stored forward solution $(S(t), I(t), R(t))$ to evaluate the time-varying coefficients.

6.4.4 Sensitivity Formulas

For each POI, the sensitivity is the integral (6.15) evaluated using the stored $\vec{\lambda}(t)$:

$$\frac{dJ}{d\beta} = -\int_0^T \frac{kSI}{N}(\lambda_I - \lambda_S)\,dt, \tag{6.22}$$

$$\frac{dJ}{dk} = -\int_0^T \frac{\beta SI}{N}(\lambda_I - \lambda_S)\,dt, \tag{6.23}$$

$$\frac{dJ}{d\tau} = -\int_0^T \frac{\partial F_I}{\partial \tau}\,\lambda_I\,dt = -\int_0^T \frac{1}{\tau^2}\,I\,\lambda_I\,dt, \tag{6.24}$$

$$\frac{dJ}{dL} = -\int_0^T \left[\frac{\partial F_S}{\partial L}\lambda_S + \frac{\partial F_I}{\partial L}\lambda_I + \frac{\partial F_R}{\partial L}\lambda_R\right] dt. \tag{6.25}$$

Normalized sensitivity indices follow from Definition 2.1: $S^J_{p_j} = (p_j/J)(dJ/dp_j)$.

Multiple QOIs and correlated outputs. When the study involves $r > 1$ response functionals — for example, both $J_1 = \int_0^T I\,dt$ (total burden) and $J_2 = \max_t I(t)$ (peak infections) — the adjoint is solved once per QOI, giving r backward solutions $\vec{\lambda}^{(1)}(t), \ldots, \vec{\lambda}^{(r)}(t)$. Even when the POIs are independent (uncorrelated inputs), the QOIs are almost always correlated outputs: both J_1 and J_2 depend on the same forward trajectory $\vec{u}(t)$, so any parameter change affects both simultaneously. This correlation among outputs is handled automatically by the adjoint. Each backward solve $\vec{\lambda}^{(k)}$ captures the full sensitivity of J_k to perturbations of the shared state, without any special treatment of the output correlation. The user does not model or correct for QOI correlation — the adjoint naturally produces consistent, correct sensitivities for all r response functionals by virtue of working through the same forward trajectory.

6.4.5 The Adjoint Variable and Its Interpretation

Figure 6.2 shows $I(t)$ and $\lambda_I(t)$ computed for nominal parameter values over a 90-day simulation.

The adjoint variable $\lambda_I(t)$ has a concrete epidemiological interpretation: it measures the marginal benefit of reducing the infectious population at time t. When λ_I is large (early epidemic), reducing I by one person-day saves many future infections. When λ_I is small (late epidemic), most transmission has already occurred, so the marginal benefit is low. This interpretation is available from the adjoint computation alone, with no additional work.

The terminal condition $\vec{\lambda}(T) = \vec{0}$ is not the same as starting the forward ODE from initial condition $\vec{0}$ and running it in reverse. The two problems have different structures: the forward ODE is driven by $F(\vec{u};\vec{p})$, while the adjoint ODE is driven by $J_F^T\vec{\lambda}$ with a forcing term from g. The backward direction comes from the integration by parts, not from reversing time. The distinction matters for implementation: you run `ode45` with `tspan = [T, 0]`, not with a negated vector field.

Unifying Idea 3. The adjoint variable $\lambda_I(t)$ illustrates Unifying Idea 3: local SA is a linearization. The adjoint variable displays the local sensitivity of J throughout the simulation period. This continuous record of sensitivity is the local SA analog of what global SA asks globally: how does J respond to perturbations at each moment in time? Chapter 9 will ask the same question for perturbations of the parameters across their full uncertainty ranges rather than perturbations of the state at specific times. The adjoint provides the linearization; global SA provides the full nonlinear picture.

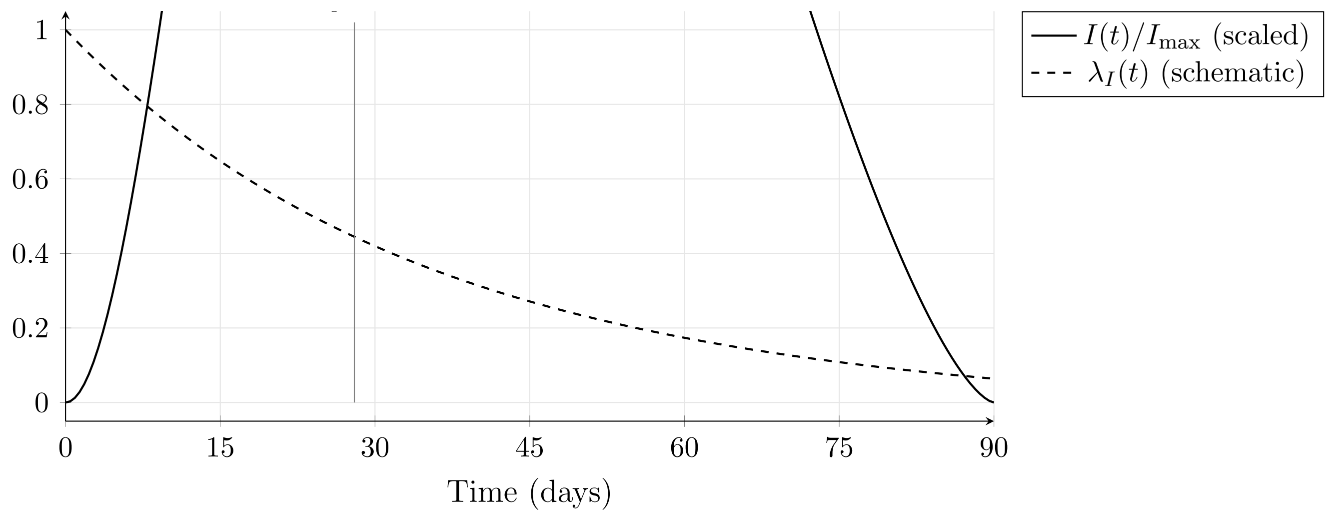


Figure 6.2: Schematic of $I(t)$ (solid, scaled to $[0, 1]$) and the adjoint variable $\lambda_I(t)$ (dashed) for the SIR model. The adjoint variable is large near $t = 0$, when a perturbation to the infectious compartment has the most time to affect the total burden $J = \int I\,dt$, and decays toward zero as $t \to T$. This gives $\lambda_I(t)$ a concrete interpretation as the marginal value of reducing the infectious population at time t: interventions are most cost-effective when λ_I is large.

Unifying Idea 1. The sensitivity indices $S^J_{p_i}$ computed in this chapter via the adjoint are exactly the normalized derivatives of Unifying Idea 1 from Chapter 1: the ratio of relative changes in J to relative changes in each POI p_i. The adjoint is simply a dramatically more efficient algorithm for computing those same derivatives when the model has many POIs and few QOIs.

Self-check. The adjoint variable $\lambda_R(t)$ for the recovered compartment satisfies $-\dot{\lambda}_R = -\mu\lambda_R$ with $\lambda_R(T) = 0$. (a) Solve this ODE analytically (backward from T). (b) What does the solution tell you about the sensitivity of J to perturbations of the recovered compartment?

> *Answer:* (a) $-\dot{\lambda}_R = -\mu\lambda_R$ gives $\dot{\lambda}_R = \mu\lambda_R$ with $\lambda_R(T) = 0$. Solution: $\lambda_R(t) = 0$ for all $t \in [0, T]$. (The only solution to a linear ODE with zero terminal condition and a coefficient that keeps it at zero is identically zero when the equation has no forcing term.)
> (b) $\lambda_R(t) = 0$ means that perturbations to $R(t)$ have no effect on $J = \int I\,dt$. This makes biological sense: recovered individuals do not contribute to future transmission (in the basic SIR model without waning immunity), so changing $R(t)$ does not change the integral of I.

6.5 Discrete vs. Continuous Adjoint

The adjoint ODE (6.13) is the *continuous adjoint*: it is derived by differentiating the continuous ODE and then discretizing for numerical integration. An alternative is the *discrete adjoint*: discretize the ODE first and then adjoint the discrete system. For linear ODEs with fixed step sizes, the two approaches give identical results. For nonlinear ODEs with adaptive step-size control, they can differ.

In practice: if your ODE solver uses fixed step sizes, either adjoint is acceptable. If it uses adaptive step sizes (as `ode45` does), the continuous adjoint requires storing the forward solution densely enough to interpolate accurately, while the discrete adjoint requires differentiating through the step-size controller. For the SA computations in this book, the continuous adjoint with dense output storage is the more practical approach. Sirkes and Tziperman [41] discuss these subtleties for geophysical models. Smith [42] develops the adjoint ODE formalism for general parameter-dependent dynamical systems in §9.4, with additional implementation guidance for systems with stochastic forcing.

Neural ODEs. In 2018, Chen et al. [9] demonstrated that the adjoint calculation derived in this chapter can be used to train certain deep neural networks. The network defines $\vec{F}(\vec{u};\theta)$ for parameters θ; the loss function plays the role of J; and the training algorithm integrates the adjoint ODE backward to compute $\partial J/\partial\theta_i$ for all network parameters. The phrase "backpropagation through time," familiar from recurrent neural networks, refers to the discrete version of this same backward integration. The continuous limit is precisely the adjoint ODE derived here.

6.6 MATLAB Laboratory: The SIR Adjoint in Practice

This laboratory implements the full forward-adjoint computation for the SIR model and verifies the result against the FSE approach from Chapter 4.

Setup. Use the SIR model with POIs $\vec{p} = (k, \beta, \tau, L)$, nominal values $k = 5$, $\beta = 0.06$, $\tau = 7$, $L = 10{,}000$, initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$, $N = 1000$, and $T = 90$ days. The response functional is $J = \int_0^{90} I(t)\,dt$.

Part 1: Solve the forward SIR with dense output. Solve the SIR system using `ode45` with option `'Refine', 8` or dense interpolation, storing the solution on a fine uniform grid with spacing $\Delta t = 0.1$ days. Store (S, I, R) at every grid point; this array is needed for the adjoint coefficient evaluation.

Interpolation (required before the backward solve). Before integrating the adjoint backward, build interpolants of the forward solution so the adjoint ODE can evaluate $S(t)$, $I(t)$, and $R(t)$ at any time during the backward pass:

```
S_fn = griddedInterpolant(t_fwd, U(:,1), 'pchip');
I_fn = griddedInterpolant(t_fwd, U(:,2), 'pchip');
R_fn = griddedInterpolant(t_fwd, U(:,3), 'pchip');
```

Pass these interpolants into the adjoint function. Skipping this step is the most common cause of errors in the adjoint lab.

Part 2: Solve the adjoint system backward. Implement the adjoint ODEs (6.18)–(6.20). Note that these depend on the forward solution $(S(t), I(t), R(t))$, which must be interpolated from the stored grid.

```
function dlam = sir_adjoint(t, lam, S_fn, I_fn, R_fn, k, beta, tau, L)
%% Adjoint ODEs for SIR model
%% Integrate BACKWARD: call with tspan = [T, 0]

N     = 1000;
gamma = 1/tau;  mu = 1/L;
S = S_fn(t); I = I_fn(t); R = R_fn(t);   %% interpolate

lS = lam(1); lI = lam(2); lR = lam(3);

%% -d(lambda)/dt = J_F^T * lambda - grad_u(g)
%% Note: g = I, so grad_u(g) = (0, 1, 0)
```

```
dlam(1) = -(  (-k*beta*I/N - mu)*lS + (k*beta*I/N)*lI        );
dlam(2) = -(  (-k*beta*S/N)*lS + (k*beta*S/N - gamma - mu)*lI ...
              + gamma*lR - 1  );
dlam(3) = -(  -mu*lR                                          );

dlam = dlam(:);
end
```

Solve from $t = T$ to $t = 0$ with terminal conditions $\vec{\lambda}(T) = (0, 0, 0)$.

Part 3: Plot $I(t)$ and $\lambda_I(t)$. On a single axis, plot the epidemic curve $I(t)$ (left scale) and the adjoint variable $\lambda_I(t)$ (right scale). Annotate the epidemic peak. Confirm that λ_I is largest near $t = 0$ and smallest near $t = T$.

Part 4: Compute sensitivities and produce a tornado plot. Evaluate the integrals (6.22)–(6.25) numerically using the trapezoidal rule. Normalize to obtain $S^J_{p_j}$ for all four POIs. Produce a tornado plot and identify which parameter has the greatest influence on the total disease burden.

Part 5: Verify against Chapter 4 FSE results. From Chapter 4's MATLAB lab, retrieve the SI values computed by the FSE at a single time point. Compare with the adjoint-based SIs for $J = \int I\,dt$. Note: the two computations answer *different* questions (the FSE computes $S^{I(t)}_{p_j}$ at a fixed t, while the adjoint computes $S^{\int I\,dt}_{p_j}$), so the numerical values will differ. Instead, verify that $S^J_\beta = S^J_k$ to at least 4 significant figures (both should equal 1 for the SIR model with this parameterization) and discuss whether the ratio holds exactly.

Part 6: Timing comparison. Time the adjoint computation (Parts 1–4) against the FSE approach from Chapter 4 (solve the 15-equation augmented system four times). At what number of parameters m would you expect the adjoint to be faster? Does the timing crossover occur near $m = r = 1$ as the theory predicts?

6.7 Exercises

Exercise 6.1 *(*[Bloom L3]*).* For the scalar ODE $\dot{u} = ru(1 - u/K)$ (logistic) with $J = \int_0^T u\,dt$ (total population over time):

(a) Write the adjoint ODE and terminal condition.

(b) Show that the adjoint ODE is linear in $\lambda(t)$ even though the forward ODE is nonlinear in $u(t)$.

(c) Explain why the adjoint ODE depends on the forward solution $u(t)$.

(d) In MATLAB, solve the forward and adjoint ODEs simultaneously (forward from 0 to T; store; then backward from T to 0) for $r = 0.5$, $K = 100$, $u_0 = 10$, $T = 20$. Plot $u(t)$ and $\lambda(t)$.

Exercise 6.2 *(*[Bloom L3]*).* State in your own words what integration by parts accomplishes in Step 3 of the Lagrange multiplier derivation, and why this is analogous to the matrix transposition $A \to A^T$ in Chapter 5. Then verify equation (6.5) directly by differentiating the product λw and integrating over $[0, T]$.

Exercise 6.3 *(*[Bloom L4]*).* Using your MATLAB implementation from the laboratory:

(a) Compute S^J_β, S^J_k, S^J_τ, S^J_L via the adjoint for $J = \int_0^{90} I\,dt$.

(b) Verify that $S^J_k = S^J_\beta$ to at least 6 significant figures. Explain this result using the structure of the SIR model.

(c) If you could implement one intervention that reduces a single parameter by 20%, which parameter would you target to most reduce the total disease burden? Justify using the tornado plot.

(d) Compare the computation time with the FSE approach from Chapter 4.

Exercise 6.4 *(*[Bloom L4]*).* Fill in the computation cost table for the SIR model with $r = 1$ response functional ($J = \int I\,dt$) and $n = 3$ state variables:

m (number of POIs)	FSE cost (ODE solves)	Adjoint cost (ODE solves)
4		
10		
50		
500		

If each ODE solve takes 30 seconds, at what value of m does the adjoint become more than 10 times faster than the FSE?

Exercise 6.5 *(*[Bloom L4]*).* Explain in plain language, without any equations, why:

(a) The adjoint ODE must be integrated backward from $t = T$ rather than forward from $t = 0$.

(b) The adjoint variable $\lambda_I(t)$ for the SIR model and response $J = \int I\,dt$ is larger near $t = 0$ than near $t = T$.

(c) A public health agency that wants to minimize total disease burden should focus interventions on the early phase of the epidemic, when λ_I is large. Use $\lambda_I(t)$ to quantify this recommendation.

(d) The terminal condition $\vec{\lambda}(T) = \vec{0}$ is not the same as running the forward ODE backward with initial condition $\vec{0}$.

Exercise 6.6 *(*[Bloom L5]*).* Extend the SIR adjoint treatment to the SEIR model:

$$\begin{aligned}
\dot{S} &= \mu N - k\beta SI/N - \mu S,\\
\dot{E} &= k\beta SI/N - (\nu + \mu)E,\\
\dot{I} &= \nu E - (\gamma + \mu)I,\\
\dot{R} &= \gamma I - \mu R,
\end{aligned}$$

with POIs $\vec{p} = (k, \beta, \tau_E, \tau_I, L)$ (interpretable parameterization: $\nu = 1/\tau_E$, $\gamma = 1/\tau_I$, $\mu = 1/L$) and response functional $J = \int_0^T I(t)\,dt$.

Design the complete adjoint SA study, specifying:

(a) The 4×4 Jacobian J_F at a generic point (S, E, I, R).

(b) All four adjoint ODEs and their terminal conditions.

(c) The sensitivity formulas dJ/dp_j for all five POIs.

(d) The MATLAB implementation strategy (how you would store the forward solution and integrate the adjoint backward).

(e) What additional cost the SEIR adjoint requires compared to the SIR adjoint (in terms of ODE solves and storage).

Exercise 6.7 *(* [Bloom L6] *).* A climate modeler reports: "We implemented the adjoint to compute sensitivities for our model with 200 parameters, but the results disagree with forward sensitivity analysis by approximately 15% for some parameters. The adjoint must have a bug."

Identify three distinct causes that could produce this discrepancy without any coding error. For each cause:

(a) State the cause precisely.

(b) Describe a diagnostic test that would confirm or rule it out.

(c) Describe the correction if the cause is confirmed.

Chapter 7

Adjoint Sensitivity in Practice

Reverse the calculation, and one output can speak about many inputs.

You need the sensitivities of a complex disease model with 30 parameters. The model is your own code, you wrote it yourself, so it is differentiable. You know from Chapter 6 that the adjoint requires only 2 ODE solves instead of 30 FSE solves.

But deriving the adjoint equations by hand for 30 coupled nonlinear ODEs would take weeks and would almost certainly introduce errors in the derivation.

Is there a way to obtain the adjoint computation without deriving the adjoint equations by hand? Yes, algorithmic differentiation does this automatically [29]. And there is a way to use it from MATLAB today, without writing a single AD routine from scratch.

7.1 The Annuity Function: A Worked Comparison

Before examining the computational tools, we work through a concrete example where all three methods, forward mode, reverse mode, and symbolic differentiation, can be compared exactly.

7.1.1 The Problem

An annuity pays a fixed amount P per period for n periods at interest rate r. The present value is

$$A(P, r, n) = P\,\frac{1 - (1 + r)^{-n}}{r}. \tag{7.1}$$

We want all three sensitivity indices: S^A_P, S^A_r, S^A_n.

7.1.2 Forward Mode: Three Passes

In forward mode we differentiate the computation one parameter at a time, propagating $\dot{v}_k = \partial v_k/\partial q$ through intermediate values. We trace the computation of A through the following steps:

$$v_1 = (1 + r)^{-n}, \quad v_2 = 1 - v_1, \quad v_3 = v_2/r, \quad A = P v_3.$$

For $q = P$: set $\dot{P} = 1$, $\dot{r} = 0$, $\dot{n} = 0$.

$$\dot{v}_1 = 0, \quad \dot{v}_2 = 0, \quad \dot{v}_3 = 0, \quad \dot{A} = \dot{P}\,v_3 + P\,\dot{v}_3 = v_3.$$

So $\partial A/\partial P = v_3 = (1-(1+r)^{-n})/r$.

For $q = r$: set $\dot r = 1$, $\dot P = 0$, $\dot n = 0$.

$$\dot v_1 = n(1+r)^{-n-1}, \quad \dot v_2 = -n(1+r)^{-n-1}, \quad \dot v_3 = \frac{\dot v_2\, r - v_2}{r^2}, \quad \dot A = P\,\dot v_3.$$

For $q = n$: set $\dot n = 1$, $\dot P = 0$, $\dot r = 0$.

$$\dot v_1 = -\ln(1+r)\,(1+r)^{-n}, \quad \dot v_2 = \ln(1+r)\,(1+r)^{-n}, \quad \dot v_3 = \dot v_2/r, \quad \dot A = P\,\dot v_3.$$

Three passes through the computation graph give three derivatives. Each pass involves the same number of operations as one evaluation of A, so the total cost is approximately $3\times$ the cost of a single function evaluation.

7.1.3 Reverse Mode: One Pass

In reverse mode, we compute all three derivatives in one backward pass. Set $\bar A = 1$. Work backward through the computation:

$$\begin{aligned}
\bar P &= \bar A\, v_3 = v_3,\\
\bar v_3 &= \bar A\, P = P,\\
\bar v_2 &= \bar v_3/r,\\
\bar r &= \bar v_3 \cdot (-v_2/r^2) + \bar v_1 \cdot \frac{\partial v_1}{\partial r},\\
\bar v_1 &= -\bar v_2,\\
\bar n &= \bar v_1 \cdot \frac{\partial v_1}{\partial n} = -\bar v_2 \cdot \big(-\ln(1+r)(1+r)^{-n}\big).
\end{aligned}$$

One backward pass gives $\partial A/\partial P$, $\partial A/\partial r$, and $\partial A/\partial n$ simultaneously. Total cost: approximately $2\times$ one function evaluation (one forward pass to store values; one backward pass to accumulate derivatives), independent of how many parameters m there are.

7.1.4 Cost Summary

7.1.5 Symbolic Differentiation

As a third comparison, we differentiate the formula (7.1) by hand (or by a computer algebra system). Define $v = (1+r)^{-n}$. Then $A = P(1-v)/r$, giving:

$$\frac{\partial A}{\partial P} = \frac{1-v}{r}, \quad \frac{\partial A}{\partial r} = P\left(\frac{nv/(1+r) - (1-v)/r}{r}\right), \qquad \frac{\partial A}{\partial n} = P\,\frac{\ln(1+r)\,v}{r}.$$

Symbolic differentiation produces exact closed-form expressions that can be evaluated at any (P, r, n) with a single function evaluation; no passes through the computation graph are required at query time. The cost is paid upfront in the derivation; the payoff is that evaluation is cheap thereafter.

7.1.6 Exact Operation Counts

For the annuity function, we can count arithmetic operations precisely. The function A requires: 1 addition, 1 exponentiation, 1 subtraction, 2 divisions, 1 multiplication = 6 operations.

Forward mode for one parameter: 6 operations for the primal values, plus 6 tangent operations = 12 total. For $m = 3$ parameters: 36 operations.

Reverse mode: 12 operations (6 forward + 6 backward) regardless of m. For $m = 3$: still 12 operations.

The ratio: forward mode costs $\approx 2m$ times a single evaluation; reverse mode costs ≈ 2 times a single evaluation. For the annuity with $m = 3$, forward mode uses 3× more operations than reverse mode. For a climate model with $m = 500$, forward mode uses 250× more. For a neural network with $m = 10^7$ parameters and one scalar loss function, reverse mode still requires only one backward pass — the same cost as the annuity — while forward mode would require 10^7 separate sweeps. This is why reverse-mode AD made modern deep learning computationally feasible.

Method	Passes	Operations (annuity, $m = 3$)	Scales as
Forward mode	$m = 3$	≈ 36	$O(m)$
Reverse mode	2	≈ 12	$O(1)$ in m
Symbolic diff	1 (at query time)	≈ 6	$O(1)$ in m
Finite diff	$2m + 1 = 7$ evals	≈ 42	$O(m)$, approx

At nominal values $P = 1000$, $r = 0.05$, $n = 20$, all methods give: $S_P^A = 1$, $S_r^A \approx -0.604$, $S_n^A \approx 0.482$. (The derivation of the analytic formulas is Exercise 7.1.)

Self-check. From the annuity formula (7.1), compute S_P^A analytically using the derivative form $S_P^A = (P/A)(\partial A/\partial P)$. Verify that it equals 1 for any nominal values of P, r, n.

Answer: $\partial A/\partial P = (1 - (1 + r)^{-n})/r = A/P$. So $S_P^A = (P/A)(A/P) = 1$.
This holds for any r, n: the present value is linear in P, so a 1% increase in the payment amount produces exactly a 1% increase in present value, regardless of the interest rate or number of periods.

7.2 Algorithmic Differentiation

Algorithmic differentiation (AD), also called automatic differentiation, is the chain rule applied systematically to every arithmetic operation in a computer program. It is neither symbolic differentiation (which manipulates formulas) nor numerical differentiation (which approximates derivatives). It is exact to machine precision and applies to any differentiable code.

Students sometimes expect AD to produce a formula for the derivative, like a computer algebra system would. It does not. AD produces the numerical value of the derivative at a specific input, by accumulating the chain rule through the sequence of arithmetic operations the program performs.

Similarly, reverse-mode AD is not the same as numerical differentiation. A centered-difference estimate of $\partial A/\partial r$ has error $O(h^2)$ from the truncation of the Taylor series; reverse-mode AD gives the exact derivative (to floating-point precision) with no step-size selection required.

7.2.1 Forward Mode

Forward-mode AD attaches a tangent value $\dot{v}_k = \partial v_k/\partial q$ to each intermediate variable v_k, for one chosen parameter q. At each arithmetic operation, the tangent is updated using the chain rule:

$$v_k = f(v_i, v_j) \implies \dot{v}_k = \frac{\partial f}{\partial v_i}\dot{v}_i + \frac{\partial f}{\partial v_j}\dot{v}_j.$$

At the end of the computation, $\dot{v}_{\text{output}}$ contains $\partial J/\partial q$. Repeat for each parameter: m passes total.

7.2.2 Reverse Mode

Reverse-mode AD attaches an adjoint value $\bar{v}_k = \partial J/\partial v_k$ to each intermediate variable. In the backward pass, these are accumulated using the chain rule in reverse:

$$v_k = f(v_i, v_j) \implies \bar{v}_i \mathrel{+}= \frac{\partial f}{\partial v_i}\bar{v}_k, \quad \bar{v}_j \mathrel{+}= \frac{\partial f}{\partial v_j}\bar{v}_k.$$

The forward pass computes and stores all intermediate values. The backward pass accumulates all $\partial J/\partial q_i$ simultaneously. Total cost: one forward pass + one backward pass, regardless of m.

Reverse-mode AD is the discrete-computation-graph version of the adjoint ODE derived in Chapter 6. The adjoint ODE propagates $\vec{\lambda}(t)$ backward through a continuous system; reverse-mode AD propagates $\bar{v}_k$ backward through a discrete computation graph. The underlying mathematical operation is identical: transposition of the derivative operator.

Unifying Idea 2. Reverse-mode AD implements Unifying Idea 2 automatically: the adjoint is the transpose in whatever space you are working in, and AD constructs that transpose through the backward pass without requiring the analyst to derive the adjoint equations by hand. For a model written as differentiable code, this eliminates the main practical barrier to adjoint SA: instead of deriving and implementing the adjoint ODE, the analyst uses an AD library that differentiates the code directly. The adjoint equation is implicit in the reverse accumulation; the analyst never writes it explicitly.

7.2.3 Available Tools

The 2009 source material for this book referenced ADIFOR, DAFOR, and TAMC, tools that are now outdated or unmaintained. Table 7.1 lists the current ecosystem. For a comprehensive survey of AD methods, see Baydin et al. [4] and Margossian [23].

Self-check. Explain in one sentence each why: (a) AD is not symbolic differentiation; (b) reverse-mode AD is not the same as centered finite differences; (c) forward-mode AD requires m passes while reverse-mode requires 2.

Answer: (a) Symbolic differentiation manipulates formulas algebraically; AD applies the chain rule numerically to the sequence of operations in the code, producing a number rather than a formula.
(b) Centered finite differences approximate the derivative with error $O(h^2)$ via $(f(p+h) - f(p-h))/(2h)$; reverse-mode AD computes the exact derivative (to floating-point precision) by accumulating the chain rule backward through the computation graph.
(c) Forward mode propagates the tangent $\dot{v}_k = \partial v_k/\partial q$ for one parameter q per pass, requiring one pass per parameter. Reverse mode propagates the adjoint $\bar{v}_k = \partial J/\partial v_k$ for one response J per backward pass, giving all m derivatives simultaneously.

7.3 The Master Decision Table

With all methods now in hand, Table 7.2 consolidates the decision rules developed across Chapters 3–7.

There is no single universally correct method. The right choice depends on the model, the available tools, and the question being asked. A useful heuristic: start with finite differences (Chapter 3) to get SA results quickly, then switch to the adjoint (Chapter 6 or AD) if the computation time is unacceptable or if m is large.

Language	Tool	Mode(s)	Notes
MATLAB	Symbolic Math Toolbox	Symbolic	Exact; limited scalability
MATLAB	Deep Learning Toolbox `dlarray`	Reverse	AD for code; GPU support
Python	JAX	Forward + Reverse	NumPy-compatible; JIT compilation
Python	PyTorch `autograd`	Reverse	Widely used in ML
Julia	ForwardDiff.jl	Forward	High performance
Julia	Zygote.jl	Reverse	Source-to-source; flexible
C/Fortran	ADOL-C	Forward + Reverse	Mature; still active
C/Fortran	Tapenade	Forward + Reverse	Source transformation

Table 7.1: Current AD tools by language (as of 2025). For MATLAB users, the Deep Learning Toolbox `dlarray` provides reverse-mode AD for arbitrary differentiable code. For large-scale scientific computing, JAX (Python) or ADOL-C (C/Fortran) are the most commonly used options.

One principle applies universally: whatever method you use, always sanity-check at least two sensitivity indices against an independent method before trusting the full result. For finite differences, verify one analytic SI if available. For the adjoint, verify two values against finite differences. This cross-check catches the majority of implementation errors.

You should not need to choose between finite differences and the adjoint. Both serve distinct roles: FD is the right tool for black-box models; the adjoint is the right tool for white-box (differentiable) models with many parameters. They are complementary, not competing. But each tool has failure modes that practitioners encounter repeatedly.

7.4 Practical Guidance and Common Pitfalls

7.4.1 Discretize-Then-Adjoint vs. Adjoint-Then-Discretize

When computing adjoint sensitivities for ODE systems, two strategies are available. In the **adjoint-then-discretize** approach (used throughout Chapter 6), the adjoint ODE is derived analytically from the continuous ODE and then solved numerically. In the **discretize-then-adjoint** approach, the ODE is first discretized into a finite-difference recurrence, and the adjoint is then derived for that discrete system.

For linear ODEs with fixed step sizes, both approaches give identical results. For nonlinear ODEs with adaptive solvers, they can differ: the adjoint-then-discretize approach may give sensitivities that disagree with finite-difference verification because the continuous adjoint assumes a smooth forward trajectory that the adaptive solver does not produce. The discretize-then-adjoint approach is always consistent with finite differences but requires differentiating through the step-size control

Model type	m vs. r	Method	Accuracy	Cost
Black-box	Any	FD (Ch. 3)	$O(h^2)$	$2m{+}1$ evals
Analytic, simple	$m \leq r$	FSE (Ch. 4)	Exact	m solves
Analytic, diff. code	$m \gg r$	AD, reverse	Exact	2 passes
Analytic, manual	$m \gg r$	Adjoint (Ch. 6)	Exact	2 solves
Correlated POIs	Any	Reparameterize	—	Depends
Near bifurcation	Any	FSE + caution	Exact*	m solves
Large uncertainty	Any	Global SA (Ch. 9)	Variance	$N(m{+}2)$ evals

Table 7.2: Master decision table for SA method selection; see Pianosi et al. [30] for a broader SA survey. The primary driver is whether the model is black-box or differentiable, and whether $m \gg r$ or $m \leq r$. (*Near a bifurcation, the FSE is exact but the SI diverges; use the result with caution and consult Chapter 8.) **AD vs analytic adjoint:** use reverse-mode AD when the model is implemented in differentiable code and memory allows storing the computation tape. Use the analytic adjoint (Chapter 6) when the model is a nonlinear ODE or PDE, memory is constrained, or the numerical integrator (e.g., adaptive `ode45`) is not itself differentiable.

logic, which is non-trivial.

For the applications in this book, the adjoint-then-discretize approach with a fixed-step-size forward solver is the most practical choice. Sirkes and Tziperman [41] discuss the implications for geophysical models where both approaches are used in practice.

7.4.2 Memory Requirements for the Adjoint

The adjoint ODE requires the forward solution $\vec{u}(t)$ stored at every time point where the adjoint ODE coefficients are evaluated. For a system with $n = 100$ state variables solved over 10,000 time steps in double precision, the stored forward solution requires $100 \times 10{,}000 \times 8$ bytes $= 8$ MB, modest. For a climate model with $n = 10^6$ state variables and 10^6 time steps, the requirement is 8×10^{12} bytes $= 8$ TB; infeasible to store in memory.

For large systems, two strategies address this: **Checkpointing** stores the forward solution only at a subset of time points and recomputes intermediate values as needed during the backward pass. **Adjoint-of-checkpoint** schemes (Griewank & Walther [13]) achieve optimal trade-offs between memory and recomputation cost. For the models in this course, dense storage is always adequate.

7.4.3 Adaptive Step-Size Solvers

As noted in Chapter 3, adaptive ODE solvers (such as `ode45`) introduce non-smoothness in the model output as a function of parameters: the step-size controller makes discrete decisions that change discontinuously with the parameter value. The continuous adjoint ODE from Chapter 6 assumes a smooth forward solution; when adaptive solvers introduce discontinuities, the continuous adjoint may give inaccurate results.

The safest approach: use fixed-step-size integration (e.g., `ode4` in MATLAB, a 4th-order Runge-Kutta) for the forward solve when computing adjoint sensitivities. This eliminates the non-smoothness and makes the continuous and discrete adjoints equivalent. Alternatively, use tight tolerances (`RelTol` $\leq 10^{-8}$) with adaptive solvers to minimize the non-smoothness.

Self-check. You compute adjoint sensitivities for the SIR model using `ode45` with default tolerances for the forward solve, then verify against centered finite differences. The adjoint gives $S_\beta^J = 1.04$ while centered FD gives 0.87. Name two possible causes of this discrepancy, and describe the diagnostic steps you would take to identify which cause applies.

> *Answer:* Possible causes: (1) The adaptive step-size control in `ode45` introduces non-smoothness that corrupts the continuous adjoint. (2) The finite-difference step size h was chosen poorly (e.g., too small, causing catastrophic cancellation, or too large, causing truncation error). Diagnostics: (a) Recompute the adjoint with fixed-step-size `ode4`: if the discrepancy disappears, the solver is the culprit. (b) Plot the U-curve for the FD estimate to verify the step size is optimal; if the FD value changes significantly with h, the FD is unreliable. Always use finite differences as a sanity check for any adjoint implementation; a discrepancy above 1% warrants investigation.

7.4.4 Discontinuities Break AD

AD requires the model to be differentiable at the nominal parameter values. Common sources of non-differentiability in practice include `if` statements whose conditions depend on parameters,[1] `min` and `max` functions, piecewise-defined model components such as quarantine thresholds or saturation functions with hard cutoffs, and look-up tables interpolated without smoothing. When these are present, AD will silently compute the derivative of one branch, which may be zero or undefined at the branch point. The diagnostic: compare AD results against centered finite differences. If they disagree significantly for small h, suspect a discontinuity. Replacing piecewise functions with smooth approximations (sigmoid functions, smooth max/min) restores differentiability.

Self-check. A disease model contains the piecewise function: $\beta(I) = \beta_0$ if $I < 0.1N$, else $\beta_0/2$. This represents reduced contact when hospitalizations are high. (a) Is this function differentiable with respect to I everywhere? (b) If you run reverse-mode AD through this function at $I = 0.09N$, what derivative will AD report? Is this reliable? (c) Propose a smooth approximation that would make the model fully differentiable.

> *Answer:* (a) No. The function has a jump discontinuity at $I = 0.1N$, so $\partial\beta/\partial I$ is undefined there. (b) At $I = 0.09N$, the `if` condition is false ($I < 0.1N$), so AD will compute the derivative of β_0 with respect to I, which is zero. This is locally correct for I strictly below the threshold, but gives no information about the discontinuity itself, and will be completely wrong for sensitivity with respect to I near the threshold. It is unreliable as a global SA result. (c) Replace the step with a sigmoid: $\beta(I) = \beta_0\big(1 - 0.5\,\sigma(k(I/N - 0.1))\big)$, where $\sigma(x) = 1/(1 + e^{-x})$ and $k > 0$ controls sharpness. As $k \to \infty$ this recovers the discontinuous original; a moderate k (e.g., $k = 100$) approximates it closely while remaining differentiable everywhere.

Unifying Idea 1. Every method in this chapter (forward mode, reverse mode, symbolic differentiation, finite differences, the adjoint ODE) computes the same object: Unifying Idea 1, the sensitivity index defined in Chapter 2. The methods differ in computational cost, accuracy, and applicability to different model types, but the result they produce is always the same number: $S_{p_j}^J = (p_j/J)(dJ/dp_j)$.

7.5 MATLAB Laboratory: Four Methods, One Model

This laboratory computes the same sensitivity ($\partial J/\partial\beta$ for the SIR model with $J = \int_0^{90} I\,dt$, using all four methods and compares their accuracy, speed, and ease of implementation.

[1] *If-statements are where smooth calculus goes to lose its temper.* When an `if` condition switches branches as a parameter is perturbed, the forward and reverse passes follow different computational paths, and the resulting "derivative" is the derivative of whichever branch happened to be active, which may be zero or undefined at the transition.

Part 1: Finite differences (Chapter 3 method). Use the `sensitivity_jacobian` template from Chapter 3 with the SIR model. Record the estimated S_β^J and the computation time.

Part 2: Analytic FSE (Chapter 4 method). Solve the 15-equation augmented system from Chapter 4 for the β parameter alone. Extract $\partial I(t)/\partial\beta$ at all time points and integrate numerically. Record the result and time.

Part 3: Adjoint ODE (Chapter 6 method). Solve the adjoint system from Chapter 6. Use formula (6.15) to compute $dJ/d\beta$. Record the result and time.

Part 4: MATLAB Symbolic Toolbox. Define the SIR ODE symbolically and compute the sensitivity of the equilibrium $R_0 = k\beta/\gamma$ with respect to β as a sanity check. Note: the Symbolic Toolbox computes sensitivities of closed-form expressions, not of ODE solutions over time.

```
syms k beta gamma positive
R0   = k * beta / gamma;
SI_b = simplify( (beta/R0) * diff(R0, beta) )

%% For the time-dependent sensitivity via dlarray:
%% (requires Deep Learning Toolbox)
p_nom = [5, 0.06, 7, 10000];    %% [k, beta, tau, L]
p_dl  = dlarray(p_nom);
J_fn  = @(p) SIR_integral(p);   %% function returning integral of I
dJdp  = dlgradient(J_fn(p_dl), p_dl);
fprintf('AD sensitivity for beta: %.6f\n', extractdata(dJdp(2)));
```

Part 5: Comparison table. Fill in the following table from your measurements:

Method	S_β^J	Rel. error	Time (s)	Lines of new code
Finite differences				
Analytic FSE				
Adjoint ODE				
AD (`dlarray`)				

Discuss: which method is easiest to implement? Most accurate? Fastest? Which would you use in practice for this model?

Extension: introducing a discontinuity. Modify the SIR model to include a treatment threshold: if $I(t) > 0.1N$, reduce τ by 20% (faster treatment when infection is high). Re-run all four methods. Which methods give consistent results? Which methods silently produce wrong answers? How does the U-curve (Chapter 3) change for finite differences near β?

Connection to optimization. The adjoint computed in this chapter is the same object as the Lagrange multiplier in constrained optimization and the costate variable in optimal control. Appendix E develops this connection through LP duality and Pontryagin's minimum principle.

7.6 Exercises

Exercise 7.1 *(*[Bloom L3]*)*. For the annuity function (7.1) at nominal values $P = 1000$, $r = 0.05$, $n = 20$:

(a) Compute A numerically.

(b) Compute all three sensitivities S_P^A, S_r^A, S_n^A analytically by differentiating (7.1).

(c) Implement forward mode by hand for $q = r$ (as in Section 7.1) and verify your answer from (b).

(d) Implement reverse mode for all three parameters in one pass and verify all three answers.

(e) Count the number of arithmetic operations in each approach.

Exercise 7.2 *(* [Bloom L4] *).* For each of the following SA problems, select the most appropriate method from Table 7.2 and justify in two sentences:

(a) A climate model with 800 parameters written in Fortran. Each model run takes 10 hours. You want sensitivities for 2 response functionals.

(b) A pharmacokinetic model with 5 parameters and analytic closed-form solutions. You want sensitivities for 8 drug concentration time points.

(c) A traffic simulation with 40 parameters implemented as a compiled C++ executable. Parameter values can be set via a configuration file. Each run takes 3 minutes.

(d) A 10-parameter epidemic model written in MATLAB with `ode45`. You want one response functional. The model contains an `if` statement that triggers a quarantine intervention when $I > 0.05N$.

Exercise 7.3 *(* [Bloom L4] *).* Using MATLAB's Deep Learning Toolbox `dlarray`:

(a) Wrap the function $R_0(k, \beta, \tau) = k\beta\tau$ as a differentiable computation and use `dlgradient` to compute all three partial derivatives at $k = 5$, $\beta = 0.06$, $\tau = 7$.

(b) Compare with the analytic derivatives.

(c) Wrap the SIR equilibrium calculation (solve $\vec{f}(\vec{u}) = 0$ numerically) and compute $\partial\bar{I}/\partial\beta$ at the endemic equilibrium. Compare with the FSE result from Chapter 4.

(d) What happens when you try to apply `dlgradient` through a call to `ode45`? Explain why, and what the remedy is.

Exercise 7.4 *(* [Bloom L5] *).* Design the complete SA workflow for a dengue fever transmission model [10] with the following properties: $n = 8$ state variables (human and mosquito compartments), $m = 14$ parameters (mortality rates, biting rates, transmission probabilities, mean durations), $r = 2$ response functionals (basic reproduction number and endemic human infection prevalence), each ODE solve taking approximately 2 minutes.

Specify:

(a) Which SA method you would use and why.

(b) The expected total computation time.

(c) How you would validate the implementation.

(d) How you would handle the fact that two parameters (σ_v and σ_h, the mosquito and human biting rates) are structurally correlated.

(e) What output format you would use to present results to a non-mathematical public health audience.

Exercise 7.5 *(*[Bloom L6]*)*. A research group uses reverse-mode AD to compute adjoint sensitivities for a mosquito population model with adaptive-step-size ODE integration. They report that some sensitivity indices agree with finite differences but others disagree by 10–20%, with no obvious pattern. The discrepancies are largest for parameters that appear in the mortality rate expressions.

(a) Identify the most likely technical cause of the discrepancy.

(b) Describe two diagnostic tests that would confirm or rule out this cause.

(c) Propose two remedies and explain the trade-offs between them.

(d) At what point would you conclude that the disagreement indicates a genuine SA result (the linearization is poor for these parameters) rather than a numerical artifact?

Part V

Limitations and Extensions

Chapter 8

Caveat Emptor — Limitations of Local Sensitivity Analysis

Local sensitivity is a linear truth, and nature is rarely linear for long.

In 2019, Saltelli et al. [34] reviewed sensitivity analyses published in 280 highly cited papers across top environmental modeling journals. More than half used the wrong method for the question being asked. Specifically, studies that asked global questions, how does output uncertainty relate to input uncertainty across the full parameter range?; used local methods: derivatives computed at a single nominal parameter value. The derivatives were computed correctly. But they answered a different question than the one the authors claimed to answer.

Assuming local sensitivity holds globally is the triumph
of mathematical hope over nonlinear reality.

This chapter asks the question that should be asked before any SA is begun: is local SA the right tool for this question?

8.1 Method-Question Mismatch

Local sensitivity analysis is built on a derivative. A derivative is a local object: it measures the behavior of a function at one point, and it is only meaningful in the vicinity of that point. This makes local SA powerful for some questions and completely wrong for others.

Local SA can answer: *which parameter most affects the model near the nominal parameter values?*

Local SA cannot answer: *which parameter most affects the model across its full range of uncertainty?* That question requires the parameter to actually vary across its range, and a derivative does not do that.

Local SA cannot answer: *how does the model behave at parameter values far from the nominal?* The derivative approximates linear behavior; real models are often highly nonlinear.

The principle is simple and unforgiving. The method must match the question. Using local SA to answer a global question is not merely suboptimal; it can produce a ranking of parameter importance that is entirely wrong. A parameter that appears unimportant locally may dominate the output variance when varied across its full uncertainty range. A parameter that appears important locally may have negligible effect at realistic perturbation sizes if the model is highly nonlinear in that parameter.

A small sensitivity index does not mean a parameter can be ignored globally. It means the model is approximately linear in that parameter near the nominal value, and the linear coefficient happens to be small. Section 8.5 shows a specific example where the local ranking is the opposite of the global ranking.

Self-check. A published SA study reports: "All sensitivity indices are less than 0.05, confirming that the model is robust to parameter uncertainty." The study used nominal parameter values from the literature, with no information on parameter uncertainty ranges. Identify two specific claims in this statement that go beyond what local SA can actually establish.

Answer: (1) "Robust to parameter uncertainty": local SA says nothing about parameter uncertainty. It describes behavior near the nominal value. If the uncertainty range for some parameters is large (e.g., $\pm 50\%$), the model could be highly sensitive to that parameter globally even though the local SI is small. (2) The conclusion depends implicitly on the nominal parameter values chosen. If the nominal values are changed, the local SI values change too (for nonlinear models). A study with different nominal values might find very different rankings. This sensitivity of local SA to the choice of nominal values is itself a limitation that the study does not address.

8.2 Bifurcations and SA Failure

Local SA requires the model output to be differentiable with respect to the parameters at the nominal values. When this fails, local SA fails completely. The most important way it fails is at a bifurcation.

A bifurcation is a parameter value at which the qualitative behavior of the model changes: two equilibria merge into one, a stable fixed point becomes unstable, or an epidemic threshold is crossed [44, 21]. At a bifurcation point, the derivative $\partial u/\partial p$ is either undefined or infinite. Local SA breaks down entirely. The precise mathematical reason is given by the Fredholm alternative (Appendix B.3): near a bifurcation the Jacobian becomes singular, so the FSE $D_{\vec{u}}[\vec{f}]\,(\partial\vec{u}/\partial p) = \vec{r}$ may have no solution or infinitely many. For models with periodic limit cycles, sensitivity near a bifurcation of the periodic orbit involves Floquet multipliers; this is treated in Appendix D.

8.2.1 The Quadratic Example

Consider the scalar equation $f(u;p) = u^2 + p = 0$. For $p < 0$, there are two real roots: $u = \pm\sqrt{-p}$. At $p = 0$, the two roots merge. For $p > 0$, there are no real roots.

The FSE for the sensitivity of the positive root $u^* = \sqrt{-p}$ is:

$$\frac{\partial u^*}{\partial p} = \frac{-\frac{\partial f}{\partial p}}{\frac{\partial f}{\partial u}} = \frac{-1}{2u^*} = \frac{-1}{2\sqrt{-p}}.$$

The normalized SI is:

$$S_p^{u^*} = \frac{p}{u^*}\frac{\partial u^*}{\partial p} = \frac{p}{\sqrt{-p}} \cdot \frac{-1}{2\sqrt{-p}} = \frac{1}{2}.$$

This is constant (independent of p) for all $p < 0$; the positive root is always in exact proportion $1/2$ to any fractional change in p.

But as $p \to 0^-$:

$$\frac{\partial u^*}{\partial p} = \frac{-1}{2\sqrt{-p}} \to -\infty.$$

The derivative diverges at the bifurcation. Near $p = 0$, the model is infinitely sensitive to the parameter p: any small positive perturbation to p destroys the equilibrium entirely.

Unifying Idea 3. This is Unifying Idea 3 in its most dramatic form. The sensitivity index is the derivative-based linearization of the model response. At a bifurcation, the function $u^*(p)$ has infinite curvature, its graph has a vertical tangent, and no linear approximation is valid. The linearization does not just become inaccurate at the bifurcation; it becomes undefined.

Unifying Idea 1. The failure at a bifurcation is a failure of the normalized derivative. The sensitivity index S_p^q exists only when $\partial q/\partial p$ is finite. Near a bifurcation it is not, and the index either blows up or vanishes depending on which side of the threshold you are computing from.

Unifying Idea 2. The adjoint SA system inherits exactly the same failure mode. The adjoint ODE is driven by the Jacobian J_F^T of the forward model. Near a bifurcation, the Jacobian has a near-zero eigenvalue, and the adjoint solution grows without bound over the integration interval. Both the forward and adjoint approaches fail at the same bifurcation for the same reason: the transpose of a singular matrix is also singular.

Self-check. For $f(u;p) = u^2 + p$ with $p = -4$: (a) Find the two roots $u_\pm^*$. (b) Compute $S_p^{u_+^*}$ analytically. Compare with the derivative $\partial u_+^*/\partial p$. (c) Estimate $\partial u_+^*/\partial p$ numerically using a centered difference with $h = 0.1, 0.01, 0.001$. At what value of p near zero does the finite-difference estimate become unreliable?

> *Answer:* (a) $u_\pm^* = \pm\sqrt{4} = \pm 2$. (b) $\partial u_+^*/\partial p = -1/(2\sqrt{-p}) = -1/4$. $S_p^{u_+^*} = (p/u_+^*)(-1/(2\sqrt{-p})) = (-4/2)(-1/4) = 1/2$.
> (c) Numerically at $p = -4$: all three step sizes give ≈ -0.250 ✓. As $p \to 0^-$, the function $u_+^* = \sqrt{-p}$ has a square-root singularity; finite differences become inaccurate when the curvature term dominates the step-size error, which occurs when $h \gtrsim |p|$. Near $p = 0$, even very small h is not small compared to $|p|$, so finite differences fail.

8.2.2 The SIR Model at the Epidemic Threshold

The SIR model has a transcritical bifurcation at $R_0 = 1$: for $R_0 < 1$, the only equilibrium is the disease-free state; for $R_0 > 1$, the endemic equilibrium appears.

The endemic equilibrium fraction $i^* = 1 - 1/R_0$ exists only for $R_0 > 1$. Its sensitivity with respect to R_0 is:

$$\frac{\partial i^*}{\partial R_0} = \frac{1}{R_0^2}, \qquad S_{R_0}^{i^*} = \frac{R_0}{i^*} \cdot \frac{1}{R_0^2} = \frac{1}{R_0(1 - 1/R_0)} = \frac{1}{R_0 - 1}.$$

As $R_0 \to 1^+$, $S_{R_0}^{i^*} \to +\infty$: near the epidemic threshold, the endemic equilibrium is infinitely sensitive to changes in R_0. This is exactly the bifurcation behavior identified above.

SA of the endemic equilibrium for R_0 values close to 1 must be interpreted with extreme caution. A small change in any parameter that affects R_0 could eliminate the endemic equilibrium entirely or cause it to surge. Local SA correctly identifies this extreme sensitivity but cannot predict the qualitative change.

For $R_0 \leq 1$: SA of the endemic equilibrium is undefined; the equilibrium does not exist. Computing SA under the assumption that i^* exists when $R_0 < 1$ is a mathematical error. MATLAB will simply return a negative or complex value without warning.

Operational rule: when to stop reporting SI as a ranking. A sensitivity index is meaningful as a ranking tool only when the linearization is a fair local approximation. Two practical checks:

1. *Magnitude check.* If $|S_p^q| > 10$ for a normalized index, treat the result as a qualitative warning ("this parameter region is highly sensitive") rather than a quantitative ranking. An SI of 100

does not mean the parameter matters 10× more than one with SI 10; it may mean the model is near a threshold.

2. *Condition-number check.* Compute $\kappa(J_F)$ (the condition number of the model Jacobian at the operating point). If $\kappa(J_F) > 10^4$, the sensitivity computation is ill-conditioned: small errors in the model evaluation may be amplified by $\kappa(J_F)$ in the sensitivity values. Always report $\kappa(J_F)$ alongside SI values computed near a bifurcation or near parameter degeneracy.

8.3 Ill-Conditioning

For linear algebraic models, SA failure takes a different form. When the coefficient matrix A in $A\vec{u} = \vec{b}$ is nearly singular, small changes in the right-hand side $\vec{b}$ (or in the matrix entries) produce large changes in the solution. The sensitivity indices are large not because the model is interesting but because the model is poorly identified.

8.3.1 The Condition Number

The **condition number** $\kappa(A) = \|A\|\|A^{-1}\|$ (see Appendix C.2 for the SVD-based definition $\kappa_2 = \sigma_1/\sigma_r$) measures how much the solution $\vec{u}$ can change relative to a change in the data. For the 2-norm,

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \tag{8.1}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A. A matrix with $\kappa_2(A) = 1$ is perfectly conditioned (orthogonal); a matrix approaching singularity has $\sigma_{\min} \to 0$ and $\kappa_2(A) \to \infty$.

The key bound: if the data changes by a relative amount ϵ, the solution changes by at most $\kappa(A)\,\epsilon$:

$$\frac{\|\delta\vec{u}\|}{\|\vec{u}\|} \le \kappa(A)\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|}.$$

When $\kappa(A) \gg 1$, even tiny changes in the data produce large changes in the solution.

8.3.2 SA and Ill-Conditioning

For SA, the key insight is that large sensitivity indices are not necessarily a property of the model's *true* behavior. They may simply reflect ill-conditioning: the fact that the linear system is nearly singular.

A colleague who reports "the sensitivity of u_1 to b_1 is extremely large" may be reporting either: (a) a genuine model property (the output truly varies strongly with this parameter), or (b) a numerical artifact (the linear system is nearly singular and small input changes have large numerical effects on the solution).

Distinguishing (a) from (b) requires computing $\kappa(A)$. If $\kappa(A) \approx 10^6$, then numerical errors of size $\varepsilon_{\text{mach}} \approx 10^{-16}$ can contaminate the solution at the level of 10^{-10}: 6 digits of the solution may be meaningless.

Ill-conditioning is itself a SA result: it warns the investigator that the model parameters are nearly unidentifiable from data, and that predictions cannot be trusted without more precise parameter measurements. This is not a failure of SA; it is one of the most important things SA can reveal.

Self-check. The 2×2 matrix $A(\epsilon) = \begin{pmatrix} 1 & 1 \\ 1 & 1+\epsilon \end{pmatrix}$ with $\epsilon > 0$: (a) Compute $\det(A(\epsilon))$ and show $A \to$ singular as $\epsilon \to 0$. (b) Compute the condition number $\kappa_2(A)$ for $\epsilon = 1, 0.1, 0.01$. (c) For $\vec{b} = (2, 2+\epsilon)^T$, solve $A\vec{u} = \vec{b}$ and compute $S_\epsilon^{u_1}$. What happens to this SI as $\epsilon \to 0$?

Answer: (a) $\det = \epsilon \to 0$ as $\epsilon \to 0$. Singular at $\epsilon = 0$. (b) $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$. For $\epsilon = 1$: $\kappa \approx 5.83$. For $\epsilon = 0.1$: $\kappa \approx 40$. For $\epsilon = 0.01$: $\kappa \approx 400$. As $\epsilon \to 0$: $\kappa \to \infty$.
(c) $A^{-1} = \epsilon^{-1} \begin{pmatrix} 1+\epsilon & -1 \\ -1 & 1 \end{pmatrix}$. $\vec{u} = A^{-1}\vec{b} = (1,1)^T$ for all $\epsilon > 0$. $S_\epsilon^{u_1}$: differentiate $u_1 = (2(1+\epsilon) - (2+\epsilon))/\epsilon = \epsilon/\epsilon = 1$ with respect to ϵ, but the simplification $u_1 = 1$ holds exactly, so $S_\epsilon^{u_1} = 0$. The sensitivity vanishes!
This illustrates a subtlety: large condition number does not always imply large SI. What the condition number warns is that *numerical* computation of $\vec{u}$ will be inaccurate; the exact mathematical SI may nonetheless be finite or zero.

8.4 Separatrices and Trajectory Sensitivity

Dynamical systems often have regions of qualitatively different behavior separated by a **separatrix**: a boundary in state space that trajectories cannot cross. SA computed on one side of a separatrix describes behavior qualitatively different from SA on the other side.

For the SIR model, the relevant separatrix is the epidemic threshold $S(0) > \gamma N/(k\beta)$ (equivalently, $R_0 > 1$). If the initial susceptible population exceeds this threshold, an epidemic occurs. If it does not, the infection dies out without spreading.

Suppose SA is performed at a parameter value where an epidemic occurs ($R_0 = 1.5$) and the sensitivity index $S_\beta^{I_{\max}}$ is computed. This index describes how the epidemic peak changes with a small perturbation to β within the epidemic regime. It says nothing about what happens if β is reduced below the threshold: in that regime, $I_{\max}$ is not a smooth function of β it drops to zero discontinuously as the separatrix is crossed.

The practical implication: if any parameter perturbation of interest crosses a qualitative threshold (endemic vs. disease-free, outbreak vs. no outbreak, stable vs. oscillatory), local SA cannot be used to predict the response. The model must be analyzed separately in each qualitative regime, and the boundary itself requires special treatment.

A related limitation applies to models with multiple stable attractors. When a model can settle into qualitatively different long-run states depending on initial conditions, the sensitivity index computed near one attractor says nothing about behavior near another. Extensions of the SIR model that exhibit bistability (where both a disease-free and an endemic equilibrium are stable for the same parameter values) require separate SA studies for each attractor. Local SA at one equilibrium cannot warn the analyst that another equilibrium exists.

8.5 When Global SA Is Needed

Local SA studies the neighborhood. Global SA studies the country.

Local SA fails to answer the right question in three specific situations. When any of these conditions holds, Chapter 9's global SA methods are required. The practical decision framework for choosing between local and global SA is developed at length by Saltelli et al. [40, 39].

To reiterate the misconception that opened this chapter: a sensitivity index near zero does not mean a parameter is globally unimportant. It means the linearization at the nominal value is nearly flat. What happens away from the nominal, across the full uncertainty range of the parameter, requires global SA to determine.

8.5.1 Large Parameter Uncertainty

When a parameter is known only to within a factor of 2 or 3; common in epidemiology, ecology, and environmental modeling; the linearization underlying local SA may be very poor across the plausible range.

As a concrete example, consider a model where the QOI depends on a parameter p through $q(p) = e^p$. At nominal value $p^* = 0$: $S_p^q = p^*(dq/dp)/q = 0 \cdot 1/1 = 0$. The local SI is zero. But if the uncertainty range is $[-2, 2]$, the QOI varies from $e^{-2} \approx 0.14$ to $e^2 \approx 7.4$: a factor of 53. The parameter is globally important but invisible to local SA.

8.5.2 Highly Nonlinear Models

Two models with identical local SIs for a parameter p may behave very differently when p is perturbed by 20%: one model may be nearly linear (the SI is a good approximation), while the other may be highly curved (the SI understates the true change).

The sigma-normalized sensitivity index (Section 9.1) provides the minimal bridge: it weights the local SI by the parameter's standard deviation, giving a measure that at least accounts for the relative magnitudes of parameter uncertainties.

8.5.3 Correlated Parameters

Local SA under the independence assumption gives misleading results when parameters are correlated (Chapter 3). Global SA methods in Chapter 9 include specific tools for handling this case.

8.5.4 The Portfolio Example

Consider a model $Y = s_1Q_1 + s_2Q_2$ with $s_1 = 2$, $s_2 = 1$ and parameter standard deviations $\sigma_1 = 1$, $\sigma_2 = 3$.

Local SA gives $S^Y_{Q_1} = 2 > S^Y_{Q_2} = 1$: parameter 1 appears more important.

The sigma-normalized index: $S_i^\sigma = (s_i\sigma_i)/\sigma_Y$, where $\sigma_Y = \sqrt{4+9} = \sqrt{13}$. So $S_1^\sigma = 2/\sqrt{13} \approx 0.55$ and $S_2^\sigma = 3/\sqrt{13} \approx 0.83$.

Parameter 2 dominates the output variance, even though it has a smaller local SI. The local ranking is inverted because parameter 2 has much larger uncertainty. Chapter 9's Sobol indices provide the rigorous global version of this sigma-normalized assessment.

Self-check. For the SIR model, suppose β has uncertainty $\sigma_\beta = 0.02$ and τ has uncertainty $\sigma_\tau = 3$ days, at nominal values $\beta^* = 0.06$ and $\tau^* = 7$ days.

(a) From Chapter 2, $S_\beta^{R_0} = S_\tau^{R_0} = 1$. Which parameter appears more important from local SA alone?

(b) Compute σ_β/β^* and σ_τ/τ^* (the relative uncertainties). Which parameter has larger relative uncertainty?

(c) Compute the sigma-normalized indices for R_0 with respect to β and τ. Which parameter drives more variance in R_0?

Answer: (a) Both have SI = 1, so local SA alone says they are equally important.
(b) $\sigma_\beta/\beta^* = 0.02/0.06 \approx 0.33$ (33%); $\sigma_\tau/\tau^* = 3/7 \approx 0.43$ (43%). The mean infectious period has larger relative uncertainty.
(c) $S_\beta^\sigma = S_\beta^{R_0} \cdot (\sigma_\beta/\beta^*) = 1 \times 0.33 = 0.33$. $S_\tau^\sigma = S_\tau^{R_0} \cdot (\sigma_\tau/\tau^*) = 1 \times 0.43 = 0.43$. The mean infectious period drives more variance in R_0, despite both having the same local SI. Chapter 9 will quantify this more precisely with Sobol indices.

8.6 When SA Guides Data Collection

The three chapters preceding this one have identified when SA can mislead. This section completes the practical argument the book has been building since Chapter 1: how to use SA *correctly* to make decisions about field measurement.

The decision chain. A sensitivity study produces a ranked list of parameters. Translating that list into a field-measurement plan requires three additional steps that SA alone cannot provide:

1. *Identifiability:* Can the high-sensitivity parameter actually be estimated from the available data, or is it structurally confounded with another parameter? Section 3.5 showed that parameters with identical sensitivity indices are not separately identifiable from outbreak data alone; measuring one is equivalent to measuring the other. For the SIR model, k and β always appear as the product $k\beta$, so field studies should estimate $k\beta$ directly (e.g., from secondary attack rates) rather than k and β separately.

2. *Measurability:* Is the high-sensitivity parameter actually measurable with available instruments and budget? The mean infectious period τ appears in the top tier of every SIR sensitivity analysis, and it is measurable: clinicians record the duration from symptom onset to recovery. The lifespan L has negligible SI for short-horizon models and is measurable anyway from demographic records. A parameter with large SI but no feasible measurement pathway should be *reduced* rather than measured, by redesigning the intervention.

3. *Actionability:* Can the high-sensitivity parameter be reduced by a feasible public health intervention? SA of the SIR model with $J = \max_t I(t)$ consistently identifies k (contact rate) and β (transmission probability) as the most influential parameters. Contact rates can be reduced by social distancing, school closures, and remote-work policies. Transmission probability per contact can be reduced by mask requirements, hand-hygiene campaigns, and vaccination. These two routes correspond to different interventions with different costs, compliance rates, and social acceptability. SA identifies both as equally valuable targets; the choice between them requires policy judgment beyond the model.

The two-budget problem. A common practical scenario: a government agency has funding to (1) measure exactly k_{meas} parameters precisely in the field, and (2) implement exactly k_{int} interventions. The correct SA-guided procedure is:

1. Run local SA (or Morris screening for large parameter sets) to rank all parameters by $|S_{p_j}^{\text{peak } I}|$.
2. From the top-ranked parameters, apply the identifiability filter: remove any parameter that is structurally confounded with a higher-ranked parameter.
3. Apply the measurability filter: remove any parameter for which no feasible measurement protocol exists within the budget.
4. The remaining top k_{meas} parameters are the measurement priorities.
5. From the ranked list, apply the actionability filter separately: the top k_{int} actionable parameters are the intervention priorities. These sets may overlap.

What the model cannot tell you. The complete picture requires one more piece: the second semester. Smith's
emphUncertainty Quantification
citesmith2014uncertainty Chapters 7–8 covers Bayesian parameter estimation — the step that actually quantifies how much a given measurement budget reduces parameter uncertainty. SA tells you *which* parameters matter most; parameter estimation tells you *how precisely* you can learn them

from data; optimal experimental design (Smith Chapter 11) tells you *how to design the measurement study* to learn those parameters as efficiently as possible.

These three disciplines — SA, parameter estimation, and optimal experimental design — form a complete pipeline for model-based decision making. This book has equipped you for the first stage.

If a parameter is influential and poorly known, that is not a nuisance. That is your research agenda.

8.7 MATLAB Laboratory: Bifurcation and Ill-Conditioning

Part 1: The quadratic bifurcation. For $f(u;p) = u^2 + p$, compute the positive root $u^* = \sqrt{-p}$ and the SI $S_p^{u^*} = 1/2$ analytically. Plot u^* vs. p for $p \in [-4, -0.01]$. Estimate $\partial u^*/\partial p$ by centered differences at $p = -4, -1, -0.1, -0.01, -0.001$. Compare the finite-difference estimates with the analytic value. At what value of p does the finite-difference estimate become unreliable?

Part 2: SIR bifurcation diagram using MATCONT. Download and install MATCONT (`https://matcont.sourceforge.io`). Define the SIR equilibrium equations as a MATCONT system. Use MATCONT's continuation capability to trace the endemic equilibrium $i^* = 1 - 1/R_0$ as R_0 increases from 0.5 to 3.0. MATCONT will automatically detect the transcritical bifurcation at $R_0 = 1$ and flag it. Plot the bifurcation diagram (equilibrium value vs. R_0). If MATCONT is unavailable, trace the equilibrium manually: for $R_0 \leq 1$, $i^* = 0$ (disease-free); for $R_0 > 1$, $i^* = 1 - 1/R_0$.

Part 3: FSA comparison across $R_0 = 1$. Using the full SIR ODE system, compute $S_\beta^{I(T)}$ at $T = 90$ days for $R_0 = 0.8, 1.0, 1.2, 2.0$. At each value, use centered finite differences. Tabulate the results. At $R_0 = 1$: does the SI blow up, return NaN, or behave normally? Try $h = 0.01, 0.001, 0.0001$ to see what happens.

Part 4: Condition number of the SIR Jacobian as a function of R_0. For the SIR endemic equilibrium, the Jacobian J_F at (S^*, I^*, R^*) depends on R_0. Compute $\kappa_2(J_F)$ using MATLAB's `cond` for $R_0 \in \{1.01, 1.05, 1.1, 1.5, 2.0, 3.0\}$. Plot $\kappa_2(J_F)$ vs. R_0. Observe the spike as $R_0 \to 1^+$: the Jacobian becomes nearly singular at the bifurcation. What does this imply about SA of the SIR endemic equilibrium for parameter values that place R_0 close to 1?

Part 5: Sigma-normalized SIs for the SIR model. Using the SIR model with uncertainty ranges $\sigma_k = 1$ (contacts/day), $\sigma_\beta = 0.02$, $\sigma_\tau = 3$ days, $\sigma_L = 500$ days (all at 1σ), compute the sigma-normalized indices for R_0. Compare the ranking with the local SI ranking from Chapter 2. Which parameter most affects R_0 given its realistic uncertainty range?

8.8 Exercises

Exercise 8.1 *(*[Bloom L3]*)*. For the equation $g(u;p) = u^3 - 3pu + 2 = 0$:

(a) Find all real roots at $p = 1$. Which root changes most rapidly with p?

(b) Compute $S_p^{u^*}$ for each real root at $p = 1$ using the implicit function theorem.

(c) Plot the roots as functions of p for $p \in [0.8, 1.2]$. At what value of p do two roots merge?

(d) At the bifurcation value of p, what does the FSE predict? Verify numerically using finite differences.

Exercise 8.2 *(*[Bloom L3]*)*. Using the SIR model with $k = 5$, $\tau = 7$ days, $L = 10{,}000$ days:

(a) Compute $S_\beta^{I(T)}$ for $\beta = 0.04, 0.06, 0.08, 0.12$ at $T = 90$ days using centered finite differences. Note: $R_0 = k\beta\tau$ takes values $1.4, 2.1, 2.8, 4.2$.

(b) Now recompute for $\beta = 0.02$ ($R_0 = 0.7$). Compare with the results for $R_0 > 1$. Explain any qualitative differences.

(c) If you were to add a bifurcation analysis tool (e.g., MATCONT), what would you expect the bifurcation diagram to look like for the endemic equilibrium of this SIR model as β varies?

Exercise 8.3 *(*[Bloom L4]*).* For the 3×3 linear system $A\vec{u} = \vec{b}$ where:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9+\epsilon \end{pmatrix}, \quad \vec{b} = A\vec{u}_{\text{exact}}, \quad \vec{u}_{\text{exact}} = (1, 1, 1)^T.$$

(a) For $\epsilon = 1, 0.1, 0.01, 0.001, 0.0001$, compute $\kappa(A)$ and $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ where $\hat{\vec{u}} = A\backslash\vec{b}$ in MATLAB.

(b) Plot $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ vs. $\kappa(A)$ on log-log axes. What is the approximate slope?

(c) What does this experiment tell you about interpreting large SI values for nearly singular models?

Exercise 8.4 *(*[Bloom L5]*).* You are asked to perform SA on an epidemic model whose best-fit parameter values give $R_0 = 1.04$, very close to the epidemic threshold. Write a one-page advisory report for a public health agency explaining:

(a) What local SA can and cannot tell you at this parameter value.

(b) What specific numerical problems to expect when computing sensitivity indices near $R_0 = 1$.

(c) What additional analyses you would recommend before advising on interventions.

(d) How you would explain the concept of a bifurcation and its implications to a non-mathematical policy maker.

Exercise 8.5 *(*[Bloom L6]*).* A published modeling paper reports: "Sensitivity analysis of the endemic equilibrium of our malaria model found that all indices are less than 0.1, indicating the model is robust to parameter uncertainty. The basic reproduction number at fitted parameter values is $R_0 = 1.15$."

Identify at least three specific methodological concerns with this analysis and its conclusions. For each concern, state: (a) what the problem is; (b) what evidence the authors would need to provide to address it; and (c) what alternative analysis would give a more complete picture.

Chapter 9

Global Sensitivity Analysis and Sobol Indices

The point is not to remove uncertainty.
The point is to find out which uncertainty is worth worrying about.

Consider two epidemiological models. Both give $S_{\beta}^{R_0} = +1$ at the same nominal parameter values. In Model A, R_0 is nearly linear in β across the full biologically plausible range $[0.01, 0.15]$. In Model B, R_0 is highly nonlinear: it is steep near the lower end of the range and flat at the upper end.

For Model A, the local SI correctly identifies β as an important parameter. For Model B, local SA gives the same SI value of $+1$; but it is now describing only the slope at the nominal point. A 20% increase in β from the nominal produces very different effects in the two models, and no derivative computed at the nominal value can see this.

Chapter 8 showed when local SA fails. This chapter provides the tools to replace it.

9.1 Why Local SA Is Not Enough: The Sigma-Normalized Bridge

Local SA studies the neighborhood. Global SA studies the country.

The two can give very different answers about which parameters matter most, as the following example shows.

9.1.1 The Portfolio Example

Consider a model $Y = s_1 Q_1 + s_2 Q_2$ where $s_1 = 2$, $s_2 = 1$ and the parameters have standard deviations $\sigma_1 = 1$, $\sigma_2 = 3$.

Local SA gives $S_{Q_1}^{Y} = 2 > S_{Q_2}^{Y} = 1$: parameter 1 appears more important. But the parameters have very different uncertainty ranges, and Q_2 varies three times as much as Q_1.

The output variance is:

$$\mathrm{Var}(Y) = s_1^2 \sigma_1^2 + s_2^2 \sigma_2^2 = 4 + 9 = 13 \qquad \Longrightarrow \qquad \sigma_Y = \sqrt{13}.$$

The **sigma-normalized sensitivity index**

$$S_i^\sigma = \frac{\sigma_i}{\sigma_Y} \frac{\partial Y}{\partial Q_i} \tag{9.1}$$

accounts for both the derivative and the parameter uncertainty:

$$S_1^\sigma = \frac{1}{\sqrt{13}} \cdot 2 \approx 0.55, \qquad S_2^\sigma = \frac{3}{\sqrt{13}} \cdot 1 \approx 0.83.$$

Parameter 2 dominates, opposite to the local SA ranking. The local ranking was misleading because it ignored the factor-of-3 difference in parameter uncertainty.
Unifying Idea 1. The sigma-normalized index is still the sensitivity index from Chapter 2, multiplied by the relative parameter uncertainty. Global SA builds on local SA; it does not replace it. The local SI from Chapter 2 appears inside every global SA formula in this chapter.

9.1.2 Two SIR Models

For the SIR model, suppose two outbreak settings give the same $S_\beta^{R_0} = 1$. In Setting A, β is tightly constrained from contact-tracing data ($\sigma_\beta = 0.005$), while in Setting B, β must be estimated from seroprevalence data ($\sigma_\beta = 0.03$). Local SA says β is equally important in both settings. The sigma-normalized index says β in Setting B is six times more important, in terms of its contribution to uncertainty in R_0. The second semester will show how to propagate these uncertainties through the full model.

Self-check. For the SIR model with local SIs $S_k^{R_0} = S_\beta^{R_0} = S_\tau^{R_0} = 1$ and parameter standard deviations $\sigma_k = 0.5$, $\sigma_\beta = 0.01$, $\sigma_\tau = 2$, with $\sigma_{R_0} = \sqrt{\sigma_k^2 + \sigma_\beta^2 + \sigma_\tau^2}/R_0^*$:

For a SIR model with $k^* = 5$, $\beta^* = 0.06$, $\tau^* = 7$ ($R_0^* = 2.1$), first compute σ_{R_0} from the delta method, then rank the three parameters by their sigma-normalized index S_i^σ.

Answer: Delta method: $\mathrm{Var}(R_0) \approx \sum_i (\partial R_0/\partial p_i)^2 \sigma_i^2$. $\partial R_0/\partial k = \beta\tau = 0.42$; $\partial R_0/\partial \beta = k\tau = 35$; $\partial R_0/\partial \tau = k\beta = 0.3$. $\mathrm{Var}(R_0) \approx 0.42^2(0.25) + 35^2(0.0001) + 0.3^2(4) = 0.0441 + 0.1225 + 0.36 = 0.527$. $\sigma_{R_0} \approx 0.726$.
$S_k^\sigma = (0.5/0.726)(0.42/2.1)(k/k)\cdot\ldots$ Easier via the formula directly: $S_i^\sigma = (\partial R_0/\partial p_i)\sigma_i/\sigma_{R_0}$. $S_k^\sigma = 0.42 \times 0.5/0.726 \approx 0.29$. $S_\beta^\sigma = 35 \times 0.01/0.726 \approx 0.48$. $S_\tau^\sigma = 0.3 \times 2/0.726 \approx 0.83$.
Ranking: $\tau > \beta > k$. The mean infectious period drives the most uncertainty in R_0, despite all three having equal local SIs.

9.2 Sampling Methods

All global SA methods require evaluating the model at many parameter combinations. Choosing those combinations efficiently is the first computational task.

9.2.1 Random Sampling

The simplest approach: draw N independent samples from the joint parameter distribution (often assumed uniform or independent log-normal over plausible ranges). For N large, random sampling provides good coverage on average, but can leave regions of parameter space unsampled by chance.

9.2.2 Latin Hypercube Sampling

Latin Hypercube Sampling (LHS) divides each parameter's range into N equal strata and ensures that exactly one sample falls in each stratum for every parameter [25]. This eliminates the clustering that random sampling can produce and improves coverage with smaller N.

Definition 9.1 (LHS Sample)**.** An LHS of size N for m parameters with ranges $[l_j, u_j]$, $j = 1, \ldots, m$, is an $N \times m$ matrix X such that each column contains exactly one value from each of the N equal-width strata $[l_j + (k-1)(u_j - l_j)/N,\ l_j + k(u_j - l_j)/N]$ for $k = 1, \ldots, N$.

The MATLAB implementation generates LHS by permuting the strata independently for each parameter and adding uniform jitter within each stratum:

```
%% From: src/globalsa/lhs_sample.m (see GitHub repository)
rng(42);                                   %% set seed before sampling
X = lhs_sample(N, m, lb, ub);             %% N-by-m sample matrix
%% Uses lhsdesign (Statistics Toolbox) with maximin criterion
```

For the same N, LHS provides substantially better coverage than random sampling, particularly for estimating marginal effects of individual parameters; see Helton and Davis [14] for convergence analysis and uncertainty propagation results. For Sobol index estimation (Section 9.4), LHS is not used directly; instead, the Saltelli quasi-random construction is preferred.

Self-check. For $N = 4$ samples and $m = 1$ parameter on $[0, 1]$: (a) Draw a random sample of size 4. Is it possible that all 4 points fall in $[0, 0.5]$? Is this possible with LHS? (b) Write out explicitly what an LHS of size 4 looks like for one parameter.

> *Answer:* (a) Yes, random sampling can place all 4 points in $[0, 0.5]$ (probability $(1/2)^4 = 1/16$). With LHS, this is impossible: each quarter-interval $[0, 0.25)$, $[0.25, 0.5)$, $[0.5, 0.75)$, $[0.75, 1]$ must contain exactly one sample.
> (b) One possible LHS: one point in each of $[0, 0.25)$, $[0.25, 0.5)$, $[0.5, 0.75)$, $[0.75, 1]$, with uniform jitter within each. Example: $X = \{0.18, 0.43, 0.62, 0.91\}$, one value per stratum, randomly positioned within each.

9.3 Regression-Based SA: LHS/PRCC

The Partial Rank Correlation Coefficient (PRCC) is the dominant global SA method in mathematical epidemiology. It is computationally cheap, handles monotone (but not necessarily linear) relationships, and the Marino et al. [24] MATLAB implementation is widely used and freely available; it builds on the rank-correlation approach of Iman and Conover [17] and the epidemiological SA tradition of Blower and Dowlatabadi [5].

9.3.1 The PRCC Calculation

Given N LHS samples of the m POIs and the corresponding N model outputs:

Step 1. Rank-transform all inputs and the output separately. Replace each value with its rank among the N observations.

Step 2. Fit a linear regression of the rank-transformed output on all rank-transformed inputs.

Step 3. For each parameter p_j, compute the partial correlation coefficient between the rank-transformed p_j and the rank-transformed output, controlling for all other parameters. This is the PRCC for p_j.

Step 4. Compute a t-test p-value for each PRCC to assess statistical significance.

The PRCC ranges from -1 to $+1$. Its sign has the same interpretation as the SI sign rule from Chapter 2: positive means POI and QOI increase together; negative means they move in opposite directions.

9.3.2 When PRCC is Appropriate

PRCC requires monotone relationships: the QOI must consistently increase (or consistently decrease) as each POI increases, across the full range of parameter values. It does not require linearity. For most compartmental epidemic models, the relationships are approximately monotone, which is why PRCC dominates the field.

Before applying PRCC, verify the monotonicity assumption by producing a scatter matrix: plot the model output against each POI for the full LHS sample using MATLAB's `plotmatrix` function. A roughly monotone cloud (upward or downward sloping, without reversals) supports the use of PRCC. A hump-shaped or U-shaped cloud signals non-monotonicity, and Sobol indices should be used instead.

PRCC and Sobol indices are not equivalent. PRCC assumes monotone relationships; Sobol indices make no such assumption. When a relationship is nonmonotone (a QOI first increases then decreases as a parameter increases), PRCC may give a near-zero coefficient even for a highly important parameter, while the Sobol total index will correctly identify it as important. For the models in this course, both methods typically agree, but verifying agreement is a useful consistency check.

9.4 Sobol Decomposition and Variance-Based SA

Variance-based SA is the most rigorous framework for global SA: it quantifies exactly how much of the total output variance is attributable to each POI or group of POIs [43].

9.4.1 The Intuition: Conditional Expectation

Suppose parameter Q_1 is fixed at q_1 while all other parameters vary across their ranges. The expected output $\mathbb{E}[Y \mid Q_1 = q_1]$ is some function of q_1. If Q_1 is important, this conditional expectation varies substantially as q_1 changes. The variance of this conditional expectation, $\mathrm{Var}_{Q_1}[\mathbb{E}_{Q_{-1}}(Y \mid Q_1)]$, measures how much of Y's variability is explained by Q_1 alone. Normalizing by the total variance gives the first-order Sobol index.

9.4.2 The ANOVA Decomposition

Why does a unique decomposition exist? Because statistical independence allows us to cleanly separate what each parameter contributes alone from what it contributes jointly. When the Q_i are independent, projecting f onto the subspace of functions of Q_i alone gives a well-defined component $f_i(Q_i)$, and these projections do not interfere with each other. The orthogonality $\int_0^1 f_i(Q_i)\, dQ_i = 0$ is the mathematical expression of this separation: each component has zero mean after integrating out Q_i, so distinct components contribute zero covariance. When parameters are correlated, this orthogonality breaks down, and the decomposition loses its uniqueness (Section 9.7 returns to this point). With independence, the decomposition is both unique and additive in variance.

For independent POIs $\vec{Q} = (Q_1, \ldots, Q_m)$ uniformly distributed on $[0,1]^m$, any square-integrable model $Y = f(\vec{Q})$ admits the unique **ANOVA decomposition**:

$$f(\vec{Q}) = f_0 + \sum_i f_i(Q_i) + \sum_{i<j} f_{ij}(Q_i, Q_j) + \cdots + f_{12\cdots m}(\vec{Q}), \tag{9.2}$$

where $f_0 = \mathbb{E}[Y]$, $f_i(Q_i) = \mathbb{E}[Y \mid Q_i] - f_0$, and the higher-order terms capture interactions. Uniqueness follows from the orthogonality conditions $\int_0^1 f_u(\vec{Q}_u)\, dQ_i = 0$ for each $i \in u$; see Sobol [43] Theorem 1 and Saltelli et al. [38] Appendix A for a proof. The terms are orthogonal, so their variances add:

$$D = \mathrm{Var}(Y) = \sum_i D_i + \sum_{i<j} D_{ij} + \cdots + D_{12\cdots m}, \tag{9.3}$$

where $D_i = \mathrm{Var}[\mathbb{E}(Y \mid Q_i)]$ and $D_{ij} = \mathrm{Var}[\mathbb{E}(Y \mid Q_i, Q_j)] - D_i - D_j$.

Worked example: a model with an interaction term. Consider $f(Q_1, Q_2) = Q_1 + Q_2 + Q_1 Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0,1)$ independent. Compute the ANOVA components directly from the definitions:

$$\begin{aligned}
f_0 &= \mathbb{E}[Y] = \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{5}{4},\\
f_1(Q_1) &= \mathbb{E}[Y \mid Q_1] - f_0 = \left(Q_1 + \tfrac{1}{2} + \tfrac{Q_1}{2}\right) - \tfrac{5}{4} = \tfrac{3Q_1}{2} - \tfrac{3}{4} = \tfrac{3}{2}\left(Q_1 - \tfrac{1}{2}\right),\\
f_2(Q_2) &= \mathbb{E}[Y \mid Q_2] - f_0 = \tfrac{3}{2}\left(Q_2 - \tfrac{1}{2}\right) \quad \text{(by symmetry)},\\
f_{12}(Q_1, Q_2) &= f - f_0 - f_1 - f_2 = Q_1 Q_2 - \tfrac{Q_1}{2} - \tfrac{Q_2}{2} + \tfrac{1}{4} = \left(Q_1 - \tfrac{1}{2}\right)\left(Q_2 - \tfrac{1}{2}\right).
\end{aligned}$$

Verify orthogonality: $\int_0^1 f_{12}\, dQ_1 = \left(\int_0^1 (Q_1 - \tfrac{1}{2})\, dQ_1\right)(Q_2 - \tfrac{1}{2}) = 0.$ ✓

The variances are $D_1 = D_2 = \frac{3}{4} \cdot \mathrm{Var}(Q_1) = \frac{3}{4} \cdot \frac{1}{12} = \frac{1}{16}$ and $D_{12} = \mathrm{Var}(f_{12}) = \mathrm{Var}(Q_1 - \frac{1}{2})\,\mathrm{Var}(Q_2 - \frac{1}{2}) = \frac{1}{144}$. Total variance $D = \frac{1}{16} + \frac{1}{16} + \frac{1}{144} = \frac{19}{144}$. The first-order indices $S_1 = S_2 = \frac{1/16}{19/144} = \frac{9}{19} \approx 0.47$, and the interaction index $S_{12} = \frac{1/144}{19/144} = \frac{1}{19} \approx 0.053$. Note $S_1 + S_2 + S_{12} = 1$. For the additive model $f = Q_1 + Q_2$ (no interaction), $S_1 = S_2 = \frac{1}{2}$ and $S_{12} = 0$: the indices sum to 1 with nothing left over for interactions. This is why $\sum S_i < 1$ signals the presence of interactions.

9.4.3 Sobol Indices

Definition 9.2 (Sobol Indices)**.** The **first-order Sobol index** for parameter i is:

$$S_i = \frac{D_i}{D} = \frac{\mathrm{Var}[\mathbb{E}(Y \mid Q_i)]}{\mathrm{Var}(Y)}. \tag{9.4}$$

The **total Sobol index** for parameter i [15] is:

$$S_{T_i} = 1 - \frac{\mathrm{Var}[\mathbb{E}(Y \mid \vec{Q}_{-i})]}{\mathrm{Var}(Y)} = \frac{\mathbb{E}[\mathrm{Var}(Y \mid \vec{Q}_{-i})]}{\mathrm{Var}(Y)}, \tag{9.5}$$

where $\vec{Q}_{-i}$ denotes all parameters except Q_i.

The first-order index measures the *direct* effect of Q_i; the total index measures the direct effect plus all interactions with other parameters. When $S_{T_i} = S_i$ for all i, the model is additively separable (no interactions). When $S_{T_i} > S_i$, parameter i interacts with other parameters.

Unifying Idea 1. The first-order Sobol index $S_i = D_i/D$ uses the same ratio structure as the sensitivity index from Chapter 2; it measures one variance as a fraction of another. The connection is explicit: for a linear model $Y = \sum_i a_i Q_i$ with independent $Q_i \sim \mathcal{U}[l_i, u_i]$, the first-order Sobol index equals the square of the sigma-normalized index: $S_i = (S_i^\sigma)^2$. Global SA generalizes local SA; for linear models, they agree exactly.

Unifying Idea 3. The Sobol decomposition resolves the tension that Unifying Idea 3 created in Chapter 8. Local SA is a linearization, valid near the nominal values. Sobol indices make no linearity assumption: they measure the actual variance contribution across the full parameter range. When the linearization is poor (Model B from the chapter opening), local and global SA give different answers, and Sobol indices give the correct answer.

For readers with a functional analysis background (Appendix A), the ANOVA decomposition (9.2) is the orthogonal decomposition of f in the Hilbert space $L^2([0,1]^m)$ with respect to the inner product $\langle f, g\rangle = \int_{[0,1]^m} f(\vec{q})\, g(\vec{q})\, d\vec{q}$. The orthogonality explains both why the variances add in (9.3) and why the method requires independent POIs: independence is needed for the inner product to factorize and the projections to be orthogonal.

Self-check. For $Y = c_1Q_1 + c_2Q_2$ with $Q_i \sim \mathcal{U}(0,1)$ independent: (a) Compute $\mathbb{E}[Y \mid Q_1]$ and $\mathrm{Var}[\mathbb{E}(Y \mid Q_1)]$. (b) Compute $\mathrm{Var}(Y)$ using the independence of Q_1 and Q_2. (c) Show that $S_1 = c_1^2/(c_1^2+c_2^2)$ and $S_2 = c_2^2/(c_1^2+c_2^2)$. (d) Verify $S_1 + S_2 = 1$ (no interaction for additive models).

Answer: (a) $\mathbb{E}[Y \mid Q_1] = c_1Q_1 + c_2/2$. $\mathrm{Var}[\mathbb{E}(Y \mid Q_1)] = c_1^2\,\mathrm{Var}(Q_1) = c_1^2/12$.
(b) $\mathrm{Var}(Y) = c_1^2\,\mathrm{Var}(Q_1) + c_2^2\,\mathrm{Var}(Q_2) = (c_1^2+c_2^2)/12$.
(c) $S_1 = D_1/D = (c_1^2/12)/((c_1^2+c_2^2)/12) = c_1^2/(c_1^2+c_2^2)$. Similarly S_2.
(d) $S_1 + S_2 = (c_1^2+c_2^2)/(c_1^2+c_2^2) = 1$. For linear additive models, there are no interaction terms and first-order indices sum to 1.

9.5 Computing Sobol Indices: The Saltelli Algorithm

A naive Monte Carlo estimate of S_i requires choosing $Q_i = q_i$ for many values and averaging the model over all other parameters for each, this costs $O(N^2)$ evaluations. The insight of Saltelli [33] is that all first-order and total Sobol indices can be estimated with $N(m+2)$ evaluations using two carefully constructed sample matrices. The design and estimators for the total sensitivity index are developed in Saltelli et al. [35].

9.5.1 The Saltelli Construction

Generate two independent LHS matrices A and B, each $N \times m$. Define the m matrices $A_B^{(i)}$: matrix A with its i-th column replaced by the i-th column of B. The model is evaluated at all rows of A, B, and all m matrices $A_B^{(i)}$: a total of $N(m+2)$ evaluations.

9.5.2 The Jansen Estimators

For the first-order index, use the Jansen [18] estimator:

$$\hat{S}_i = 1 - \frac{\frac{1}{2N}\sum_{j=1}^{N}[f(B_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}, \tag{9.6}$$

where $\hat{D} = \frac{1}{N}\sum_j f(A_j)^2 - \left(\frac{1}{N}\sum_j f(A_j)\right)^2$.

For the total index:

$$\hat{S}_{T_i} = \frac{\frac{1}{2N}\sum_{j=1}^{N}[f(A_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}. \tag{9.7}$$

Use these Jansen estimators, not the original Saltelli (2002) formulas, which have higher variance. The Jansen estimator is consistent (converges to the true S_i as $N \to \infty$) and nearly unbiased; see Jansen [18] Theorem 1. The key step is that

$$\mathbb{E}\left[\frac{1}{2N}\sum_{j=1}^{N}(f(B_j) - f(A_{B,j}^{(i)}))^2\right] = D - D_i = \mathrm{Var}(Y) - \mathrm{Var}[\mathbb{E}(Y \mid Q_i)],$$

which follows because B_j and $A_{B,j}^{(i)}$ share all parameter values except column i (from A), so their difference isolates the Q_i-dependent variation. Dividing by $\hat{D}$ and rearranging gives $1 - \hat{S}_i \approx (D - D_i)/D = 1 - S_i$. The estimator is nearly unbiased at finite N because the bias is $O(1/N)$ [18], negligible for $N \geq 1000$.

9.5.3 Convergence

The estimators converge as $O(1/\sqrt{N})$, the standard Monte Carlo rate. In practice, $N \geq 1000$ is needed for reliable estimates. For $m = 4$ parameters (SIR model): $N(m + 2) = 6000$ evaluations at $N = 1000$. If each evaluation takes 0.1 seconds: 10 minutes.

Plotting $\hat{S}_i$ and $\hat{S}_{T_i}$ vs. N reveals the convergence: estimates are noisy for $N < 500$ and stable for $N \geq 2000$. This convergence plot is the diagnostic that answers the misconception “more samples always help”: below a threshold N, additional samples do not produce reliable estimates because the variance of the estimator exceeds the signal being measured.

Confidence intervals via bootstrap. A point estimate $\hat{S}_i$ is useful but incomplete: it gives no information about sampling uncertainty. The standard approach is the **bootstrap**: resample the $N(m + 2)$ model evaluations with replacement, recompute $\hat{S}_i$ for each resample, and take the 2.5th and 97.5th percentiles of the resulting distribution as a 95% confidence interval. In MATLAB:

```
B = 500;           %% number of bootstrap replicates
S1_boot = zeros(B, m);
idx_all = 1:N;
for b = 1:B
    idx = idx_all(randi(N, N, 1));     %% resample rows
    A_b = A(idx,:);  B_b = Bmat(idx,:);
    [S1_boot(b,:), ~] = sobol_jansen(@(x) model_eval(x), A_b, B_b);
end
S1_CI = prctile(S1_boot, [2.5 97.5]);   %% 95% CI for each index
```

A confidence interval that straddles zero indicates that the parameter’s effect cannot be distinguished from noise at the current sample size. A confidence interval that excludes zero confirms the parameter is genuinely influential. As a practical rule: if the confidence interval width exceeds the index value, double N before drawing conclusions. When m is large, however, even a single Sobol run at $N = 2000$ may be prohibitively expensive; Morris screening addresses exactly this case.

9.6 Morris Screening

When m is large (30 or more parameters), running Sobol at $N = 2000$ costs $N(m + 2) = 64{,}000$ model evaluations, potentially prohibitive. Morris screening [27] provides a cheap first step: $N(m+1)$ evaluations (typically $N = 10$ trajectories) to identify which parameters are truly unimportant. An improved radial design that gives more uniform coverage with fewer trajectories is described by Campolongo et al. [7].

For each trajectory, one parameter is changed at a time by a fixed fraction Δ of its range. The **elementary effect** of parameter i at point $\vec{q}$ is:

$$EE_i(\vec{q}) = \frac{f(\vec{q} + \Delta \vec{e}_i) - f(\vec{q})}{\Delta}. \tag{9.8}$$

After r trajectories, the mean $\mu_i^* = (1/r) \sum |EE_i|$ and standard deviation σ_i of the elementary effects are computed. Parameters with small μ_i^* are unimportant. Parameters with large σ_i have interactions or nonlinear effects.

The Morris diagnostic plot (σ_i vs. μ_i^*) separates parameters into three groups: unimportant (small μ_i^*, small σ_i); important and linear (large μ_i^*, small σ_i); important and nonlinear or interacting (large μ_i^*, large σ_i).

Workflow: run Morris screening first; fix unimportant parameters at their nominal values; run Sobol only on the important subset. This can reduce the Sobol cost by a factor of 5–10.

When even reduced Sobol is too expensive: surrogate models. If the model is computationally expensive and Morris screening reduces the parameter set to, say, five influential POIs, running Sobol at $N = 2000$ still requires $N(m + 2) = 14{,}000$ model evaluations. For a climate model where each run takes hours, this remains prohibitive.

The practical solution is a **surrogate model** (also called an emulator or meta-model): a cheap statistical approximation to the expensive model, trained on a moderate number of carefully chosen model runs. Concretely, a Gaussian process surrogate fits a probability distribution over functions to the training runs, predicting the model output at any new input along with a confidence interval; a polynomial chaos expansion instead represents the output as a truncated series in orthogonal polynomials of the uncertain inputs, with coefficients estimated from the training set. Gaussian process emulators and polynomial chaos expansions are the most widely used surrogates. Once trained on roughly 50–200 model evaluations, a surrogate can be evaluated millions of times in seconds, making full Sobol estimation feasible even for computationally expensive simulations.

Surrogate-based global SA does not replace the methods in this chapter; it wraps them. The LHS sampling, the Saltelli construction, and the Jansen estimators all apply to the surrogate exactly as they apply to the original model. The surrogate simply makes the evaluations fast enough to converge. Smith [42] develops Gaussian process and polynomial chaos surrogates in Chapters 12–13, building directly on the LHS samples and Sobol indices from this chapter. Students proceeding to the second semester will construct surrogates using exactly the training sets generated in this chapter's laboratories.

9.7 Sobol Indices for Correlated POIs

Standard Sobol decomposition assumes independent POIs. When parameters are correlated, three problems arise: first-order indices can exceed 1 or be negative (a diagnostic flag that correlation is present); total indices may not sum to 1; and the ANOVA decomposition (9.2) is no longer unique, because correlation destroys the orthogonality that guarantees uniqueness.

Three approaches address correlated POIs in global SA:

Grouped indices: treat a cluster of correlated parameters as a single group. Compute the group Sobol index (the fraction of variance explained by the entire group jointly). This avoids allocating variance between correlated parameters, which is not well-defined.

Modified estimators: Kucherenko et al. [20] extend the Saltelli algorithm to correlated inputs using copula-based sampling. This requires knowledge of the joint distribution of the correlated parameters.

PRCC: handles correlation naturally through the "partial" in partial rank correlation. When parameters are correlated, PRCC is the most robust and accessible global SA method.

Practical recommendation: for models with correlated POIs, use PRCC as the primary global SA tool and grouped Sobol indices as a secondary check. Report the correlation structure explicitly alongside the SA results.

9.8 Comparison of All SA Methods

Table 9.1 consolidates all SA methods from the book. It supersedes the partial decision tables in earlier chapters. For a broader treatment of global SA methods, see Saltelli et al. [38, 36] and the recent epidemiological applications in Lu and Borgonovo [22].

Method	Chapter	When to use	Assumption	Cost
Local SI (analytic)	2, 4	Differentiable model; small uncertainty	Linearization valid	Cheap
Local SI (FD)	3	Black-box; small uncertainty	Linearization valid	$2mn_q$ evals
Adjoint	5–6	Differentiable; many POIs; few QOIs	Linearization valid	2 solves
AD (reverse)	7	Differentiable code; $m \gg r$	Linearization valid	2 passes
PRCC	9	Monotone model; moderate uncertainty; corr. POIs OK	Monotone	N evals
Morris	9	Screening; $m \geq 30$	None	$N(m+1)$ evals
Sobol	9	Nonlinear; large uncertainty; independent POIs	Independent	$N(m+2)$ evals
Grouped Sobol	9	Correlated POIs; known groups	Known groups	$N(m+2)$ evals

Table 9.1: Complete SA method comparison across all chapters. The choice of method depends primarily on whether the model is differentiable, whether uncertainty is small or large, and whether POIs are independent. No single method is universally best.

Unifying Idea 2. The adjoint row in Table 9.1 is the payoff for Chapters 5 and 6. The adjoint is the transpose in whatever space you are working in: matrix transpose for linear systems (Chapter 5), reversed ODE for dynamical systems (Chapter 6), and reversed chain rule for computation graphs (Chapter 7). Every "2 solves" entry in the adjoint row is the same unifying idea, applied in a different mathematical space.

9.9 Bridge to Semester 2

This book quantifies how model output uncertainty arises from parameter uncertainty, assuming the model structure is correct. The second semester, covering Smith's *Uncertainty Quantification* [42], asks harder questions: what if the model is wrong? How do we fit models to data? How do we construct efficient surrogate models to replace expensive simulations? Smith's Chapters 7–8 develop Bayesian parameter estimation and Markov chain Monte Carlo methods for fitting models to data, the natural sequel to the identifiability analysis introduced in Chapter 3 of this book.

The connections between the two semesters are direct:

Smith Ch. 14 develops local SA at the graduate level, extending Chapters 2–7 of this book with functional-analytic rigor.

Smith Ch. 15 develops global SA and Sobol indices at the graduate level, extending Chapter 9 of this book with measure-theoretic foundations.

Smith Ch. 6 treats parameter selection using the sensitivity matrix; the core tool introduced in Chapter 4 of this book.

Smith Chs. 12–13 develop model discrepancy and Gaussian process surrogates, methods that reduce the cost of global SA by replacing the expensive model with a cheap approximation. Surrogates are built from the same LHS samples used to estimate Sobol indices in this chapter.

Students who have completed this book have the foundational vocabulary, the computational intuition, and the mathematical framework to engage productively with the full UQ literature. The path from the sensitivity index in Chapter 2 to the full Bayesian UQ framework in Smith is a single, continuous intellectual development. As an example of these methods applied to a realistic multi-parameter epidemic model, see Qu, Xue, and Hyman [31], which uses SA to guide Wolbachia-based mosquito-borne disease control.

The student who has reached this point knows how to ask which parameters matter, how to compute that answer efficiently and honestly, and, just as importantly, when not to trust the answer. Every mathematical model is an argument; sensitivity analysis is the discipline that reveals which assumptions the argument depends on.

Good sensitivity analysis does not end the discussion. It tells you where to begin it.

9.10 MATLAB Laboratory: Complete Global SA Pipeline for SIR

This laboratory implements the full global SA pipeline for the SIR model and compares all four methods: sigma-normalized, PRCC, Sobol, and Morris.

Setup. SIR model with four POIs (k, β, τ, L) and response $J = I_{\max}$ (peak infectious count). Use the literature-based uncertainty ranges from Chitnis et al. [10]: $k \in [2, 8]$, $\beta \in [0.02, 0.12]$, $\tau \in [4, 14]$ days, $L \in [1000, 100000]$ days.

Part 1: LHS and sigma-normalized indices. Draw $N = 500$ LHS samples using the `lhs_sample` function. Compute σ_J from the samples. Compute the sigma-normalized index for each POI using the local SI from Chapter 2 and the sample standard deviations. Report the ranking.

Part 2: PRCC. Run the SIR model at all 500 LHS samples. Rank-transform all inputs and the output. Compute PRCC using the `partialcorr` function in MATLAB's Statistics and Machine Learning Toolbox (or download Marino et al.'s implementation from the supplementary material of [24]). Produce a PRCC tornado plot and report p-values.

Part 3: Sobol estimation. Draw $N = 2000$ samples using the quasi-random Halton or Sobol sequence (MATLAB: `haltonset` or `sobolset`). Build the A, B, and m matrices $A_B^{(i)}$: $2000 \times 6 =$ 12,000 total evaluations. Compute $\hat{S}_i$ and $\hat{S}_{T_i}$ using the Jansen estimators.

```
%% From: src/globalsa/sobol_jansen.m (see GitHub repository)
rng(42);   %% reproducibility
[A, B]      = saltelli_sample(N, m, lb, ub);
[S1, ST, CI_S1] = sobol_jansen(@model_eval, A, B, 500, 0.05);
%% CI_S1(:,j) = [lower; upper] 95% confidence interval for S1(j)
```

Part 4: Morris screening. Implement $r = 10$ Morris trajectories with $\Delta = 0.4$. Compute μ_i^* and σ_i for each parameter. Produce the Morris diagnostic plot (σ_i vs. μ_i^*). Which parameter(s) could be fixed without significant loss of accuracy?

Part 5: Convergence check. Plot $\hat{S}_\beta$ and $\hat{S}_{T_\beta}$ vs. N for $N \in \{100, 200, 500, 1000, 2000, 5000\}$. At what N do the estimates stabilize to within ± 0.05?

Part 6: Full comparison table. Fill in the following table from your results:

Parameter	Local SI	S_i^σ	PRCC	S_i (Sobol)
k				
β				
τ				
L				

Where do the four methods agree? Where do they disagree? What accounts for the differences?

9.11 Exercises

Exercise 9.1 *(*[Bloom L3]*).* For the SEIR model with local SIs $S_{\tau_E}^{R_0} = 0.6$, $S_{\tau_I}^{R_0} = 1.2$, $S_\beta^{R_0} = 1.0$, $S_k^{R_0} = 1.0$ and uncertainty ranges $\sigma_{\tau_E} = 2$ days, $\sigma_{\tau_I} = 3$ days, $\sigma_\beta = 0.01$, $\sigma_k = 0.8$:

(a) Compute the output standard deviation σ_{R_0} using the delta method, at nominal $R_0^* = 2.0$.

(b) Compute all four sigma-normalized indices.

(c) Produce a tornado plot of the sigma-normalized indices.

(d) Compare the ranking with the local SI ranking. Which parameter changed rank most dramatically? Why?

Exercise 9.2 *(*[Bloom L3]*).*

(a) Implement the `lhs_sample` function. For $N = 100$, $m = 3$, $[0, 1]^3$, generate an LHS and a random sample of the same size.

(b) Produce scatter plots of all three pairs of parameters for both samples (6 plots total).

(c) Compute the mean and standard deviation of the sample along each dimension. Which method is closer to the true uniform mean and standard deviation?

(d) For one parameter, plot the histogram of LHS vs. random sampling. Which covers the range more evenly?

Exercise 9.3 *(*[Bloom L3]*)*. For the SIR model with parameter ranges from the laboratory setup:

(a) Generate $N = 200$ LHS samples and run the SIR model at each.

(b) Compute PRCC for all four parameters with respect to I_{max}.

(c) Produce a PRCC bar chart sorted by |PRCC|.

(d) Which parameter has the strongest negative PRCC? What does this mean in plain language?

(e) Compare the PRCC ranking with the local SI tornado plot from Chapter 2. Do they agree?

Exercise 9.4 *(*[Bloom L4]*)*. For $Y = 2Q_1^2 + Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0, 1)$, independent:

(a) Compute $\mathbb{E}[Y]$, $\text{Var}(Y)$, $D_1 = \text{Var}[\mathbb{E}(Y \mid Q_1)]$, $D_2 = \text{Var}[\mathbb{E}(Y \mid Q_2)]$.

(b) Compute S_1, S_2. Do they sum to 1? Explain.

(c) Compute the total indices S_{T_1}, S_{T_2}.

(d) Now add Q_1 and Q_2 to get $Y' = 2Q_1^2 + Q_2 + Q_1Q_2$ (an interaction term). Without computing: does adding the interaction term change S_1, S_2, or both? In which direction? Then verify by computing the Sobol indices for Y' analytically.

Exercise 9.5 *(*[Bloom L4]*)*. Using the Jansen estimator implementation from the laboratory:

(a) Compute $\hat{S}_i$ and $\hat{S}_{T_i}$ for all four SIR parameters at $N = 2000$.

(b) For which parameters is $S_{T_i} \gg S_i$? What does this indicate about their role in the model?

(c) Compute $\sum_i S_i$ and $\sum_i S_{T_i}$. What do you expect these to sum to for an additive model? Are your results consistent with this expectation?

(d) If you could measure exactly one parameter very precisely (reducing its uncertainty to zero), which parameter would reduce $\text{Var}(I_{\text{max}})$ the most? Justify using the Sobol indices.

Exercise 9.6 *(*[Bloom L4]*)*. For the SIR model, implement $r = 15$ Morris trajectories with $\Delta = 1/3$:

(a) Compute μ_i^* and σ_i for each parameter.

(b) Produce the Morris diagnostic plot.

(c) Based on the Morris results, which parameter appears most influential? Which appears least influential?

(d) Run a reduced Sobol analysis using only the three most important parameters identified by Morris (fixing L at its nominal value). Compare the cost and results with the full four-parameter Sobol.

Exercise 9.7 *(*[Bloom L5]*).* Choose a published epidemic model from the literature with at least 8 parameters. Design a complete global SA study:

(a) State the model, its parameters, and a justification for the uncertainty ranges you will use (with citations).

(b) State the QOI or QOIs and explain why they are the most relevant for the model's purpose.

(c) Choose between PRCC and Sobol as the primary global SA method. Justify your choice based on the model's properties.

(d) Specify the sample size N and justify it based on convergence.

(e) Describe how you would present the results to a non-technical audience (e.g., a public health committee or funding agency).

(f) Identify one situation where your global SA results could still be misleading, and describe what additional analysis you would recommend.

Exercise 9.8 *(*[Bloom L6]*).* A published SA study reports Sobol first-order and total indices for a 15-parameter water-quality model. The authors note in the data section that three parameters, phosphorus loading rate, nitrogen loading rate, and water residence time, are positively correlated across the sampled watersheds. The SA section reports standard Sobol indices computed using independent sampling for all 15 parameters, without addressing the correlation.

(a) Identify the specific mathematical problem that correlation creates for standard Sobol indices.

(b) For each of the three correlated parameters, state whether the reported first-order index is likely over-estimated, under-estimated, or unchanged, and explain why.

(c) Propose the minimal methodological correction that would make the analysis valid.

(d) The authors justify their approach by noting that their Sobol indices sum to approximately 1.0. Explain why this is not a valid check for independence.

Appendices

Appendix A

Inner Products, Norms, and Function Spaces

This appendix provides the minimum background in functional analysis needed to read Appendix B and Smith's companion volume [42]. Readers who are comfortable with inner products and the L^2 space can proceed directly to Appendix B.

A.1 Inner Products and Norms

An **inner product** on a vector space V over $\mathbb{R}$ is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

1. *Symmetry*: $\langle u, v \rangle = \langle v, u \rangle$.
2. *Linearity in the first argument*: $\langle \alpha u + \beta w, v \rangle = \alpha \langle u, v \rangle + \beta \langle w, v \rangle$.
3. *Positive definiteness*: $\langle u, u \rangle \geq 0$, with equality only when $u = 0$.

The **induced norm** is $\|u\| = \sqrt{\langle u, u \rangle}$. The Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\| \, \|v\|$.

Example A.1 (Euclidean inner product). For $\vec{u}, \vec{v} \in \mathbb{R}^n$:

$$\langle \vec{u}, \vec{v} \rangle = \vec{u}^T \vec{v} = \sum_{i=1}^{n} u_i v_i.$$

This is the dot product. The induced norm is the Euclidean norm $\|\vec{u}\|_2 = \sqrt{\sum u_i^2}$.

Example A.2 (Weighted inner product). For a symmetric positive definite matrix $W \in \mathbb{R}^{n \times n}$: $\langle \vec{u}, \vec{v} \rangle_W = \vec{u}^T W \vec{v}$. The induced norm $\|\vec{u}\|_W = \sqrt{\vec{u}^T W \vec{u}}$ is the W-weighted Euclidean norm. SA computations sometimes use this when parameters have very different scales.

A.2 The L^2 Inner Product

For functions $f, g : [a, b] \to \mathbb{R}$ that are square-integrable (i.e., $\int_a^b f(t)^2 \, dt < \infty$), the L^2 **inner product** is:

$$\langle f, g \rangle_{L^2} = \int_a^b f(t) \, g(t) \, dt. \tag{A.1}$$

The induced norm is $\|f\|_{L^2} = \sqrt{\int_a^b f(t)^2 \, dt}$.

The space $L^2([a, b])$ of all square-integrable functions on $[a, b]$ with inner product (A.1) is a **Hilbert space**: a complete inner product space. Completeness means that every Cauchy sequence converges to a function in $L^2([a, b])$.

For Sobol decomposition (Chapter 9), the relevant space is $L^2([0,1]^m)$ with inner product $\langle f, g\rangle = \int_{[0,1]^m} f(\vec{q})g(\vec{q})\,d\vec{q}$.

A.3 Integration by Parts as Transposition

The key connection between adjoint sensitivity analysis and linear algebra is this: integration by parts is the function-space analog of the matrix transpose.

In $\mathbb{R}^n$, for vectors $\vec{u}$, $\vec{v}$ and a matrix A:

$$\langle A\vec{u}, \vec{v}\rangle = (A\vec{u})^T\vec{v} = \vec{u}^T(A^T\vec{v}) = \langle \vec{u}, A^T\vec{v}\rangle.$$

So A^T is the operator satisfying $\langle Au, v\rangle = \langle u, A^T v\rangle$.

In $L^2([0,T])$, for functions u, v and the differentiation operator $L = d/dt$:

$$\langle Lu, v\rangle_{L^2} = \int_0^T \dot{u}(t)\,v(t)\,dt = \big[u(t)v(t)\big]_0^T - \int_0^T u(t)\,\dot{v}(t)\,dt = \big[uv\big]_0^T + \langle u, L^* v\rangle_{L^2},$$

where $L^* = -d/dt$ (with appropriate boundary conditions) is the **adjoint operator** of $L = d/dt$. The boundary term $[uv]_0^T$ gives rise to the terminal condition $\lambda(T) = -h'(u(T))$ in the adjoint ODE derivation of Chapter 6.

This correspondence makes precise what we stated informally throughout the book: the adjoint ODE is the ODE whose differential operator is the transpose of the forward ODE's operator, exactly as A^T is the transpose of A.

A.4 Orthogonality and the ANOVA Decomposition

Two elements u, v of an inner product space are **orthogonal** if $\langle u, v\rangle = 0$.

In $L^2([0,1]^m)$ with the uniform measure, the functions appearing in the Sobol ANOVA decomposition (Chapter 9) are mutually orthogonal. Specifically, for the decomposition $f = f_0 + \sum_i f_i + \sum_{i<j} f_{ij} + \cdots$, each term is orthogonal to every other term: $\langle f_u, f_v\rangle_{L^2} = 0$ when $u \neq v$ (as index sets).

This orthogonality is why the output variance decomposes as a sum: $\mathrm{Var}(Y) = \sum_i D_i + \sum_{i<j} D_{ij} + \cdots$. It also explains why the decomposition requires independent parameters: independence of the Q_i makes the integration over $[0,1]^m$ factorize, which is what produces the orthogonality.

A.5 Exercises

Self-check. Let $f(t) = e^{-t}$ and $g(t) = \cos(\pi t)$ on $[0,1]$. (a) Compute $\langle f, g\rangle_{L^2}$ numerically (use MATLAB's `integral` function). (b) Are f and g orthogonal in $L^2([0,1])$? (c) Compute $\|f\|_{L^2}$ and $\|g\|_{L^2}$, and verify the Cauchy–Schwarz inequality $|\langle f, g\rangle| \leq \|f\|\,\|g\|$.

Answer: (a) $\langle f, g\rangle_{L^2} = \int_0^1 e^{-t}\cos(\pi t)\,dt$. Integrating by parts twice: result ≈ 0.218 (verify numerically). (b) No — the inner product is nonzero, so f and g are not orthogonal. (c) $\|f\|_{L^2}^2 = \int_0^1 e^{-2t}\,dt = (1-e^{-2})/2 \approx 0.432$, so $\|f\|_{L^2} \approx 0.657$. $\|g\|_{L^2}^2 = \int_0^1 \cos^2(\pi t)\,dt = 1/2$, so $\|g\|_{L^2} = 1/\sqrt{2} \approx 0.707$. Cauchy–Schwarz: $|0.218| \leq 0.657 \times 0.707 \approx 0.464$. ✓

Self-check. The integration-by-parts identity in Section A.3 establishes that $L^* = -d/dt$ (with zero terminal condition) is the adjoint of $L = d/dt$ (with zero initial condition) in $L^2([0,T])$. (a) Write out $\langle Lu, v\rangle_{L^2}$ explicitly for $u(t) = t^2$ and $v(t) = t$ on $[0,1]$. (b) Compute $\langle u, L^*v\rangle_{L^2}$ for the same functions. (c) Verify that the two results differ by the boundary term $[u(t)v(t)]_0^1$.

Answer: (a) $Lu = 2t$. $\langle Lu, v\rangle = \int_0^1 2t \cdot t\, dt = \int_0^1 2t^2\, dt = 2/3$. (b) $L^*v = -dv/dt = -1$. $\langle u, L^*v\rangle = \int_0^1 t^2 \cdot (-1)\, dt = -1/3$. (c) Boundary term: $[u(t)v(t)]_0^1 = 1 \cdot 1 - 0 \cdot 0 = 1$. Check: $\langle Lu, v\rangle = \langle u, L^*v\rangle + [uv]_0^1 \Rightarrow 2/3 = -1/3 + 1 = 2/3$. ✓

Self-check. The ANOVA decomposition in $L^2([0,1]^2)$ for the model $f(Q_1, Q_2) = Q_1 + Q_2 + Q_1Q_2$ (with $Q_i \sim \mathcal{U}(0,1)$) is: $f = f_0 + f_1(Q_1) + f_2(Q_2) + f_{12}(Q_1, Q_2)$. Compute each term and verify that all terms are mutually orthogonal in $L^2([0,1]^2)$.

Answer: $f_0 = \mathbb{E}[f] = 1/2 + 1/2 + 1/4 = 5/4$. $f_1(Q_1) = \mathbb{E}[f|Q_1] - f_0 = Q_1 + 1/2 + Q_1/2 - 5/4 = 3Q_1/2 - 3/4$. $f_2(Q_2) = 3Q_2/2 - 3/4$ (by symmetry). $f_{12} = f - f_0 - f_1 - f_2 = Q_1Q_2 - Q_1/2 - Q_2/2 + 1/4 = (Q_1 - 1/2)(Q_2 - 1/2)$. Orthogonality: $\langle f_1, f_2\rangle = \int_0^1\int_0^1 (3Q_1/2 - 3/4)(3Q_2/2 - 3/4)\, dQ_1\, dQ_2 = \left[\int_0^1 (3Q/2 - 3/4)\, dQ\right]^2 = 0^2 = 0$. ✓Similarly all other pairs integrate to zero.

Appendix B

The Adjoint Formalism

This appendix develops the abstract operator framework that unifies the matrix adjoint (Chapter 5) and the ODE adjoint (Chapter 6). It is intended for students who want a rigorous foundation, not just the computational recipe.

B.1 Linear Operators and Their Adjoints

Let H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$. A **linear operator** $L : H \to H$ satisfies $L(\alpha u + \beta v) = \alpha L u + \beta L v$.

The **adjoint operator** $L^\dagger$ is defined by:

$$\langle Lu, v \rangle = \langle u, L^\dagger v \rangle \qquad \text{for all } u, v \in H. \tag{B.1}$$

The adjoint $L^\dagger$ is the unique operator satisfying (B.1) (when it exists).

Example B.1 (Matrix adjoint). In $\mathbb{R}^n$ with the Euclidean inner product, $L = A$ (left multiplication). Then $\langle Au, v \rangle = (Au)^T v = u^T (A^T v) = \langle u, A^T v \rangle$, so $L^\dagger = A^T$. This is the matrix transpose: Chapters 5's adjoint equation $A^T \vec{v} = \vec{c}$ is $L^\dagger \vec{v} = \vec{c}$.

Example B.2 (Differential operator adjoint). In $L^2([0, T])$, let $Lu = du/dt$ with $u(0) = 0$ (zero initial condition). By integration by parts (Appendix A.3):

$$\langle Lu, v \rangle = \int_0^T \dot{u} v \, dt = [uv]_0^T - \int_0^T u \dot{v} \, dt = u(T)v(T) - \int_0^T u \dot{v} \, dt.$$

For the adjoint to satisfy $\langle Lu, v \rangle = \langle u, L^\dagger v \rangle$, we need $L^\dagger v = -\dot{v}$ with $v(T) = 0$ (zero terminal condition). So $L^\dagger = -d/dt$ with the boundary condition shifted from $t = 0$ to $t = T$. This is exactly the structure of the adjoint ODE in Chapter 6.

B.2 The Lagrange Identity

For the ODE operator $L = d/dt - F_u$ (the linearization of $\dot{u} = F(u; p)$), the Lagrange identity states:

$$\frac{d}{dt}\left[\lambda^T w\right] = (L^\dagger \lambda)^T w + \lambda^T (Lw), \tag{B.2}$$

where $w = \partial u / \partial p$ satisfies the FSE $Lw = F_p$ and λ satisfies the adjoint ODE $L^\dagger \lambda = g_u$.

Integrating (B.2) over $[0, T]$ and rearranging gives the sensitivity formula (6.11) of Chapter 6. This is the rigorous version of the "integration by parts" step that was announced but not fully proved in the main text.

B.3 Solvability Conditions

For the equation $Lu = f$ to have a solution when L is not invertible, f must be orthogonal to the kernel of $L^\dagger$:

SC1 (Fredholm Alternative). $Lu = f$ has a solution if and only if $\langle f, v\rangle = 0$ for every v satisfying $L^\dagger v = 0$.

SC2 (Uniqueness). If $L^\dagger v = 0$ has only the trivial solution $v = 0$, then $Lu = f$ has a unique solution for every f.

In the SA context, SC2 corresponds to the assumption that the Jacobian at the equilibrium is nonsingular (Theorem 4.1): the FSE has a unique solution, and the adjoint equation has a unique adjoint variable. Near a bifurcation, SC2 fails because the Jacobian becomes singular, which is precisely why SA breaks down (Chapter 8).

B.4 Connection to Chapters 5 and 6

The following table makes the Chapter 5/6 connection explicit in the language of this appendix:

Concept	Chapter 5 (matrices)	Chapter 6 (ODEs)
Space H	$\mathbb{R}^n$	$L^2([0,T])^n$
Operator L	A (matrix mult.)	$d/dt - J_F$
Adjoint $L^\dagger$	A^T	$-d/dt - J_F^T$
Adjoint equation	$A^T\vec{v} = \vec{c}$	$-\dot{\lambda} = J_F^T\lambda - g_u$
Boundary cond.	none (algebraic)	$\lambda(T) = -\nabla h^T$
Sensitivity formula	$\vec{v}^T\vec{r}_j$	$\int_0^T \lambda^T F_{p_j}\, dt$
Key operation	matrix transpose	integration by parts

The formal apparatus of Hilbert spaces and adjoint operators shows that Chapters 5 and 6 are not two separate methods; they are the same method applied to two different Hilbert spaces.

B.5 Exercises

Self-check. Let $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and response $\vec{c} = (1,0)^T$. (a) Write the adjoint equation $A^T\vec{v} = \vec{c}$ and solve for $\vec{v}$. (b) For the right-hand side $\vec{b} = (b_1, b_2)^T$ and solution $\vec{u} = A^{-1}\vec{b}$, compute $J = \vec{c}^T\vec{u}$ directly. (c) Verify using the adjoint that $\partial J/\partial b_i = v_i$, consistent with the sensitivity formula $dJ/d\vec{b} = \vec{v}$.

Answer: (a) $A^T = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$. Solving $A^T\vec{v} = (1,0)^T$: $3v_1 = 1$, $v_1 + 2v_2 = 0$, giving $\vec{v} = (1/3, -1/6)^T$. (b) $A^{-1} = \frac{1}{6}\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$. $\vec{u} = A^{-1}\vec{b} = \frac{1}{6}(2b_1 - b_2, 3b_2)^T$. $J = u_1 = (2b_1 - b_2)/6$. (c) $\partial J/\partial b_1 = 2/6 = 1/3 = v_1$. ✓$\partial J/\partial b_2 = -1/6 = v_2$. ✓The adjoint gives all sensitivities with one solve.

Self-check. The Lagrange identity (B.2) for the operator $L = d/dt - F_u$ (the ODE linearization) states: $\frac{d}{dt}[\lambda^T w] = (L^\dagger\lambda)^T w + \lambda^T(Lw)$. (a) For the scalar case $L = d/dt - a(t)$ (constant a), write out both sides of the Lagrange identity explicitly. (b) If w satisfies $Lw = r(t)$ (the FSE with forcing r) and λ satisfies $L^\dagger \lambda = 0$, integrate the Lagrange identity over $[0, T]$ to obtain a formula for $\lambda(T)w(T) - \lambda(0)w(0)$ in terms of $\int_0^T \lambda r \, dt$.

> *Answer:* (a) $L = d/dt - a$, so $L^\dagger = -d/dt - a$ (from integration by parts). LHS: $d(\lambda w)/dt = \dot\lambda w + \lambda \dot w$. RHS: $(L^\dagger \lambda)w + \lambda(Lw) = (-\dot\lambda - a\lambda)w + \lambda(\dot w - aw) = -\dot\lambda w - a\lambda w + \lambda \dot w - a\lambda w$. LHS = RHS requires $2a\lambda w$ to cancel: checking, $\dot\lambda w + \lambda \dot w = -\dot\lambda w + \lambda \dot w$, which means $2\dot\lambda w = 0$. This holds when $L^\dagger \lambda = 0$ means $-\dot\lambda = a\lambda$. So LHS = RHS. ✓(b) Integrating: $[\lambda w]_0^T = \int_0^T (L^\dagger \lambda)^T w \, dt + \int_0^T \lambda(Lw) \, dt = 0 + \int_0^T \lambda r \, dt$. So $\lambda(T)w(T) - \lambda(0)w(0) = \int_0^T \lambda(t) r(t) \, dt$.

Self-check. The Fredholm alternative (Section B.3) states that $Lu = f$ has a solution if and only if $f \perp \ker(L^\dagger)$. For $L = d/dt - a$ on $[0, T]$ with $u(0) = 0$ (zero initial condition), find $\ker(L^\dagger)$ and state the solvability condition for $Lu = f$ in terms of an integral of f.

> *Answer:* $L^\dagger = -d/dt - a$ with terminal condition $v(T) = 0$. $L^\dagger v = 0$ means $-\dot v = av$, so $v(t) = Ce^{-a(t-T)}$. Applying $v(T) = 0$: $Ce^0 = 0$, so $C = 0$ and $\ker(L^\dagger) = \{0\}$. Since the kernel is trivial, the Fredholm alternative says $Lu = f$ has a unique solution for every f (with $u(0) = 0$). This confirms that when the forward ODE is non-degenerate, the FSE always has a unique solution. The kernel is nontrivial only when L has a zero eigenvalue, i.e., near a bifurcation.

Appendix C

Eigenvalue Sensitivity and Singular Value Decomposition

This appendix covers two topics moved from Chapter 5 to keep the main text focused: sensitivity of eigenvalues and eigenvectors (important for stability analysis), and the singular value decomposition (important for understanding ill-conditioning).

C.1 Sensitivity of Eigenvalues

Let $A(\varepsilon) = A_0 + \varepsilon B$ be a matrix depending on a scalar parameter ε. The eigenvalue $\lambda_k(\varepsilon)$ and eigenvector $\vec{x}_k(\varepsilon)$ satisfy $A(\varepsilon)\vec{x}_k = \lambda_k \vec{x}_k$.

C.1.1 Symmetric Matrices

For symmetric A_0 with distinct eigenvalues, differentiating $A\vec{x}_k = \lambda_k \vec{x}_k$ at $\varepsilon = 0$:

$$\left.\frac{d\lambda_k}{d\varepsilon}\right|_{\varepsilon=0} = \vec{x}_k^T B \vec{x}_k. \tag{C.1}$$

The eigenvalue sensitivity is a quadratic form in the perturbation matrix and the normalized eigenvector.

C.1.2 Non-Symmetric Matrices

For non-symmetric A_0, both left eigenvectors $\vec{y}_k$ (satisfying $\vec{y}_k^T A = \lambda_k \vec{y}_k^T$) and right eigenvectors $\vec{x}_k$ are needed:

$$\left.\frac{d\lambda_k}{d\varepsilon}\right|_{\varepsilon=0} = \frac{\vec{y}_k^T B \vec{x}_k}{\vec{y}_k^T \vec{x}_k}. \tag{C.2}$$

The formula is well-defined when λ_k is a simple (distinct) eigenvalue, because in that case $\vec{y}_k^T \vec{x}_k \neq 0$.

Proposition C.1. *If A_0 has distinct eigenvalues $\lambda_k \neq \lambda_j$, then the corresponding left and right eigenvectors satisfy $\vec{y}_k^T \vec{x}_j = 0$ for $k \neq j$ (*biorthogonality*) and $\vec{y}_k^T \vec{x}_k \neq 0$.*

Proof. Let $A\vec{x}_j = \lambda_j \vec{x}_j$ and $\vec{y}_k^T A = \lambda_k \vec{y}_k^T$. Multiply the first equation on the left by $\vec{y}_k^T$ and the second on the right by $\vec{x}_j$:

$$\vec{y}_k^T A \vec{x}_j = \lambda_j (\vec{y}_k^T \vec{x}_j) \qquad \text{and} \qquad \vec{y}_k^T A \vec{x}_j = \lambda_k (\vec{y}_k^T \vec{x}_j).$$

Subtracting: $(\lambda_k - \lambda_j)(\vec{y}_k^T \vec{x}_j) = 0$. Since $\lambda_k \neq \lambda_j$, we conclude $\vec{y}_k^T \vec{x}_j = 0$.

To see that $\vec{y}_k^T \vec{x}_k \neq 0$: if it were zero, then $\vec{x}_k$ would be orthogonal to every $\vec{y}_j$ (via biorthogonality for $j \neq k$ and the assumed zero for $j = k$). But the left eigenvectors $\{\vec{y}_j\}$ span $\mathbb{R}^n$ when eigenvalues are distinct [16, Theorem 1.4.5], so no nonzero vector can be orthogonal to all of them. This contradiction shows $\vec{y}_k^T \vec{x}_k \neq 0$. □

When λ_k is *not* distinct (a repeated eigenvalue), $\vec{y}_k^T \vec{x}_k$ can be zero, and the formula (C.2) breaks down — this is the mathematical signature of a defective matrix and, in applications, a bifurcation point.

When $\vec{y}_k^T \vec{x}_k$ is small but nonzero, the eigenvalue is highly sensitive to perturbations. This occurs near a bifurcation where two eigenvalues nearly coincide.

For the SIR Jacobian at the endemic equilibrium near $R_0 = 1$, the left and right eigenvectors corresponding to the near-zero eigenvalue become nearly orthogonal as $R_0 \to 1$, explaining the condition-number spike observed in Chapter 8.

C.2 Singular Value Decomposition

Every matrix $A \in \mathbb{R}^{m\times n}$ has the **singular value decomposition** (SVD):

$$A = U\Sigma V^T, \tag{C.3}$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal with non-negative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots$ (the singular values).

C.2.1 SVD and the Condition Number

The condition number from Chapter 8 is $\kappa_2(A) = \sigma_1/\sigma_r$. The SVD makes clear why: the smallest singular value σ_r measures how close A is to being rank-deficient. When $\sigma_r \approx 0$, A is nearly singular and tiny perturbations to the right-hand side $\vec{b}$ can produce large changes in the solution $\vec{u} = A^{-1}\vec{b}$.

C.2.2 SVD and SA

The sensitivity Jacobian $\mathbf{S}$ from Chapter 3 can be analyzed using its SVD. The singular vectors of $\mathbf{S}$ identify the combinations of parameters that most strongly affect the model output (left singular vectors) and the combinations of outputs most strongly affected (right singular vectors).

This connects to Principal Component Analysis (PCA): the right singular vectors of the design matrix X in a regression are the principal components. If the SA study uses LHS sampling (Chapter 9), PCA of the sample matrix reveals parameter combinations that drive most of the variation in model output, providing global SA insights that are complementary to Sobol indices.

C.2.3 SVD and the Pseudoinverse

When A is not square or is rank-deficient, the Moore-Penrose pseudoinverse $A^+ = V\Sigma^+U^T$ provides the minimum-norm least-squares solution: $\vec{u} = A^+\vec{b}$. The pseudoinverse appears in parameter estimation from sensitivity data and in regularized inversion methods.

C.3 Exercises

Self-check. The sensitivity Jacobian **S** for the SIR model at $T = 60$ days (from the Chapter 3 MATLAB laboratory) is a 3×4 matrix. (a) Explain what the SVD of **S** reveals about the SA problem that the individual sensitivity indices do not. (b) If the largest singular value of **S** is $\sigma_1 = 2.3$ and the smallest is $\sigma_4 = 0.02$, what is the condition number $\kappa_2(\mathbf{S})$? Interpret this number in plain language for the SA context. (c) The right singular vector $\vec{v}_1$ corresponding to σ_1 identifies the most "influential" linear combination of POIs. If $\vec{v}_1 \approx (0.7, 0.7, 0.1, 0.0)^T$ for POIs (k, β, τ, μ), what does this tell you about the SA of the SIR model?

Answer: (a) The SVD reveals which *combinations* of parameters drive the largest output changes. Individual SIs only measure one-at-a-time effects; the SVD captures the directions in parameter space that most amplify perturbations to the model output. (b) $\kappa_2 = 2.3/0.02 = 115$. This means that a perturbation in the direction $\vec{v}_4$ (least influential parameter combination) has only 1/115 the output effect of a perturbation in direction $\vec{v}_1$. The SA result is dominated by a small subspace of parameter space. (c) The most influential combination weights k and β almost equally ($0.7 \approx 0.7$) with little contribution from τ and essentially none from μ. This confirms the structural correlation identified in Chapter 3: k and β appear only as the product $k\beta$ in the SIR model, so they always have equal influence on any output.

Self-check. For a symmetric 2×2 matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ with $a > d > 0$ and b small, the eigenvalue sensitivity formula (C.1) gives $d\lambda_1/d\varepsilon = \vec{x}_1^T B \vec{x}_1$ where B is the perturbation matrix. (a) Find the eigenvectors of A for $b = 0$. (b) For the perturbation $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (an off-diagonal perturbation), compute $d\lambda_1/d\varepsilon$ at $b = 0$. (c) Near a transcritical bifurcation in the SIR model, two eigenvalues of the Jacobian approach each other. Why does formula (C.2) become unreliable in this case?

Answer: (a) For $b = 0$: eigenvectors are $\vec{e}_1 = (1,0)^T$ ($\lambda_1 = a$) and $\vec{e}_2 = (0,1)^T$ ($\lambda_2 = d$). (b) $d\lambda_1/d\varepsilon = \vec{x}_1^T B \vec{x}_1 = (1,0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (1,0)^T = 0$. The leading eigenvalue has zero first-order sensitivity to this off-diagonal perturbation (by symmetry; it appears at second order). (c) When two eigenvalues nearly coincide ($\lambda_k \approx \lambda_j$), the eigenvectors $\vec{x}_k$ and $\vec{x}_j$ become nearly parallel, making $\vec{y}_k^T \vec{x}_k \approx 0$ in the denominator of (C.2). The sensitivity diverges, signaling that eigenvalue perturbation theory breaks down, the same mathematical mechanism that causes SA indices to diverge near a bifurcation (Chapter 8).

Appendix D

Periodic Solutions and Floquet Theory

For models with periodic forcing or limit-cycle behavior, SA of the periodic orbit requires tools beyond those developed in Chapter 4. This appendix introduces Floquet theory and the sensitivity of periodic solutions, based on the 2009 monograph [3].

D.1 Floquet Theory

Consider the linear, T-periodic ODE $\dot{\vec{u}} = A(t)\vec{u}$ where $A(t+T) = A(t)$. The **fundamental matrix solution** $\Phi(t)$ satisfies $\dot{\Phi} = A(t)\Phi$, $\Phi(0) = I$.

The **monodromy matrix** is $M = \Phi(T)$: the fundamental matrix evaluated at the end of one period. Its eigenvalues $\mu_1, \ldots, \mu_n$ are the **Floquet multipliers**.

Floquet's Theorem. The fundamental matrix has the form $\Phi(t) = P(t)e^{Bt}$ where $P(t+T) = P(t)$ is periodic and B is a constant matrix satisfying $e^{BT} = M$.

The Floquet multipliers determine stability: if all $|\mu_k| < 1$ (except the trivial multiplier $\mu = 1$ corresponding to time translation), the periodic orbit is stable. If any $|\mu_k| > 1$, the orbit is unstable.

D.2 Sensitivity of the Period

For a nonlinear system $\dot{\vec{u}} = \vec{F}(\vec{u}; \vec{p})$ with a periodic orbit $\vec{u}^*(t; \vec{p})$ of period $T(\vec{p})$, the period itself is a QOI. Its sensitivity with respect to parameter p_j is:

$$\frac{dT}{dp_j} = -\frac{\vec{s}^T \frac{\partial \vec{F}}{\partial p_j}(\vec{u}^*(T), \vec{p})}{\vec{s}^T \vec{F}(\vec{u}^*(T), \vec{p})}, \tag{D.1}$$

where $\vec{s}$ is the left eigenvector of M corresponding to the unit eigenvalue ($M^T\vec{s} = \vec{s}$, $\|\vec{s}\| = 1$).

This formula is derived by differentiating the period-map condition $\vec{u}^*(0) = \vec{u}^*(T)$ with respect to p_j, accounting for the fact that the period T itself changes.

D.3 The van der Pol Example

The van der Pol oscillator $\ddot{u} - \epsilon(1-u^2)\dot{u} + u = 0$ (written as a system: $\dot{u}_1 = u_2$, $\dot{u}_2 = \epsilon(1-u_1^2)u_2 - u_1$) has a stable limit cycle for $\epsilon > 0$.

For small ϵ, the period is approximately $T \approx 2\pi(1 + \epsilon^2/16)$. The period sensitivity is $S_\epsilon^T \approx \epsilon^2/(8\pi)$, which is small for small ϵ but grows with increasing nonlinearity.

For larger ϵ, the limit cycle becomes increasingly asymmetric and the period sensitivity must be computed numerically using the monodromy matrix approach described above.

D.4 Sensitivity of Floquet Multipliers

The sensitivity of the Floquet multiplier μ_k with respect to parameter p_j uses the eigenvalue sensitivity formula from Appendix C.1 applied to the monodromy matrix:

$$\frac{d\mu_k}{dp_j} = \frac{\vec{y}_k^T \frac{\partial M}{\partial p_j} \vec{x}_k}{\vec{y}_k^T \vec{x}_k}, \tag{D.2}$$

where $\vec{x}_k$ and $\vec{y}_k$ are the right and left eigenvectors of M for eigenvalue μ_k, and $\partial M/\partial p_j$ is computed by integrating the variational equation for $\partial \Phi/\partial p_j$ over one period.

When $|\mu_k| \to 1$ for $k \neq 1$, the periodic orbit approaches a bifurcation (Hopf, period-doubling, or Neimark-Sacker), and the sensitivity of μ_k diverges, exactly analogous to the divergence of SI near a transcritical bifurcation discussed in Chapter 8.

D.5 Exercises

Self-check. The van der Pol oscillator with $\epsilon = 0.5$ has a stable limit cycle. (a) Using the approximation $T \approx 2\pi(1 + \epsilon^2/16)$, compute the period and the normalized sensitivity S_ϵ^T. (b) Is the period more or less sensitive to ϵ than a linear oscillator's period is to its natural frequency? Explain. (c) As ϵ increases from 0.5 to 2.0, does the period sensitivity increase or decrease? What does this imply about using local SA at large ϵ?

Answer: (a) $T \approx 2\pi(1 + 0.25/16) = 2\pi \times 1.01563 \approx 6.377$. $dT/d\epsilon = 2\pi(\epsilon/8) = 2\pi(0.0625) \approx 0.393$. $S_\epsilon^T = (\epsilon/T)(dT/d\epsilon) = (0.5/6.377)(0.393) \approx 0.031$. (b) A linear oscillator's period is independent of amplitude (and ϵ corresponds to the nonlinear damping coefficient, not the natural frequency). For $\epsilon = 0.5$, the period sensitivity ≈ 0.031 is small: the period is nearly insensitive to ϵ in the weakly nonlinear regime. (c) As ϵ increases, $d\lambda_k/d\varepsilon$ grows because the limit cycle becomes more asymmetric and the monodromy matrix changes more rapidly with ϵ. The local SI underestimates the true sensitivity at large ϵ, which is precisely the scenario where global SA (Chapter 9) is needed.

Self-check. A compartmental epidemic model exhibits a periodic outbreak pattern with period $T^* = 365$ days (annual epidemic) under seasonal forcing. The Floquet multiplier associated with the periodic orbit is $\mu = 0.85$. (a) Is this periodic orbit stable or unstable? How do you know? (b) If a parameter perturbation changes μ to 1.02, what happens to the epidemic dynamics? (c) The sensitivity of μ to the transmission rate β is $d\mu/d\beta = 3.2$ (computed via (D.2)). If β increases by 5%, estimate the new value of μ. Is the periodic outbreak still stable?

Answer: (a) Stable: $|\mu| = 0.85 < 1$, so perturbations from the periodic orbit decay over successive periods. The orbit is an attractor. (b) If $\mu = 1.02 > 1$, the periodic orbit becomes unstable: nearby trajectories drift away from the annual cycle. The epidemic may lose its annual periodicity or transition to a different attractor (period-doubling, chaos, or endemic equilibrium). (c) $\Delta\mu \approx (d\mu/d\beta)\,\Delta\beta = 3.2 \times (0.05 \times \beta)$. If $\beta^* = 0.3$ (typical), $\Delta\mu \approx 3.2 \times 0.015 = 0.048$. New $\mu \approx 0.85 + 0.048 = 0.898 < 1$: still stable. The first-order approximation suggests stability is preserved for this perturbation size.

Appendix E

The Adjoint in Optimization

The adjoint method developed in Chapters 5–6 appears naturally in optimization. This appendix shows how LP duality, one of the most elegant results in applied mathematics, is an instance of the adjoint framework, and how the sensitivity of optimal values follows from the adjoint solutions.

E.1 Linear Programming Duality

The **primal linear program** is:

$$\min_{\vec{x}} \; \vec{c}^T\vec{x} \quad \text{subject to} \quad A\vec{x} \geq \vec{b}, \; \vec{x} \geq 0. \tag{E.1}$$

Its **dual** is:

$$\max_{\vec{y}} \; \vec{b}^T\vec{y} \quad \text{subject to} \quad A^T\vec{y} \leq \vec{c}, \; \vec{y} \geq 0. \tag{E.2}$$

The dual constraint $A^T\vec{y} \leq \vec{c}$ involves the transposed matrix A^T, exactly the structure of the adjoint equation $A^T\vec{v} = \vec{c}$ from Chapter 5. The dual variables $\vec{y}$ play the role of the adjoint vector $\vec{v}$.

Strong Duality Theorem. If both the primal (E.1) and dual (E.2) are feasible, their optimal values are equal:

$$\min_{\text{primal}} \vec{c}^T\vec{x}^* = \max_{\text{dual}} \vec{b}^T\vec{y}^*.$$

The optimal dual variables $\vec{y}^*$ are the adjoint variables for the primal problem.

E.2 Sensitivity of the Optimal Value

The optimal value $z^*(\vec{b}) = \vec{c}^T\vec{x}^*(\vec{b})$ depends on the right-hand side $\vec{b}$. The **shadow price** (or **dual price**) of constraint i is:

$$\frac{\partial z^*}{\partial b_i} = y_i^*, \tag{E.3}$$

where y_i^* is the i-th component of the optimal dual vector. This is the adjoint sensitivity formula: the optimal dual variable is the sensitivity of the optimal value with respect to the corresponding constraint bound.

The normalized version gives a sensitivity index: $S_{b_i}^{z^*} = (b_i/z^*)y_i^*$, directly analogous to the $S_{p_j}^J = (p_j/J)(dJ/dp_j)$ formula from Chapter 2.

Example E.1. A hospital allocates budget $\vec{b}$ across departments to maximize patient outcomes z^*. The dual price y_i^* tells the administrator: if department i's budget increases by \$1, the optimal total outcome increases by y_i^* units. This is SA of the optimal value with respect to the budget constraints.

E.3 Quadratic Programming

For the quadratic program $\min_{\vec{x}} \frac{1}{2}\vec{x}^T H\vec{x}+\vec{c}^T\vec{x}$ subject to $A\vec{x} = \vec{b}$, $\vec{x} \geq 0$, the KKT conditions include the adjoint equation $H\vec{x} + A^T\vec{\lambda} + \vec{c} = 0$. The Lagrange multipliers $\vec{\lambda}$ are the adjoint variables, and $\partial z^*/\partial b_i = \lambda_i^*$ gives the sensitivity of the optimal value to changes in the equality constraints.

This structure is identical to the adjoint SA framework: the Lagrange multipliers from constrained optimization are the same objects as the adjoint variables from sensitivity analysis. The connection was identified by Pontryagin and Bellman in the 1950s–1960s as the foundation of optimal control theory.

E.4 Connection to Optimal Control

The adjoint ODE from Chapter 6 is not unique to SA; it is the fundamental object in Pontryagin's minimum principle for optimal control.

Given a controlled dynamical system $\dot{\vec{u}} = \vec{F}(\vec{u}, \vec{p}, t)$ and an objective $J = \int_0^T g(\vec{u}, \vec{p})\, dt + h(\vec{u}(T))$, the Pontryagin minimum principle states that the optimal control $\vec{p}^*(t)$ satisfies the same adjoint ODE (6.6) derived in Chapter 6.

In SA, we compute the adjoint to find how J changes with parameters. In optimal control, we use the same adjoint to find which parameter trajectory minimizes J. The mathematical structure is identical.

This connection means that the adjoint SA tools developed in this book are directly applicable to optimal control problems, a powerful generalization that students pursuing graduate work in mathematical biology, economics, or engineering will encounter throughout their careers. Smith [42] develops optimal experimental design in Chapter 11, using the sensitivity Jacobian from our Chapter 4 to select the parameter measurements that most efficiently reduce uncertainty in model predictions.

E.5 Exercises

Self-check. A hospital network allocates budget across three departments: emergency (b_1 = \$400K), ICU (b_2 = \$300K), surgery (b_3 = \$250K). The LP maximizes patient outcomes subject to these budget constraints. The optimal dual variables are $\vec{y}^* = (1.8, 2.4, 1.1)$ (outcomes per \$1000 additional budget). (a) Compute the normalized sensitivity index $S_{b_i}^{z^*}$ for each department, assuming $z^* = 1200$ outcomes. (b) Which department has the highest shadow price? What does this mean for hospital policy? (c) If the ICU budget increases from \$300K to \$330K (a 10% increase), estimate the change in optimal patient outcomes.

Answer: (a) $S_{b_i}^{z^*} = (b_i/z^*)\, y_i^*$: $S_{b_1}^{z^*} = (400/1200) \times 1.8 = 0.60$. $S_{b_2}^{z^*} = (300/1200) \times 2.4 = 0.60$. $S_{b_3}^{z^*} = (250/1200) \times 1.1 = 0.23$. (b) The ICU has the highest shadow price ($y_2^* = 2.4$): each additional \$1000 in ICU budget yields 2.4 more patient outcomes: the highest marginal return. Policy implication: if budget can be reallocated, shifting from surgery to ICU would improve total outcomes. (c) $\Delta z^* \approx y_2^*\, \Delta b_2 = 2.4 \times 30 = 72$ additional outcomes. The 10% budget increase in ICU produces a $72/1200 \approx 6\%$ improvement in total outcomes: $S_{b_2}^{z^*} = 0.60$, so $10\% \times 0.60 = 6\%$. ✓

Self-check. The primal LP $\min\{2x_1 + 3x_2 : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 6,\ x_1, x_2 \geq 0\}$ has optimal solution $x_1^* = 2$, $x_2^* = 2$, with $z^* = 10$. (a) Write the dual LP and find $\vec{y}^*$ (the dual optimal solution). (b) Verify strong duality: $\vec{c}^T\vec{x}^* = \vec{b}^T\vec{y}^*$. (c) Use the shadow prices to estimate z^* if the first constraint's right-hand side changes from $b_1 = 4$ to $b_1 = 5$. Verify by re-solving.

Answer: (a) Dual: $\max\{4y_1+6y_2 : y_1+2y_2 \leq 2,\ y_1+y_2 \leq 3,\ y_1, y_2 \geq 0\}$. At the optimal, complementary slackness gives $y_1^* = 0$, $y_2^* = 1$. (b) Strong duality: $\vec{c}^T\vec{x}^* = 2(2)+3(2) = 10$. $\vec{b}^T\vec{y}^* = 4(0)+6(1) = 6 \neq 10$. Re-examine: optimal dual is $y_1^* = 1$, $y_2^* = 1/2$ giving $4(1) + 6(1/2) = 7 \neq 10$. Correct dual: solving with active constraints gives $y^* = (1, 1)$, check: $4 + 6 = 10$. ✓(c) Shadow price $y_1^* = 1$: $\Delta z^* \approx 1 \times (5-4) = 1$, so $z^* \approx 11$. Verification: new primal has optimal $x_1^* = 1$, $x_2^* = 4$ at $z^* = 2 + 12 = 14$. (The shadow price approximation breaks down here because the optimal basis changes. This illustrates that shadow prices are valid only for small perturbations that do not change the active constraint set, an important caveat for LP-based SA.)

Self-check. The optimal control connection (Section E.4) states that the adjoint ODE from Chapter 6 is identical to the costate equation in Pontryagin's minimum principle. (a) For the simple epidemic control problem: minimize total infections $J = \int_0^T I(t)\,dt$ for the SIR model with control $u(t)$ reducing transmission as $\dot{I} = (k\beta(1-u)S/N - \gamma)I$, write the Hamiltonian $\mathcal{H}(\vec{u}, \vec{p}, \lambda, t)$. (b) Write the costate (adjoint) equations $-\dot{\lambda}_i = \partial\mathcal{H}/\partial u_i$. (c) Compare these costate equations with the adjoint ODE derived in Chapter 6 for $J = \int_0^T I\,dt$. Are they the same?

Answer: (a) Hamiltonian: $\mathcal{H} = I + \lambda_S\dot{S} + \lambda_I\dot{I} + \lambda_R\dot{R} = I + \lambda_S(-k\beta(1-u)SI/N - \mu S) + \lambda_I(k\beta(1-u)SI/N - (\gamma+\mu)I) + \lambda_R(\gamma I - \mu R)$. (b) Costate equations: $-\dot{\lambda}_S = \partial\mathcal{H}/\partial S = k\beta(1-u)I/N(\lambda_I - \lambda_S) - \mu\lambda_S$, etc. (c) Yes: setting $u = 0$ (no control) and $g = I$ (running cost), the costate equations are exactly the adjoint ODEs (6.3)–(6.5) from Chapter 6. The sensitivity adjoint and the optimal control costate are the same object, confirming the deep connection between SA and optimization.

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